Causal Inference I

Mixtape Session



Roadmap

Introduction

Foundational ideas

Identification and Assumptions

Estimators

Advice

Welcome!

- I'm Scott Cunningham, professor of economics at Baylor University, visiting professor at Harvard Kennedy School, author of Causal Inference: the Mixtape and Causal Inference: the Remix
- I work on a variety of eclectic topics – sex work, methamphetamine drug policy, abortion policy, and more recently, detecting and treating self harm in prisons and jails and the behavior of AI agents when undertaking empirical research
- I'll be talking today about matching and weighting, and more generally unconfoundedness

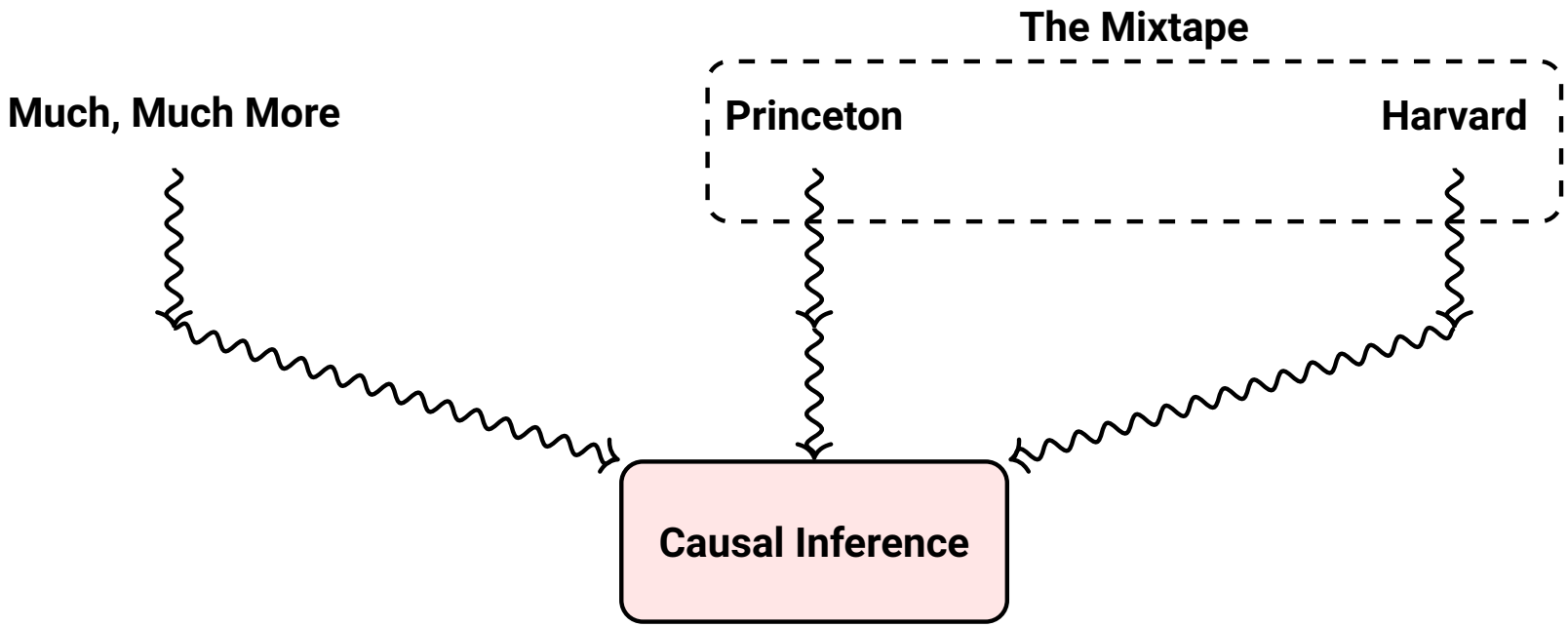
About me

- This was the conference where I learned more about causal inference, having only a background in econometrics, but minimally statistics
- Speakers were Alberto Abadie, Justin McCrary, and the rest I cannot remember, but Abadie taught then what I’m teaching today

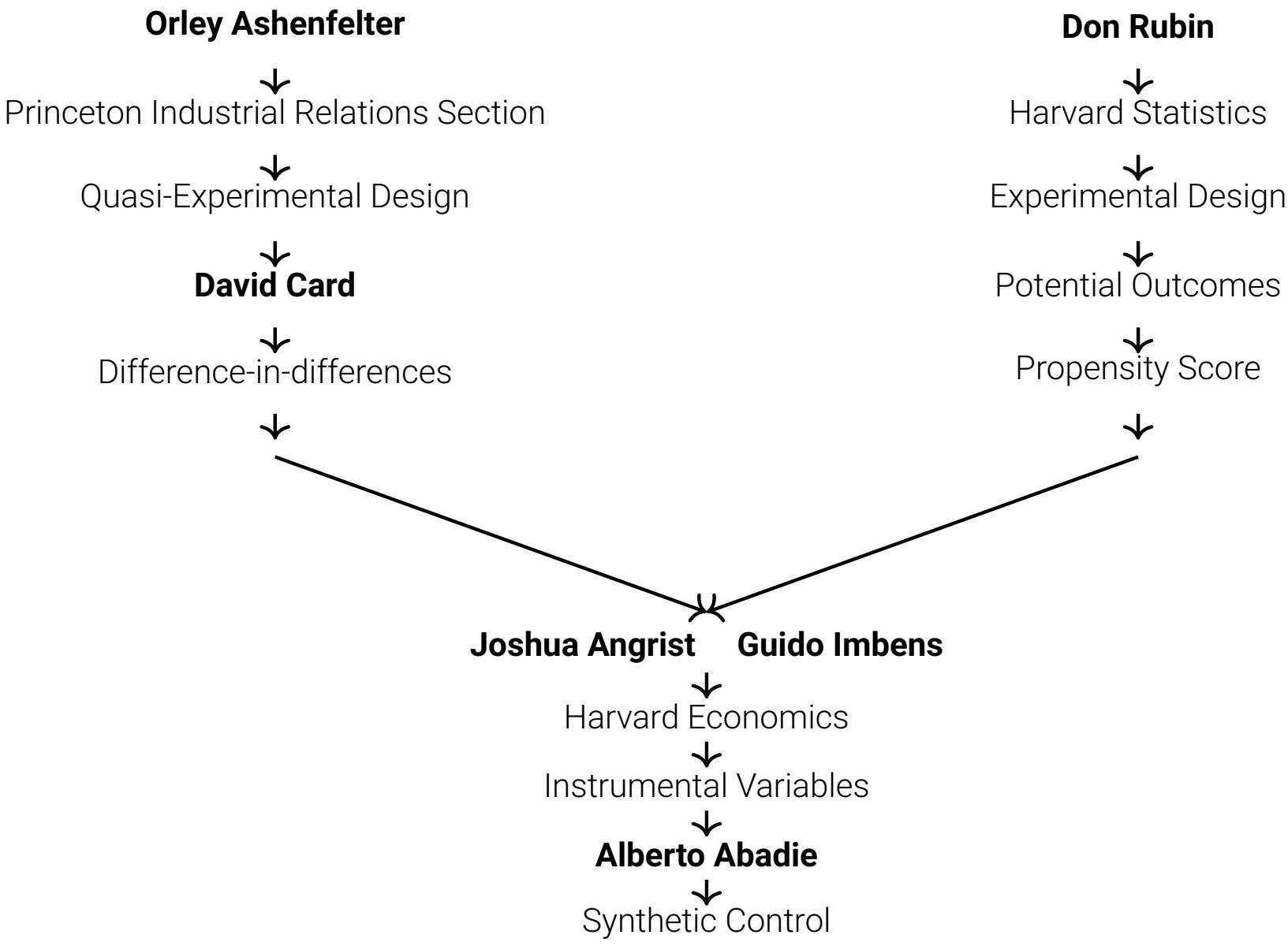
Expectations

- This workshop will review potential outcomes, various definitions and decompositions, introduce unconfoundedness, and review standard estimators
  - → Biases we typically are focused on removing
  - → Causal parameters we want to recover
  - → Assumptions needed to do it called ”identification”
- My teaching style is to try and not leave any one behind so I emphasize a lot of basics, applications, simple things repeatedly and then extend it in more advanced material
- Estimators will review weighting, matching, bias adjustment, as well as regression adjustment
- I emphasize heterogenous treatment effects and I tend to be very applied and tend to use code to break down concepts

Three Tributaries of Causal Inference



Two rivers into causal inference



Roadmap

Introduction

Foundational ideas

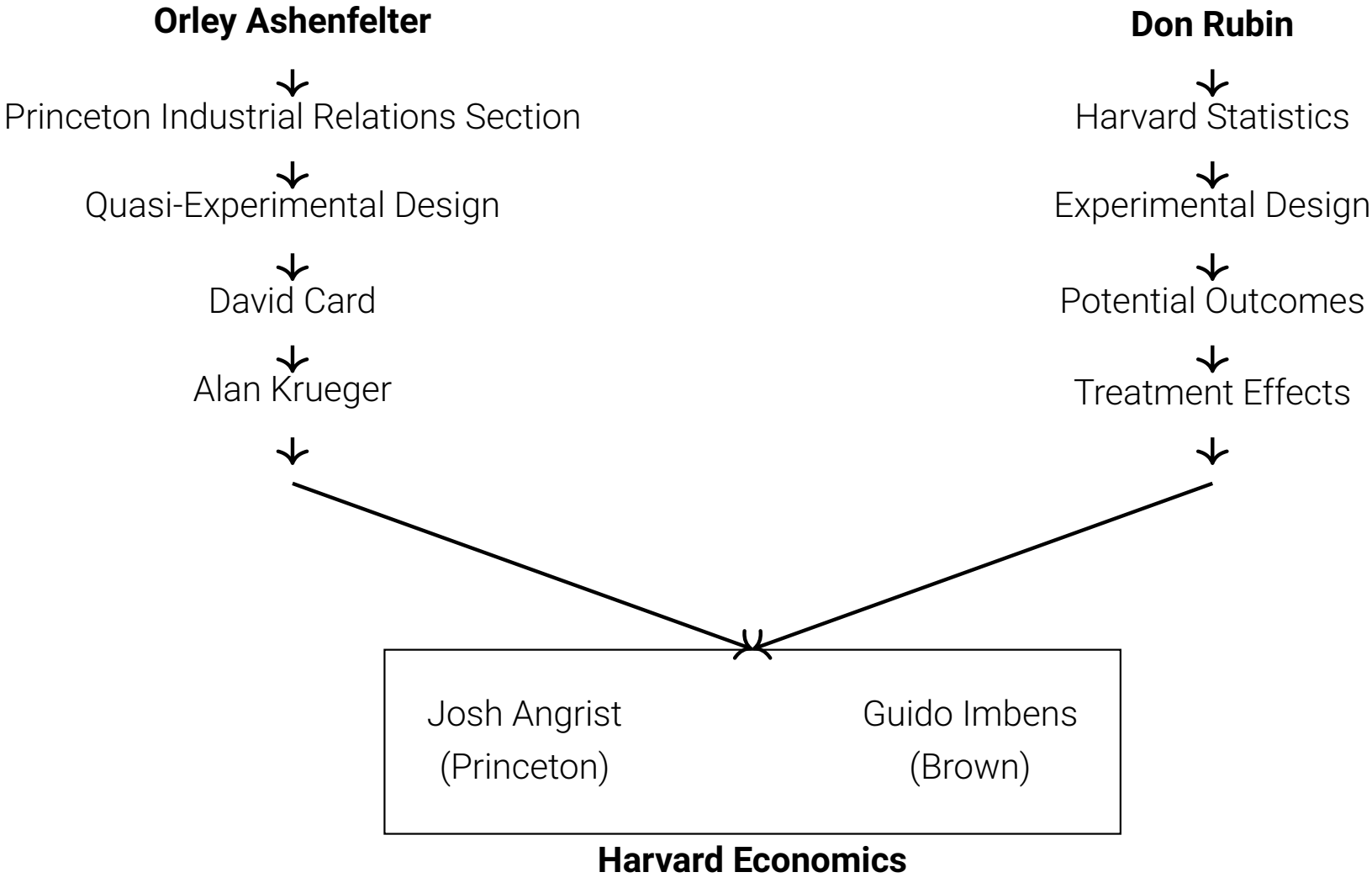
- Experimental Design Fuses with Quasi-Experimental Design
- Parameters and Estimands

Identification and Assumptions

Estimators

Advice

Two rivers into causal inference



Background I: Harvard Stats and Potential Outcomes

- Don Rubin, former chair of Harvard stats, is the main source of potential outcomes, building on Jerzy Neyman's 1923 work.
- Rubin's influential 1970s papers advocated for causal inference using contrasts of $Y(1)$ and $Y(0)$.
- Neyman's notation, initially in Polish, was translated into English in 1990, likely due to Rubin.
- Rubin expanded Neyman's ideas from experiments to observational studies, leading to developments like matching and propensity score methods.

Background II: Princeton Industrial Relations Section

- Princeton Industrial Relations Section, highly rigorous, nonpartisan and empirical, is very involved in overturning the apple cart in economics
  - → Key faculty are Orley Ashenfelter, David Card, Alan Krueger.
  - → Key students include Bob Lalonde, Josh Angrist, Steve Pischke, John Dinardo, Janet Currie, Anne Case, and many more.
- Late 1970s and early 1980s, empirical labor crisis in macroeconomics (Lucas critique) and empirical labor economics
- Orley is coming to the conclusion over the 1970s that the empirical work in labor economics is severely flawed ("not credible"), very focused on studying large job trainings programs
- Listen to David Card:
  https://youtu.be/1soLdywFb_Q?si=BCVqYeRz6jYiwHTQ&t=1580

Background III: Princeton Industrial Relations Section

Example of Princeton Paradigm Emerging

- Lalonde (1986) wanted to see how well econometric methods performed outside of randomized assignment (see Guido Imbens and Yiqing Xu, 2024)
  - → Lalonde analyzed an RCT on a job training program, found an average treatment effect of +$800, replaced the experimental control group with survey data, reran econometric methods, and estimates range from -$16,000 to +$3,000, casting doubt on the credibility of quantitative labor economics at the time
- Rubin's work, despite its prominence in statistics in the 1970s and early 1980s, is not part of the pantheon of methods LaLonde uses
- Later it does, though, with Dehejia and Wahba's work (1999, 2002), who were two of Rubin and Imbens' PhD students at Harvard

Potential outcomes notation

Let the treatment be a binary variable:

$$D_{i,t} = \begin{cases} 1 \text{ if placed on ventilator at time } t \\ 0 \text{ if not placed on ventilator at time } t \end{cases}$$

where i indexes an individual observation, such as a person

Potential outcomes notation

Potential outcomes:

$$Y_{i,t}^{j} = \begin{cases} 1 \text{ health if placed on ventilator at time } t \\ 0 \text{ health if not placed on ventilator at time } t \end{cases}$$

where j indexes a potential treatment status for the same i person at the same t point in time

Potential outcomes

- Prior to being placed on the ventilator, there are multiple possibilities that could happen
  - → They could be placed on the ventilator, call it Y^1, and something could happen, or
  - → They could not be placed on the ventilator, call it Y^0, and something could happen
- There are only two things that could happen because the treatment only has two sides to it – you either are on it or you aren't
- Neyman, on the other hand, had an urn consisting of balls representing a large number of treatments (fertilizer types) and so had many potential outcomes

Potential outcomes vs realized outcomes

- Potential outcomes are ”real” in that either of those beforehand could happen – Rubin ”a priori fixed but unknown” – but can also be “counterfactual”
- Once the decision to put the patient on the ventilator happens, one potential outcome is chosen, the other eliminated
- That’s the difference between potential outcomes (Y^1, Y^0) and realized outcomes (Y, the data)
- Potential outcomes are more or less describing what would happen under different treatments, but realized outcomes are describing what did in fact happen for a given treatment and exist in a dataset

Treatment Assignment Mechanism

- The second part of causal inference in the design tradition has to do with the reason that a person gets on a ventilator
- Did a doctor choose to do it? Did the patient choose to do it? Did an algorithm choose to do it?
- But it's more than that – *why* was the choice made, not *who* made it
- For instance, a doctor might always make it, but it matters if she flipped a coin vs read the chart and made the "best decision"

Important definitions

Definition 1: Individual treatment effect

The individual treatment effect, δ_i, associated with a ventilator is equal to $Y_i^1 - Y_i^0$.

Important definitions

Definition 2: Switching equation

An individual's realized health outcome, Y_i, is determined by treatment assignment, D_i which selects one of the potential outcomes:

$$
\begin{aligned}
Y_i &= D_i Y_i^1 + (1 - D_i) Y_i^0 \\
Y_i &= \begin{cases} Y_i^1 \text{ if } D_i = 1 \\ Y_i^0 \text{ if } D_i = 0 \end{cases}
\end{aligned}
$$

Not the same as treatment assignment mechanism. Treatment assignment mechanism describes why one patient but not another was put on the ventilator

Missing data problem

Definition 3: Fundamental problem of causal inference

If you need both potential outcomes to know causality with certainty, then since it is impossible to observe both Y_i^1 and Y_i^0 for the same individual, δ_i, is *unknowable*.

Missing data problem

- Fundamental problem of causal inference is deep and impossible to overcome – not even with more data
- Why doesn't more data solve the problem? Because you need two potential outcomes for inferring causal effects and only have one
- Causal inference is a missing data problem
- This is, though, good news – statistics had been working on filling in missing data for a long time (including work by Rubin)
- All of causal inference involves imputing missing counterfactuals – but not all imputations are equal

Research designs identify estimands

> *A research design is an identification strategy: it reveals the invisible target parameter hidden inside a calculation.*

1.

A *calculation*

(the difference in means)

+

2.

An *assumption*

(unconfoundedness)

Average Treatment Effects

Definition 4: Average treatment effect (ATE)

The average treatment effect is the population average of all i individual treatment effects

$$\begin{aligned} E[\delta] &= E[Y^1 - Y^0] \\ &= E[Y^1] - E[Y^0] \end{aligned}$$

Aggregate parameters based on individual treatment effects are *summaries* of individual treatment effects

Cannot be calculated because Y_i^1 and Y_i^0 do not exist *for the same unit i* due to switching equation

Conditional Average Treatment Effects

Definition 5: Average Treatment Effect on the Treated (ATT)

The average treatment effect on the treatment group is equal to the average treatment effect conditional on being a treatment group member:

$$\begin{aligned} E[\delta|D=1] &= E[Y^1 - Y^0|D=1] \\ &= E[Y^1|D=1] - E[Y^0|D=1] \end{aligned}$$

Cannot be calculated because Y_i^1 and Y_i^0 do not exist *for the same unit i* due to switching equation.

Conditional Average Treatment Effects

Definition 6: Average Treatment Effect on the Untreated (ATU)

The average treatment effect on the untreated group is equal to the average treatment effect conditional on being untreated:

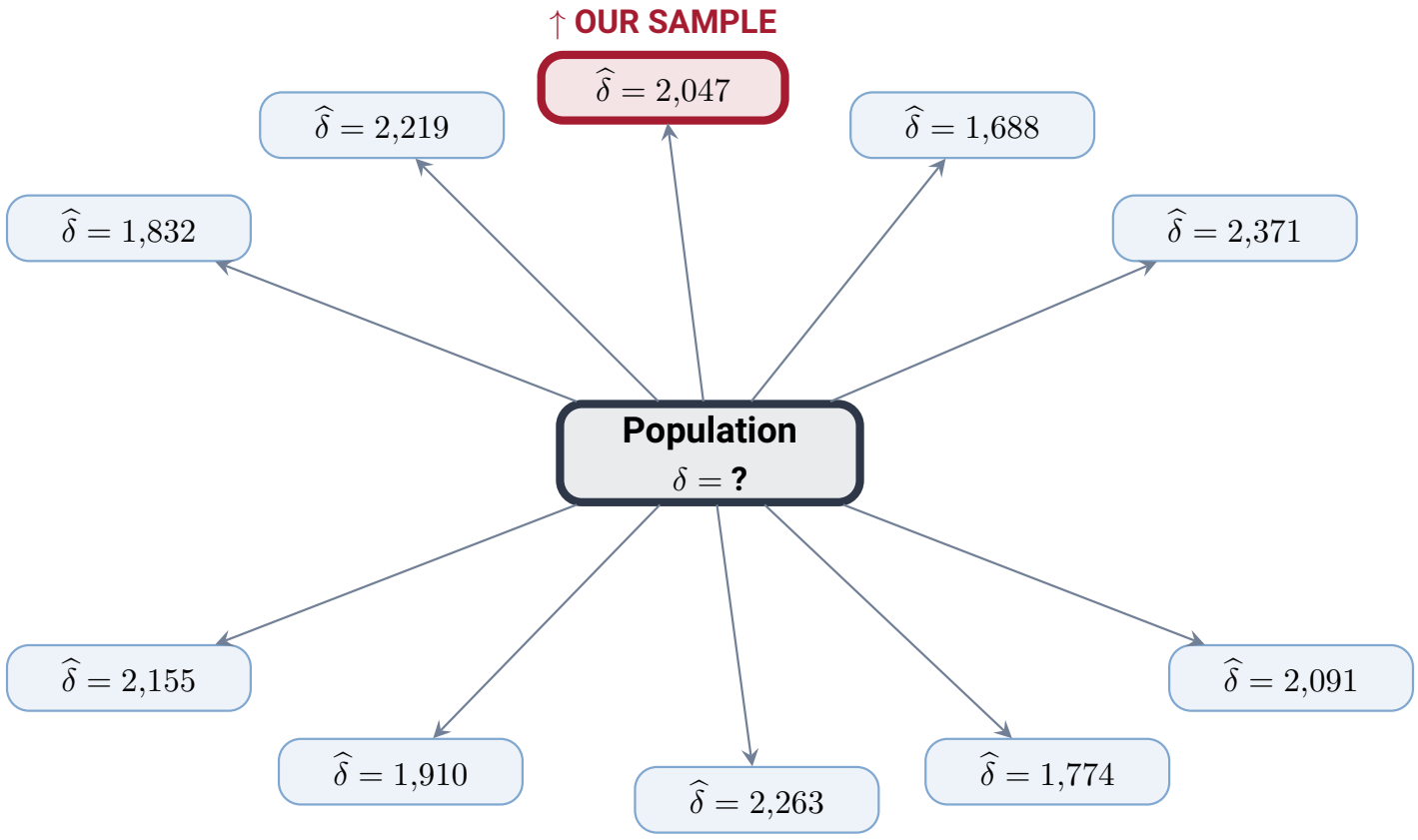
$$\begin{aligned} E[\delta|D=0] &= E[Y^1 - Y^0|D=0] \\ &= E[Y^1|D=0] - E[Y^0|D=0] \end{aligned}$$

Cannot be calculated because Y_i^1 and Y_i^0 do not exist *for the same unit i* due to switching equation

Average Treatment Effects are Simple Summaries

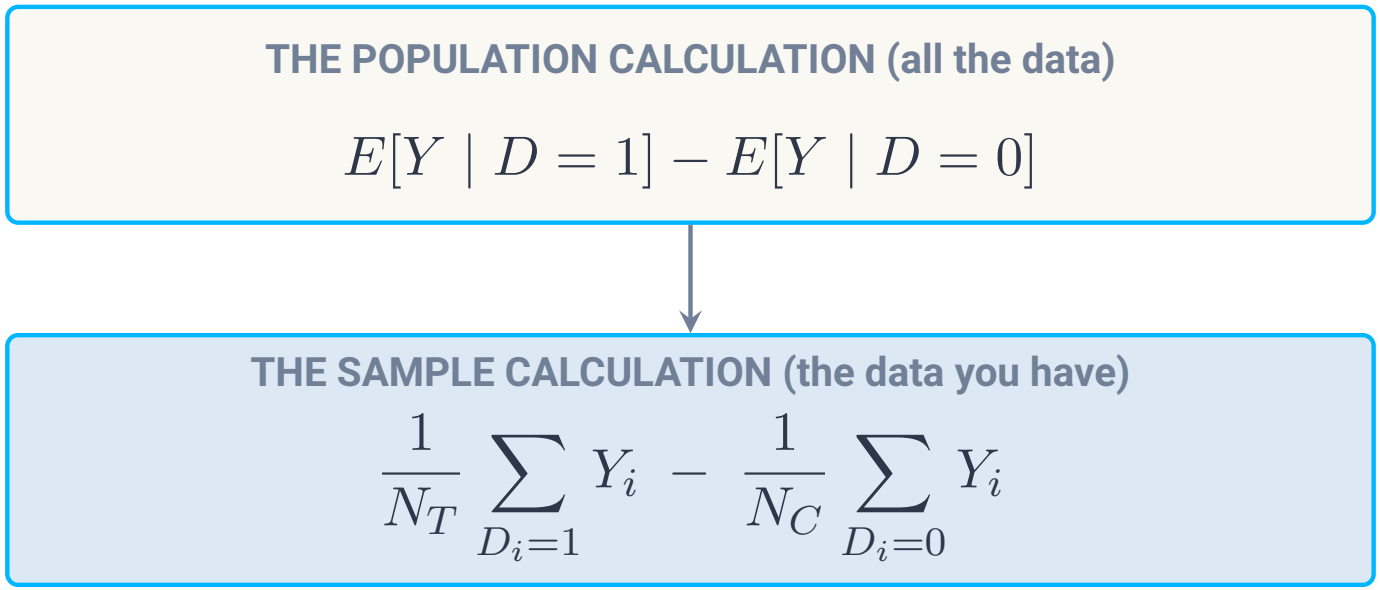
- Notice how in all three of these, all we did was take the defined treatment effect at the individual and aggregate
- Because aggregate causal parameters are *summaries* of individual treatment effects, each of which cannot be calculated, the aggregates cannot be calculated either
- Missing data in this context isn't missing your car keys – it's missing unicorns and fire breathing dragons (fictional vs real data)
- While we cannot measure average causal effects, we can estimate them, but only in situations and we review one – randomization

Your estimate is one of many you could have drawn



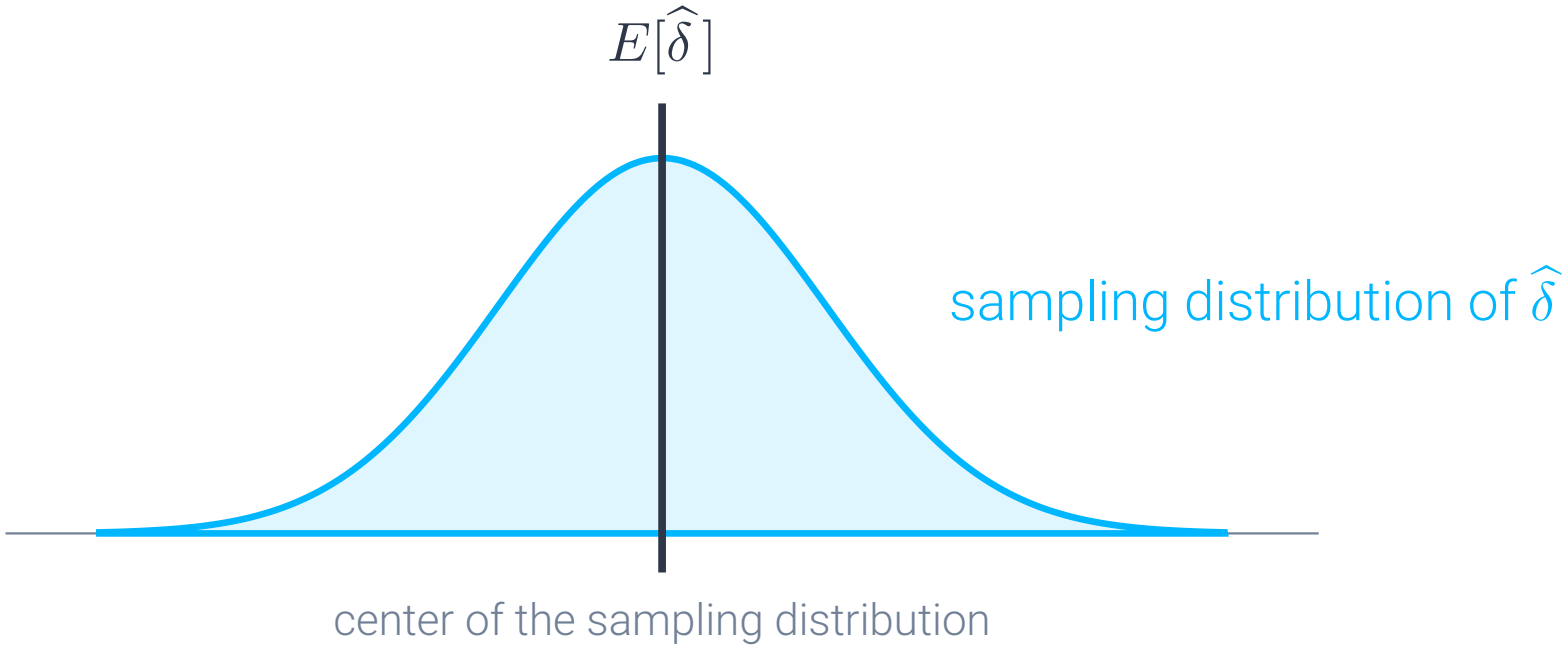
Every sample gives a different $\widehat{\delta}$ — all of them are estimating the same population number.

Two calculations, two notations



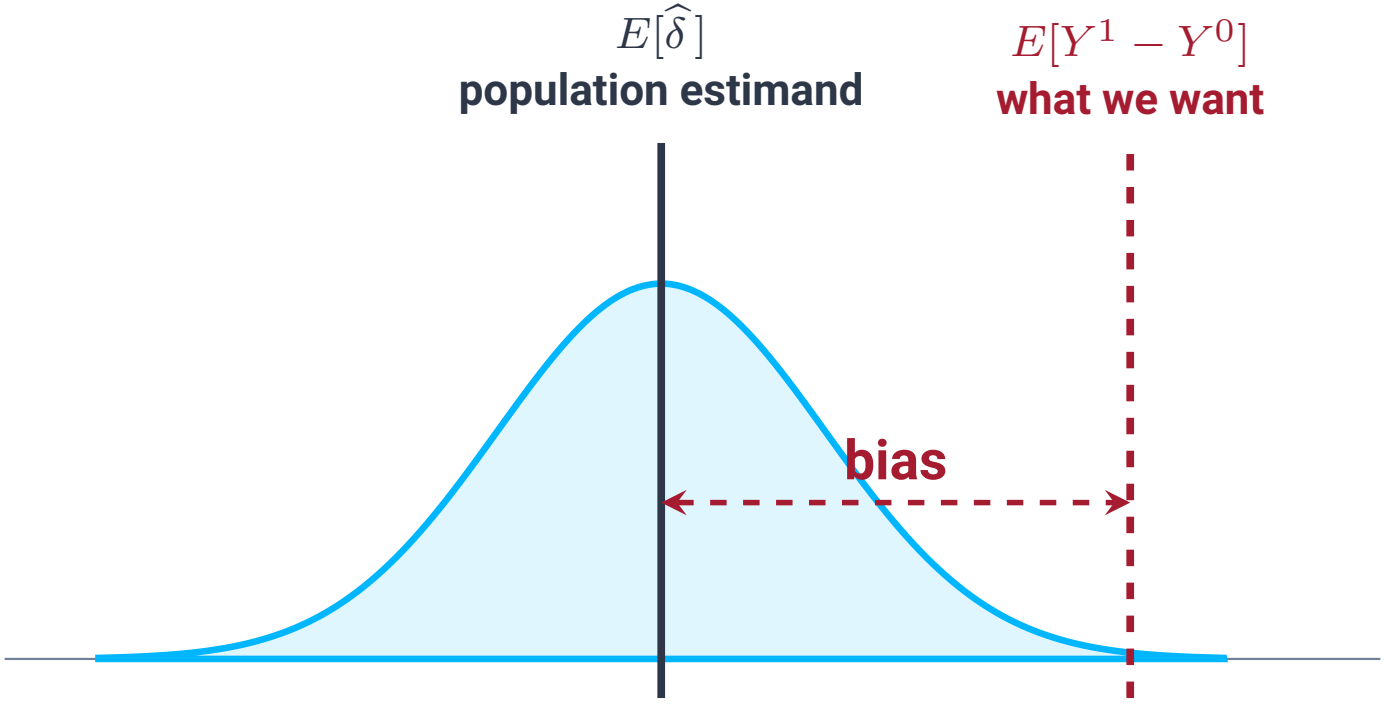
- Expectations $E[\cdot]$ upstairs, sample averages downstairs — the notation change is the whole point
- The top one is the **estimand**: what you would calculate if you had every unit
- Even with "all" your data, treat it as a sample from an infinite super-population

Your calculation has a distribution, and it has a center



- Law of large numbers and the CLT: that center *is* the population estimand
- So the sample calculation is an honest estimator *of the population calculation*

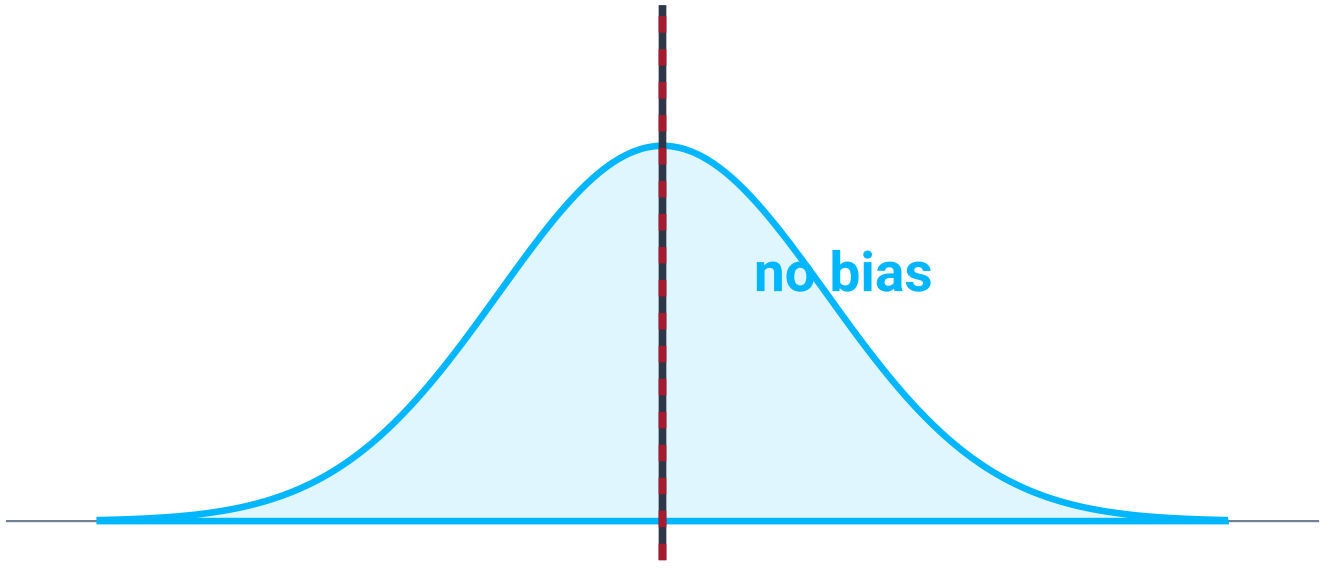
But is the population estimand the causal effect?



- Nothing about sampling closes this gap — more data shrinks the *spread*, not the *gap*
- The gap is selection bias plus heterogeneity bias, which we decompose next

Identification is what closes the gap

$$E[\widehat{\delta}] = E[Y^1 - Y^0]$$



- Randomization does it for free — that is why Fisher recommended it
- Outside the experiment we buy it with assumptions: **unconfoundedness** and **common support**
- That is the entire subject of today

Simple Comparisons

Definition 7: Simple difference in mean outcomes (SDO)

A simple difference in mean outcomes (SDO) can be approximated by comparing the sample average outcome for the treatment group ($D = 1$) with a comparison group ($D = 0$)

$$SDO = E[Y^1|D = 1] - E[Y^0|D = 0]$$

SDO is not a causal parameter because it's comparing Y^1 and Y^0 for different units, not the same units, so what is it measuring?

Decomposition of the SDO

Decomposition of the SDO

The SDO is made up of three things:

$$\begin{aligned} E[Y^1|D=1] - E[Y^0|D=0] &= ATE \\ &\quad + E[Y^0|D=1] - E[Y^0|D=0] \\ &\quad + (1-\pi)(ATT - ATU) \end{aligned}$$

where π is the share of units in the treatment group. Now let's see how we get here.

Two ways to rewrite the ATE

- Before we get started, let's look closely at the definition of the ATE
  - → We can express the ATE as the weighted average of the ATT and the ATU, and ...
  - → We can express the ATE as the sum of four conditional means multiplied by corresponding weights (Law of iterated expectations)
- They are in fact the exact same formula once you write down the definition of the ATT and the ATU
- Let's do it together before we get started:
  https://docs.google.com/spreadsheets/d/10DuQqGtH_Ewea7zQoLTFYHbnvqaTVDhn2GDzq3Oa6EQ/edit?usp=sharing

Begin with ATE definition

Rewrite the definition of the ATE

$$
\begin{aligned}
\text{ATE} &= E[Y^1] - E[Y^0] \\
&= \pi ATT + (1-\pi) ATU \\
&= \pi E[Y^1|D=1] - \pi E[Y^0|D=1] \\
&\quad +(1-\pi)E[Y^1|D=0] - (1-\pi)E[Y^0|D=0] \\
\text{ATE} &= \{\pi E[Y^1|D=1] + (1-\pi)E[Y^1|D=0]\} \\
&\quad -\{\pi E[Y^0|D=1] + (1-\pi)E[Y^0|D=0]\}
\end{aligned}
$$

Let's make this easier to read by replacing the last row with letters

Change notation

Substitute letters for expectations

$$
\begin{aligned}
E[Y^1|D=1] &= a \\
E[Y^1|D=0] &= b \\
E[Y^0|D=1] &= c \\
E[Y^0|D=0] &= d \\
\text{ATE} &= e
\end{aligned}
$$

Rewrite ATE definition

Rewrite ATE

$$e = \{\pi a + (1-\pi)b\} - \{\pi c + (1-\pi)d\}$$

Simple manipulation of ATE definition

$$
\begin{aligned}
e &= \{\pi a + (1-\pi)b\} - \{\pi c + (1-\pi)d\} \\
e &= \pi a + b - \pi b - \pi c - d + \pi d \\
e &= \pi a + b - \pi b - \pi c - d + \pi d + (\mathbf{a} - \mathbf{a}) + (\mathbf{c} - \mathbf{c}) + (\mathbf{d} - \mathbf{d}) \\
0 &= e - \pi a - b + \pi b + \pi c + d - \pi d - \mathbf{a} + \mathbf{a} - \mathbf{c} + \mathbf{c} - \mathbf{d} + \mathbf{d} \\
\mathbf{a} - \mathbf{d} &= e - \pi a - b + \pi b + \pi c + d - \pi d + \mathbf{a} - \mathbf{c} + \mathbf{c} - \mathbf{d} \\
\mathbf{a} - \mathbf{d} &= e + (\mathbf{c} - \mathbf{d}) + \mathbf{a} - \pi a - b + \pi b - \mathbf{c} + \pi c + d - \pi d \\
\mathbf{a} - \mathbf{d} &= e + (\mathbf{c} - \mathbf{d}) + (1-\pi)a - (1-\pi)b + (1-\pi)d - (1-\pi)c \\
\mathbf{a} - \mathbf{d} &= e + (\mathbf{c} - \mathbf{d}) + (1-\pi)(a-c) - (1-\pi)(b-d)
\end{aligned}
$$

Carry forward from previous slide

$$\mathbf{a - d} \quad = \quad e + (\mathbf{c - d}) + (1 - \pi)(a - c) - (1 - \pi)(b - d)$$

Replace letters with original terms

$$\begin{aligned} E[Y^1|D = 1] - E[Y^0|D = 0] \quad = \quad & \text{ATE} \\ & +(E[Y^0|D = 1] - E[Y^0|D = 0]) \\ & +(1 - \pi)(\underbrace{\{E[Y^1|D = 1] - E[Y^0|D = 1]\}}_{\text{ATT}} \\ & -(1 - \pi)(\underbrace{\{E[Y^1|D = 0] - E[Y^0|D = 0]\}}_{\text{ATU}} \end{aligned}$$

Purple terms are based on missing counterfactuals and therefore cannot be calculated.

This is an *identity*

Decomposition of the SDO

Decomposition of the SDO

$$
\begin{aligned}
E[Y^1|D=1] - E[Y^0|D=0] \quad &= \quad ATE \\
&\quad +(E[Y^0|D=1] - E[Y^0|D=0]) \\
&\quad +(1-\pi)(ATT - ATU)
\end{aligned}
$$

Although we started with π (the share of units in treatment), note we have weighted the heterogeneity bias term by $1-\pi$ (the share of units in control)

Estimate SDO with sample averages

$$\underbrace{E_N[Y|D=1] - E_N[Y|D=0]}_{\text{Estimate of SDO}} = \underbrace{E[Y^1] - E[Y^0]}_{\text{Average Treatment Effect}}$$
$$+ \underbrace{E[Y^0|D=1] - E[Y^0|D=0]}_{\text{Selection bias}}$$
$$+ \underbrace{(1-\pi)(ATT - ATU)}_{\text{Heterogenous treatment effect bias}}$$

Using the switching equation and sample averages, we can calculate $E_N[Y|D=1] \rightarrow E[Y^1|D=1]$, $E_N[Y|D=0] \rightarrow E[Y^0|D=0]$ and $(1-\pi)$ is the share of the population in the control group.

Illustrating selection bias with spreadsheets

- Eliminating selection bias requires understanding the selection mechanism – why did units end up treated but not others?
- Perfect Doctor exercise: assume a doctor sees patients, knows each person's treatment effects, despite counterfactuals, and assigns treatment based on whether gains are positive or not
- Illustrate decomposition together using numerical example: https://docs.google.com/spreadsheets/d/10DuQqGtH_Ewea7zQoLTFYHbnvqaTVDhn2GDzq3Oa6EQ/edit?usp=sharing

Roadmap

Introduction

Foundational ideas

Identification and Assumptions

- Independence and Selection Bias
- Unconfoundedness
- Unconfoundedness and common support
- Why economists resist this
- How long does credibility take?

Estimators

Advice

Selection bias and Design

- Selection bias in causal inference is when one or both mean potential outcome differ by treatment status
- Source of the bias is caused by the why people get treated, or what's called the "treatment assignment *mechanism*"
- But as we will see, if the treatment was assigned randomly (exogeneity), the implications differ than if they were assigned "optimally" (endogeneity)

Independence

Independence assumption

Treatment is assigned to a population independent of that population's potential outcomes

$$(Y^0, Y^1) \perp\!\!\!\perp D$$

This is random or quasi-random assignment and ensures mean potential outcomes for the treatment group and control group are the same. Also ensures other variables are distributed the same for a large sample.

$$\begin{aligned} E[Y^0|D=1] &= E[Y^0|D=0] \\ E[Y^1|D=1] &= E[Y^1|D=0] \end{aligned}$$

Random Assignment Solves the Selection Problem

$$\underbrace{E_N[y_i|d_i=1] - E_N[y_i|d_i=0]}_{\text{SDO}} = \underbrace{E[Y^1] - E[Y^0]}_{\text{Average Treatment Effect}} + \underbrace{E[Y^0|D=1] - E[Y^0|D=0]}_{\text{Selection bias}} + \underbrace{(1-\pi)(ATT - ATU)}_{\text{Heterogenous treatment effect bias}}$$

- If treatment is independent of potential outcomes, then swap out equations and **selection bias** zeroes out:

$$E[Y^0|D=1] - E[Y^0|D=0] = 0$$

Random Assignment Solves the Heterogenous Treatment Effects

- How does randomization affect heterogeneity treatment effects bias from the third line? Rewrite definitions for ATT and ATU:

$$\text{ATT} = E[Y^1|D=1] - E[Y^0|D=1]$$
$$\text{ATU} = E[Y^1|D=0] - E[Y^0|D=0]$$

- Rewrite the third row bias after $1-\pi$:

$$\begin{aligned} ATT - ATU &= \textbf{E[Y}^1 \mid \textbf{D=1]} - E[Y^0|D=1] \\ &\quad -\textbf{E[Y}^1 \mid \textbf{D=0]} + E[Y^0|D=0] \\ &= 0 \end{aligned}$$

- If treatment is independent of potential outcomes, then:

$$\begin{aligned} E_N[y_i|d_i=1] - E_N[y_i|d_i=0] &= E[Y^1] - E[Y^0] \\ SDO &= ATE \end{aligned}$$

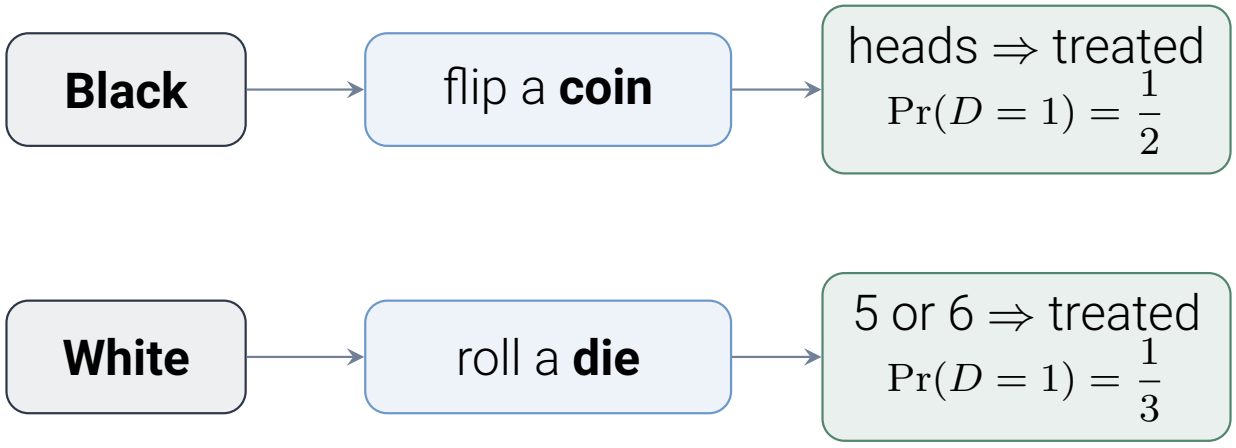
Identification with Full Independence

$$\underbrace{E_N[Y_i|D_i=1] - E_N[Y_i|D_i=0]}_{\text{Estimate of SDO}} = \underbrace{E[Y^1] - E[Y^0]}_{\text{Average Treatment Effect}} + \underbrace{0}_{\text{Selection bias}} + \underbrace{0}_{\text{Heterogenous treatment effect bias}}$$

SDO is unbiased estimate of ATE with randomized treatment assignment because it sets selection bias to zero and $ATT = ATU$.

Randomization, but not the same coin for everyone

Imagine an experimentalist who randomizes — but uses a *different device* depending on your race.



- This *is* random assignment — nobody chose their own treatment
- But independence holds only **within** race, not across it

The population

Group	N	$E[Y^0\|X]$	$E[Y^1\|X]$	ATE	Treated	Control
Black	120	6	14	**8**	60	60
White	120	20	24	**4**	40	80
All	240	$E[Y^0] = 13$	$E[Y^1] = 19$	**6**	100	140

- Read it *across*: $14 - 6 = 8$ for blacks, $24 - 20 = 4$ for whites, then average $\rightarrow 6$
- Read it *down*: $E[Y^1] - E[Y^0] = 19 - 13 = 6$
- Same answer either way — that's the law of iterated expectations
- Notice race does two jobs — it changes $\Pr(D = 1)$ *and* it changes $E[Y^0|X]$ (6 vs 20)

Within each group, the simple comparison is exactly right

$$E[Y|D=1, \text{Black}] - E[Y|D=0, \text{Black}] = 14 - 6 = \mathbf{8}$$
$$E[Y|D=1, \text{White}] - E[Y|D=0, \text{White}] = 24 - 20 = \mathbf{4}$$

- Each recovers that group's ATE exactly
- Because *within* the group, the coin (or the die) made treatment independent of potential outcomes
- Same independence result as before — applied one group at a time
- $$(Y^0, Y^1) \perp\!\!\!\perp D \mid X$$

Now pool them and it breaks

$$E[Y|D=1] = \frac{60(14)+40(24)}{100} = \frac{1800}{100} = 18$$
$$E[Y|D=0] = \frac{60(6)+80(20)}{140} = \frac{1960}{140} = 14$$

SDO		TRUE ATE
$18-14=$ **4**	$\neq$	**6**

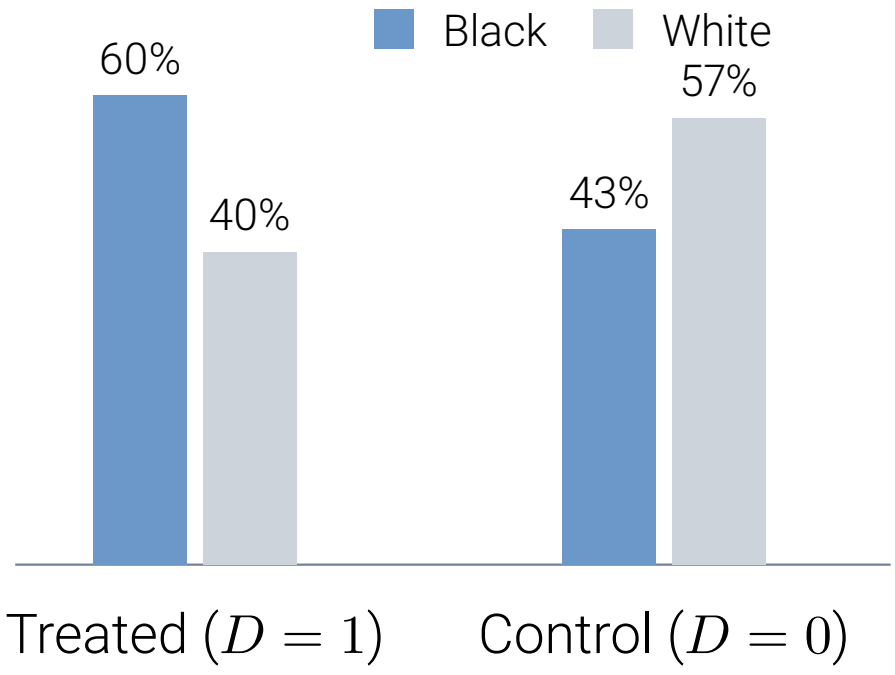
The comparison is biased even though *every single unit* was randomized.

Why? The treated and control groups are different mixtures

	Black	White	$E[Y^0]$ of that group
Treated ($N = 100$)	60%	40%	11.6
Control ($N = 140$)	43%	57%	14.0

- The die sent fewer whites into treatment, so the controls skew white
- And whites have a much higher Y^0 (20 vs 6)
- So the control group is a bad stand-in for the treated group's missing Y^0
- Exactly our selection bias term:
  $E[Y^0|D = 1] - E[Y^0|D = 0] = 11.6 - 14 = -2.4$

“Mixtures” has a name: imbalance



- The two arms are not the same population
- That gap is **covariate imbalance** — and it is measurable

Putting a number on it: normalized difference

Normalized difference (Imbens and Rubin 2015)

$$\Delta = \frac{\overline{X}_1 - \overline{X}_0}{\sqrt{(s_1^2 + s_0^2)/2}}$$

For our "black" indicator:

$$\Delta = \frac{0.60 - 0.43}{\sqrt{(0.240 + 0.245)/2}} = \frac{0.171}{0.492} = \mathbf{0.35}$$

- Rule of thumb from Imbens and Rubin: $|\Delta| > 0.25$ is severe imbalance
- Past that, linear regression adjustment gets sensitive to specification

Why not just use a t-statistic?

- A t-stat mostly tells you your sample size
- Double N and the same imbalance gets “more significant”
- The normalized difference is scale-free and doesn’t move with N
- Imbens and Rubin (2015) are emphatic about this — balance is a property of the *design*, not of your power

One covariate is easy. K covariates are not

- Here we had one binary covariate, so we could just look at it
- With K covariates you need balance on *every* dimension and every interaction
- You could report K normalized differences — people do
- But we'll see a better move: collapse all K into one number and check balance on *that*

That number is the **propensity score**, and we come back to it this afternoon.

Imbalance is the engine. It drives both bias terms

- Imbalance alone is harmless. It has to interact with something
- Imbalance $\times$ ($E[Y^0|X]$ varies with X) $\rightarrow$ selection bias $= -2.4$
- Imbalance $\times$ (ATE varies with X) $\rightarrow$ heterogeneity bias $= +0.4$

$$\underbrace{4}_{SDO} = \underbrace{6}_{ATE} \underbrace{-2.4}_{\text{selection}} \underbrace{+0.4}_{\text{heterogeneity}}$$

Balance the covariate and both terms go to zero.

The law of iterated expectations puts it back together

Condition first, then average over the covariate:

$$
\begin{aligned}
ATE &= E[\,E[Y^1 - Y^0 \mid X]\,] \\
&= \sum_x \Big(E[Y|D=1, X=x] - E[Y|D=0, X=x]\Big) \Pr(X=x) \\
&= \underbrace{8}_{\text{Black}} \cdot \frac{1}{2} + \underbrace{4}_{\text{White}} \cdot \frac{1}{2} \;=\; \mathbf{6} \quad \checkmark
\end{aligned}
$$

- Compare *within* race, then weight by how common each race is
- Weight by $\Pr(X=x)$ and you get the ATE; weight by $\Pr(X=x|D=1)$ and you get the ATT

That is the whole idea

- **Unconfoundedness**: there is a covariate X such that, within its values, treatment is as good as randomly assigned

$$(Y^0, Y^1) \perp\!\!\!\perp D \mid X$$

- **Common support**: every value of X has units in *both* arms, or you cannot make the within-group comparison at all

$$0 < \Pr(D = 1|X) < 1$$

- Here both held by construction — the experimentalist told us the design
- In observational work nobody tells us. We have to argue for X
- Everything today is a way of making that comparison and taking that weighted average

Missing Counterfactuals

- Every causal parameter is missing counterfactual potential outcomes (e.g., ATT is missing $E[Y^0|D=1]$)
- If we are going to estimate the ATT, then it must be we are estimating $E[Y^0|D=1]$ somehow, but how?
- If we are using observed confounders, then we are imputing the missing counterfactual using covariates in the comparison group
- All of causal inference uses one group to impute the missing counterfactuals for another group – but this is only valid for some groups under some assumptions

Now let’s write them down properly

- Two sites and a coin got us there by hand
- Everything today rests on the same two assumptions, stated generally
- Both are strong. One of them you can’t even check

Assumption I: Unconfoundedness

$(Y^0, Y^1) \perp\!\!\!\perp D|X$. There exists a set X of known and quantified confounders such that after adjusting for them, treatment assignment is *independent of potential outcomes*.

$$
\begin{aligned}
E[Y^0|D=1, X=x] &= E[Y^0|D=0, X=x] \\
E[Y^1|D=1, X=x] &= E[Y^1|D=0, X=x]
\end{aligned}
$$

Unconfoundedness justifies substituting units in treatment for control based on $X = x$, but only for units with *exactly* the same characteristics X; you'll need this for the ATE, ATT and the ATU

Selection on Observables and Random Choices

- Ask yourself ”Who chooses the treatment? What do they know when they did it? Why did they do it? What did they hope would happen?”
- If you are solving causal inference problems with covariates, you are saying this:
  - → Covariates chose the treatments for people
  - → Exactly identical people were only flipping coins after that
- Selection on observables is a ”strong assumption”, so sometimes we weaken it both because it’s more plausible, and because weaker assumptions may be all you need to get your target parameter

Assumption: Weak Unconfoundedness

$(Y^0) \perp\!\!\!\perp D|X$. There exists a set X of known and quantified confounders such that after adjusting for them, treatment assignment is *independent* of Y^0

$$E[Y^0|D=1, X=x] \quad = \quad E[Y^0|D=0, X=x]$$

Weak unconfoundedness will mean we can replace missing $E[Y^0|D=1, X=x]$ with observed $E[Y^0|D=0, X=x]$ so long as you have exact matches. Weak unconfoundedness matters for the ATT but "strong" unconfoundedness matters for ATE, ATT and the ATU

Identifying assumption II: Common support

For ranges of X, there is a positive probability of being both treated and untreated

- There exists units in treatment and control with same values of X – you can't make the substitutions otherwise
- Dimension k means every specific combination of the conditioning set (e.g., not males and old, but adult males, adult females, youth male, youth female)
- Testable because common support is observable unlike unconfoundedness, but as you can imagine if the dimensions of X gets large (and with a continuous covariate it's infinite!) then it won't hold in any finite sample!

Assumptions combined

But if we have them both (represented below), we can even outside of an RCT estimate the ATE through nonparametric matching

1. $(Y^1, Y^0) \perp\!\!\!\perp D|X$ (strong unconfoundedness)
2. $0 < Pr(D = 1|X) < 1$ with probability one (common support)

Comparing groups of individuals *who have the same values of* X, treatment is no longer based gains, δ.

The second term implies we have people in treatment and control for every strata of X

Estimating ATE with Assumptions

- Unconfoundedness means you can replace the missing $E[Y^0|D=1,X=x]$ with the observed $E[Y^0|D=0,X=x]$ because they are in fact the same anyway

$$
\begin{aligned}
E[Y^1-Y^0|X=x] &= E[Y^1-Y^0|X=x] \\
&= E[Y^1|X=x]-E[Y^0|X=x] \\
&= \underbrace{E[Y^1|D=1,X=x]}_{E[Y^1] \text{ same for both}} - \underbrace{E[Y^0|D=0,X=x]}_{E[Y^0] \text{ same for both}} \\
&= \underbrace{E[Y|D=1,X=x]-E[Y|D=0,X=x]}_{\text{Switching equation from potential outcomes to realized outcomes}}
\end{aligned}
$$

- Not testable (requires DAG), but if true, those steps lead you to an estimator

Estimating ATE with Assumptions

- Common support acts as a bridge connecting two groups potential outcomes through common covariates which allows you to take weighted averages of differences in means

$$
\begin{aligned}
ATE &= E[Y^1 - Y^0] = E\Big[E[Y^1 - Y^0 \mid X]\Big] \\
&= \int E[Y^1 - Y^0 | X, D = 1] dPr(X) \\
&= \int \left(E[Y|X, D = 1] - E[Y|X, D = 0]\right) dPr(X)
\end{aligned}
$$

- You're taking weighted averages of difference in means over the distribution of X

Maybe You Want the ATT

- In the social sciences it's rare you want the ATE — and if you did, you probably can't get it
- Vaccines are meant for everyone, so there you do want the ATE
- Remedial summer school is aimed at struggling students, so you want the ATT
- Discrimination studies are ATT questions — the effect on the group discriminated against

ATT Identification

We can modify those assumptions and weaken both which helps a lot

1. $Y^0 \perp\!\!\!\perp D|X$ (weak unconfoundedness)
2. $Pr(D = 1|X) < 1$ (with $Pr(D = 1) > 0$) (weak support)

We don't need full common support because we don't need to find counterfactuals for the control group – we only need units in the control group that match with our treatment group

Selection is weaker too, like I said – they are not entirely irrational, but who knows if it helps you

Estimating ATT

Weighted averages under both assumptions:

$$ATT = \int (E[Y|X, D = 1] - E[Y|X, D = 0])\, dPr(X|D = 1)$$

We match units in treatment and control because under weak unconfoundedness they're substitutable, and we use weak common support so that we can actually do it, then we take weighted averages over the differences.

Similar logic for the ATU, but oddly enough, I've never seen a study estimate the ATU even though presumably that would be useful

So why isn’t everyone doing this?

- Random within X, and covariate adjustment gets the ATE exactly
- Outside economics that’s been ordinary practice for decades (statistics, epidemiology, political science)
- Inside economics it’s been treated with suspicion for forty years
- Two reasons — one empirical, one theoretical

The empirical reason: LaLonde (1986)

- LaLonde took an experiment, threw away the experimental controls, replaced them with survey data, and ran the econometric methods of the day
- Estimates ranged from $-\$16{,}000$ to $+\$3{,}000$, against an experimental benchmark of about $+\$900$
- Wrong magnitude. Often wrong sign.

LaLonde became the founding trauma

- The lesson economists drew: *selection on observables does not work*
- It is a large part of why a generation went looking for natural experiments instead
- We will come back to LaLonde's own data later today and see what Dehejia and Wahba were able to do with it

The theoretical reason: people are not coin flippers

- Economics models people as choosing on *expected gains* — utility maximization
- Roy (1951): workers sort into occupations on comparative advantage
- Selection on the gain itself
- Heckman and Vytlacil call it **essential heterogeneity** — agents sorting on their own δ_i

This is just the Perfect Doctor again

- Selection on δ — the hardest of our three cases
- And if people really do it, no covariate adjustment saves you
- The thing they're sorting on is unobserved to you
- Especially for big decisions: college, marriage, moving, treatment

Imbens (2004) calls that view “unduly narrow”

> *In this debate it has been argued that agents’ optimizing behavior precludes their choices being independent of the potential outcomes. . . This seems an unduly narrow view.*
>
> — Imbens, ReStat 2004

He offers three arguments in reply. Let’s take them one at a time.

Imbens, argument one: it is the natural first move

- The starting point for evaluating any program is comparing average outcomes for treated and control units
- The logical next step is adjusting that difference for background characteristics

> *Such an analysis may not lead to the final word. . . but its absence would seem difficult to rationalize in a serious attempt to understand the evidence.*

Imbens, argument two: you are comparing units anyway

- Every evaluation compares treated units to *some* untreated units
- The question is never *whether* to compare — only *which units* best represent the treated had they not been treated

- Economic theory earns its keep here: it tells you which variables enter the utility function and which enter the constraints
- Unconfoundedness then just asserts that you *observed* those variables
- That is an empirical question, not a matter of principle

Imbens, argument three: agents don't know their own δ_i

- Selection on gains requires the agent to know their *own* idiosyncratic treatment effect
- But people optimize on *expected* gains given the information they actually have

- If that information set is in your data, unconfoundedness can hold even for a fully optimizing agent
- Often the decision maker isn't even the unit — a caseworker, a judge, a doctor, an algorithm — and their information is far more observable

Which moves the whole question onto balance

- Unconfoundedness is **untestable**. You don't check it — you argue for it
- But it has an observable shadow: **balance**
- Same X, exchangeable units. So the covariate distributions should look alike after adjustment

Balance is the part you can be held to

- If the arms aren’t balanced, you didn’t build the experiment you claimed to approximate
- And nothing downstream fixes it
- So the rest of today really comes down to two pictures
- A *balance* table and an *overlap* plot

Check one: balance

- Are the covariate distributions of the two arms actually alike?
- Report **normalized differences** in covariate means (Imbens and Rubin 2015)

- Not t-statistics — a t-stat mostly tells you your sample size, and a huge N will flag differences that don't matter
- A normalized difference is scale-free and doesn't move with N

Check two: overlap

- Plot the propensity score by arm and look at where the two histograms live
- Common support is **observable**, even though unconfoundedness is not — so there is no excuse for not showing it

- Where the treated have no comparable controls, you have no evidence, only extrapolation
- We'll do exactly this with the LaLonde data later today

Check three: falsification

- Estimate the “effect” of treatment on something it cannot possibly have affected
- The most useful case is a **lagged outcome** — earnings before the program started
- If you find an effect there, your design is finding something other than the treatment

Check four: bring a second control group

- Rosenbaum (1987). Find two comparison groups, C_1 and C_2
- Example: LaLonde's trainees could be compared to CPS controls or to PSID controls
- **Neither group was treated**
- So the "effect" of being in C_1 rather than C_2 is **zero by construction**
- Condition on X and go estimate it anyway

What the test is actually asking

Unconfoundedness says all three of these are the same thing:

$$\underbrace{E[Y^0|X, C_1]}_{\text{CPS controls}} = \underbrace{E[Y^0|X, C_2]}_{\text{PSID controls}} = \underbrace{E[Y^0|X, D = 1]}_{\text{what we actually need}}$$

- The first equality you *can* check — both sides are observed
- The one you need is the one you can never see
- If CPS and PSID controls disagree after conditioning on X, your X isn't doing its job
- And then there's no reason to believe the equality you're relying on

What you learn, and what you don't

- **They disagree**: unconfoundedness fails with the X you have. You do *not* learn which control group is right — probably neither
- **They agree**: encouraging, but not proof. Both could share the same bias

- So the test only has teeth if the two groups would be biased *differently*
- Heckman, Ichimura and Todd (1997): eligible nonparticipants vs. ineligibles
- One group selected itself out. The other was screened out by rule
- Different selection stories, so agreement between them actually means something

Check five: sensitivity analysis

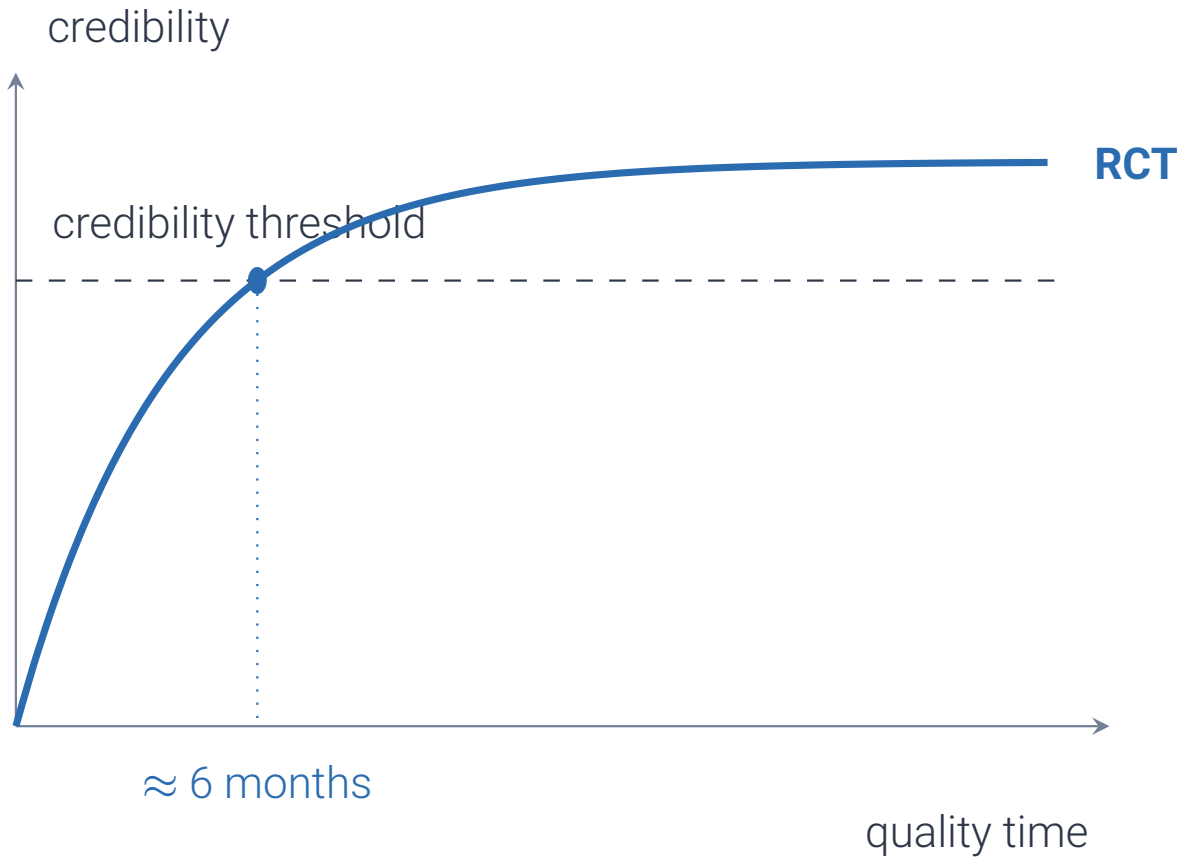
- You cannot test unconfoundedness, but you can ask how *fragile* your answer is
- How much unobserved confounding would it take to overturn the result? (Rosenbaum and Rubin 1983; Imbens 2003)
- In its extreme form this becomes a bounds analysis (Manski 1990)
- Honest framing: not “this is unconfounded” but “here is what it would take for me to be wrong”

Read this paper

Rubin, Donald B. (2008). “For Objective Causal Inference, Design Trumps Analysis.” *Annals of Applied Statistics* 2(3): 808–840.

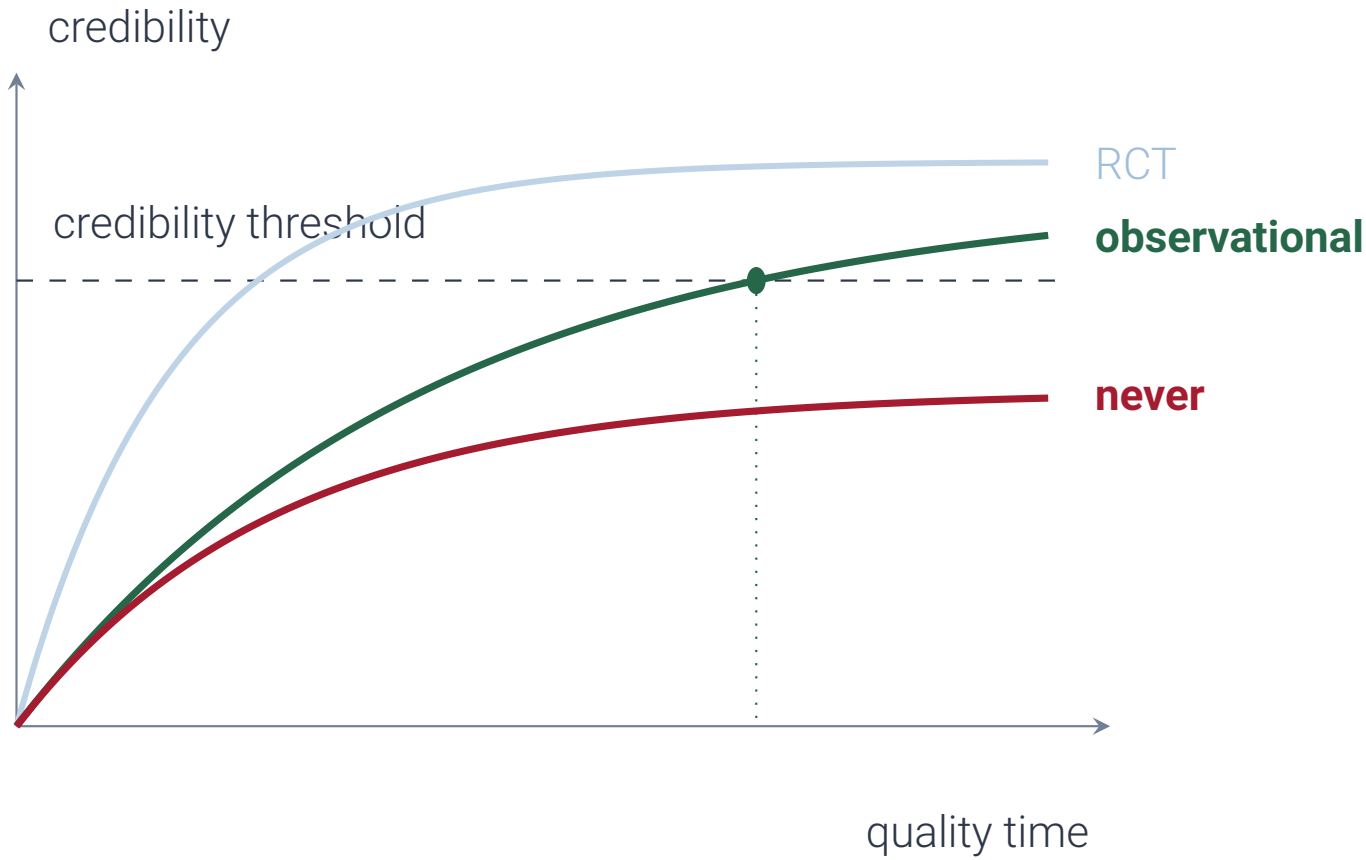
- Short, and the best argument I know for why the analysis is the easy part
- I’m not going to walk you through it
- I’d rather show you the picture underneath it

How long until an RCT is credible?



- Protocol, instruments, fielding, power, money, IRB, partners
- Half a year before a single unit is randomized
- Nobody calls that overkill. They call it doing an experiment

Now take away the randomization



- Not field staff — falsifications, balance, overlap, sensitivity
- Different work. **More of it.** And it can get there
- But some designs flatten below the line no matter what you spend

Craft, not just method

- The threshold is reachable without randomization. It just costs more quality time
- Which means the honest question isn't "which estimator?"
- It's **how much am I willing to spend, and will it be enough?**

- If your curve flattens below the line, you'll find out either way
- Better to find out before you've written the paper

So do we write the second paper at all?

- Under what conditions is an observational study worth doing?
- What would have to be true about the assignment mechanism?
- What would you need to observe? What falsification would you run?
- And what would make you walk away?

The goal is to learn something. Not to publish something.

Bernie — what do you think?

Where we're headed

- Four scenarios, four sets of assumptions:
  1. **Matching and weighting**: unconfoundedness and common support
  2. **Bias correction**: unconfoundedness, weak support
  3. **Standard regression**: unconfoundedness, constant effects, functional form
  4. **Regression adjustment**: unconfoundedness and functional form
- The goal is knowing which to use when, and what each one costs you

Roadmap

Introduction

Foundational ideas

Identification and Assumptions

Estimators

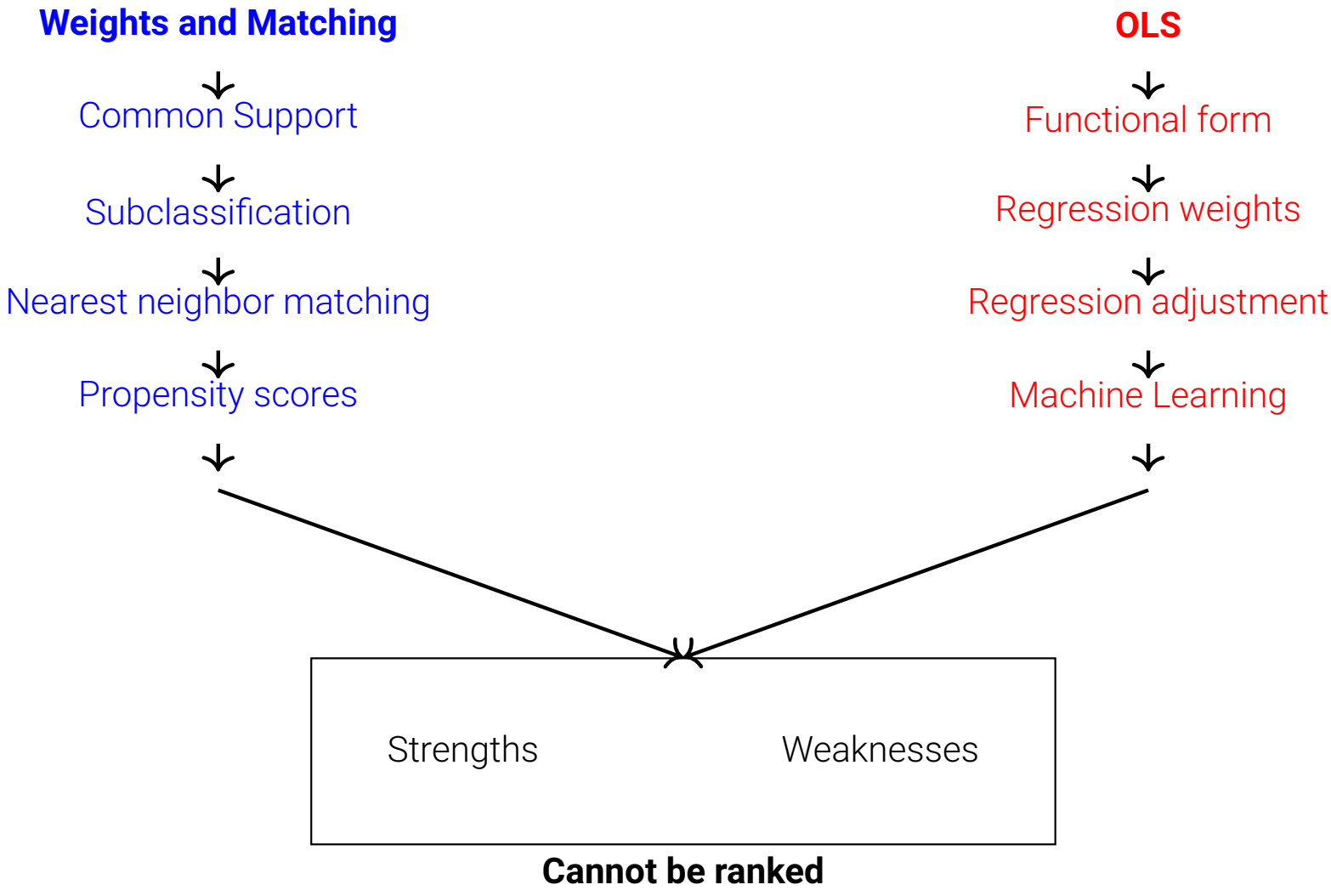
- Subclassification
- Exact and Inexact Matching
- Propensity scores
- Regressions
- What is your coefficient an average of?
- Sloczynski and the reversal
- So what do you actually run?

Advice

Estimators

- Now we will explore estimators that for lack of a better word “use covariates” to estimate average treatment effect parameters
- I will be discussing them around these topics: weighting and matching, matching biases (and corrections), regression
- Themes about heterogeneous treatment effects, common support and correct (and incorrect) regression specifications will be common

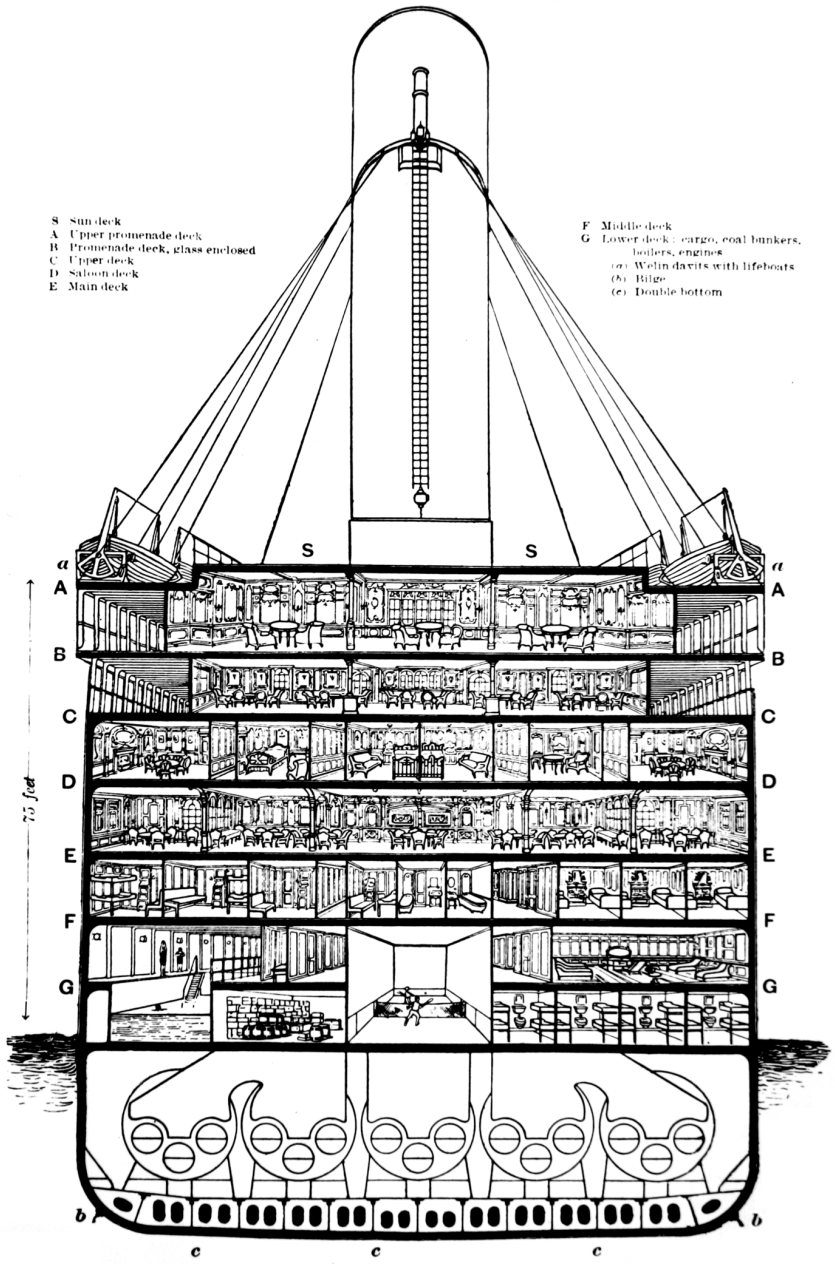
Two approaches to unconfoundedness



Subclassification method

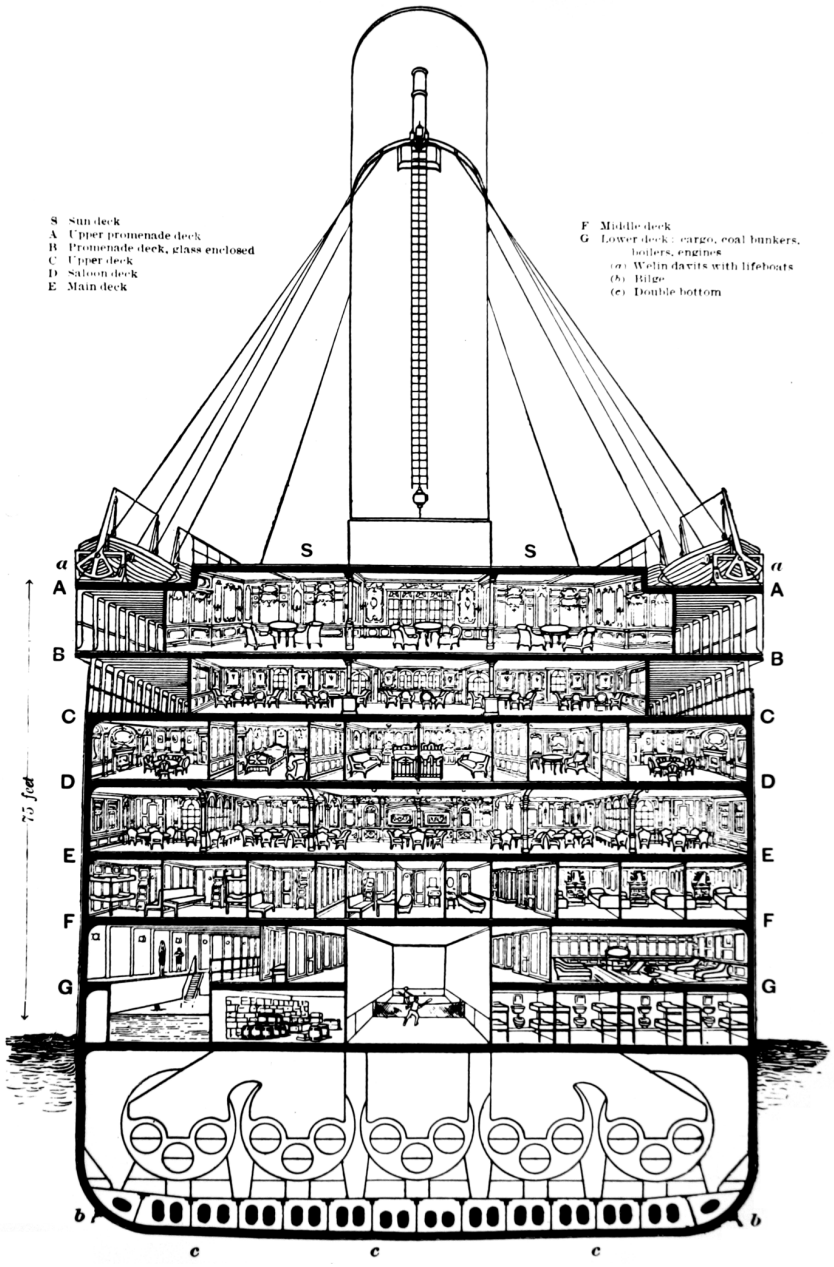
- Wanting to show subclassification as an early effort to tackle covariates, but which ultimately cannot handle large features due to common support issues before moving into regression
- Titanic sank on maiden voyage April 15, 1912 after hitting an iceberg in North Atlantic
- 2200 on board, but only 700 survived, despite 20 lifeboats with 60 capacity (1200 potential lives could've been saved)
- Women and children first was a maritime rule to ration lifeboats, but there were different cabins (1st class, 2nd class, etc.) on different levels with different proximity to boats
- What was the causal effect of 1st class on survival adjusting for W and C?

Titanic layout



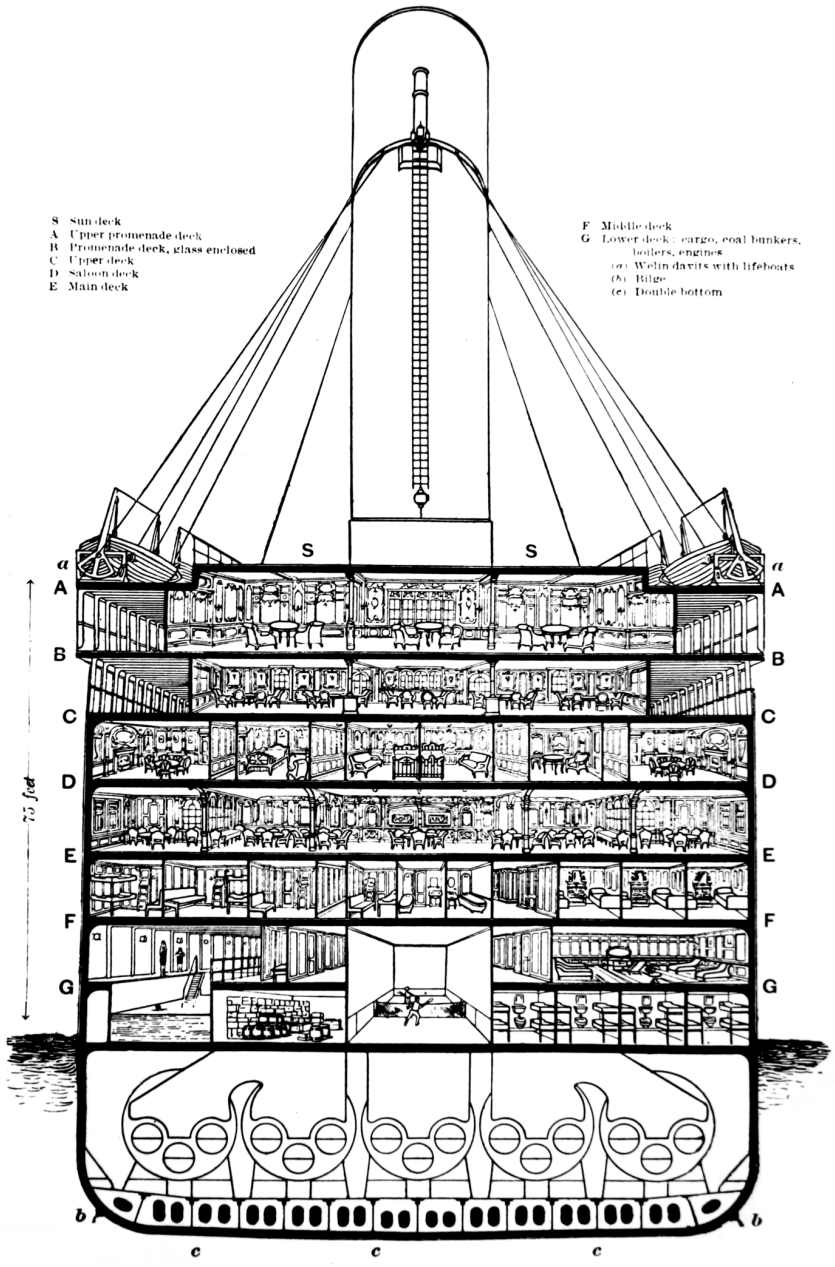
First Class: Decks B-C

Titanic layout



Second Class: Decks D-E

Titanic layout

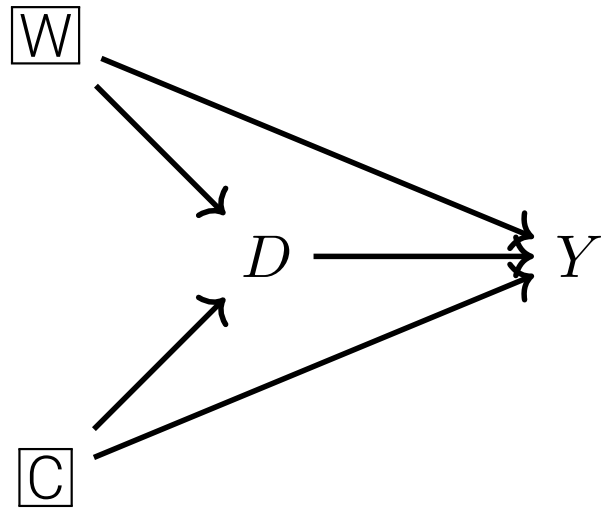


Third Class: Decks F and G

↓

Exercise: Titanic DAG

Figure: Women W and children C first maritime rule is a confounder for estimating first class D effect on surviving Y



Backdoor criterion can be satisfied by blocking on W and C. These are our known confounders. Now we just need data to see if it's quantified.

Titanic exercise

1. **Stratify the confounders**: Our age and sex variables are both binary, so we can only create four strata: male children, female children, male adults, female adults
2. **Calculate differences within strata**: Calculate average survival rates for each group within each of the four strata and difference within strata
3. **Calculate probability weights**: Count the number of people in each strata and divide by the total number of souls aboard (crew and passengers)
4. **Aggregate differences across strata using weights**: Estimate the ATE by aggregating the difference in survival rates over the four strata with each strata-specific difference weighted by that strata's weight

Table 1: Typical balance table

Table: Differences in female and adult passengers by first class status on the Titanic.

Variable name	**First class**		**All other classes**	
	Obs	Mean	Obs	Mean
Percent adult	325	98.2%	1,876	94.5%
Percent female	325	44.6%	1,876	17.3%

Table 2: Stratified sample

Table: Counts and Titanic survival rates by strata and first class status.

	First class		All other classes		
Strata	Obs	Mean	Obs	Mean	Total
Male adult	175	0.326	1,492	0.188	1,667
Female adult	144	0.972	281	0.626	425
Male child	5	1	59	0.407	64
Female child	1	1	44	0.613	45
Total observations	325		1,876		2,201

Table 3: Estimates of aggregate parameters

Table: Differences in survival rates, stratification weights, and estimates of parameters

Strata	**Differences in Survival Rates**	**Weight**$_{k,ATE}$	**Weight**$_{k,ATT}$	**Weight**$_{k,ATU}$
Male adult	0.138	0.76	0.54	0.80
Female adult	0.346	0.19	0.44	0.15
Male child	0.593	0.03	0.02	0.03
Female child	0.387	0.02	0.00	0.02
	No stratification	**Stratification weighted estimates**		
	$\widehat{\textbf{SDO}}$	$\widehat{\textbf{ATE}}$	$\widehat{\textbf{ATT}}$	$\widehat{\textbf{ATU}}$
Estimated coefficient	0.35	0.20	0.24	0.19

Curse of dimensionality

- We were able to estimate all three causal effect parameters because for all four strata there were units in both treatment and control
- But if we dropped the only female child in first class from the data, we'd be in trouble bc there wouldn't be any way to calculate a difference for that group
- This happens when you have many covariates or some are continuous – it's called curse of dimensionality
- If we do not have at least one observation in both treatment and control for *all covariate strata*, then some of parameters cannot be estimated due to losing support

Table: Counts and Titanic survival rates by strata and first class status.

	First class		**All other classes**		
Strata	Obs	Mean	Obs	Mean	Total
Male adult	175	0.326	1,492	0.188	1,667
Female adult	144	0.972	281	0.626	425
Male child	5	1	59	0.407	64
Female child	0	n/a	44	0.613	44
Total observations	324		1,876		2,200

ATT is the only one we can get

$$\begin{aligned} \widehat{\delta}_{ATT} &= (0.137 \times 0.54) + (0.346 \times 0.44) + (0.593 \times 0.02) \\ &= 0.24 \text{ or } 24 \text{ percentage points} \end{aligned} \tag{1}$$

Table: Differences in survival rates, stratification weights, and estimates of parameters without perfect stratification

Strata	**Differences in Survival Rates**	**Weight**$_{k,ATE}$	**Weight**$_{k,ATT}$	**Weight**$_{k,ATU}$
Male adult	0.137	0.76	0.54	0.80
Female adult	0.346	0.19	0.44	0.15
Male child	0.593	0.03	0.02	0.03
Female child	n/a	n/a	n/a	0.02

	No stratification	**Stratification weighted estimates**		
	$\widehat{\textbf{SDO}}$	$\widehat{\textbf{ATE}}$	$\widehat{\textbf{ATT}}$	$\widehat{\textbf{ATU}}$
Estimated coefficient	0.35	n/a	0.24	n/a

Differences in survival rates, stratification weights, and estimated parameters. All coefficients should be multiplied by 100 to get a percentage point change in survival rate as a result of having a first class cabin. Note that the SDO is a simple difference in mean outcomes and therefore *not* a weighted average over the strata differences. But the estimated ATE, ATT and ATU parameters are weighted averages in difference in means using corresponding stratification weights.

Curse of Dimensionality

- Assume we have k covariates and we divide each into 3 coarse categories (e.g., age: young, middle age, old; income: low, medium, high, etc.)
- The number of strata is 3^k. For $k = 10$, then it's $3^{10} = 59,049$
- The curse of dimensionality is based on the slices of all interactions of the covariates, not just the covariates, and that explodes fast

Empty cells

- When some strata is empty, it means there's no one in that cell – so maybe you don't have any female children in first class, but you do have them in other cabins
- When you have empty cells, you can't match within that dimension of the data, which is the "curse of dimensionality" – more covariates makes it happen
- Matching methods really force us to see these curses; they're often hidden from OLS because OLS (as we'll see) overcomes the problem by just assuming a linearity (it doesn't match; it extrapolates)

Common support vs no common support

- Common support is what will determine unbiasedness in nonparametric methods that satisfy unconfoundedness
- Strictly speaking common support will mean that we have *exact* matches
- When we have inexact matches, then nonparametric matching methods may become biased
- Some bias adjustment methods can improve estimation
- We'll review these all now

Exact matching



Exact Matching

Matching goes back at least to Rubin's early work on the propensity score, but we will start with nearest neighbor matching as there's ideas there we draw upon later with synth I want to emphasize

Matching will match a treated unit to a comparison unit that is identical on the known and quantified confounders

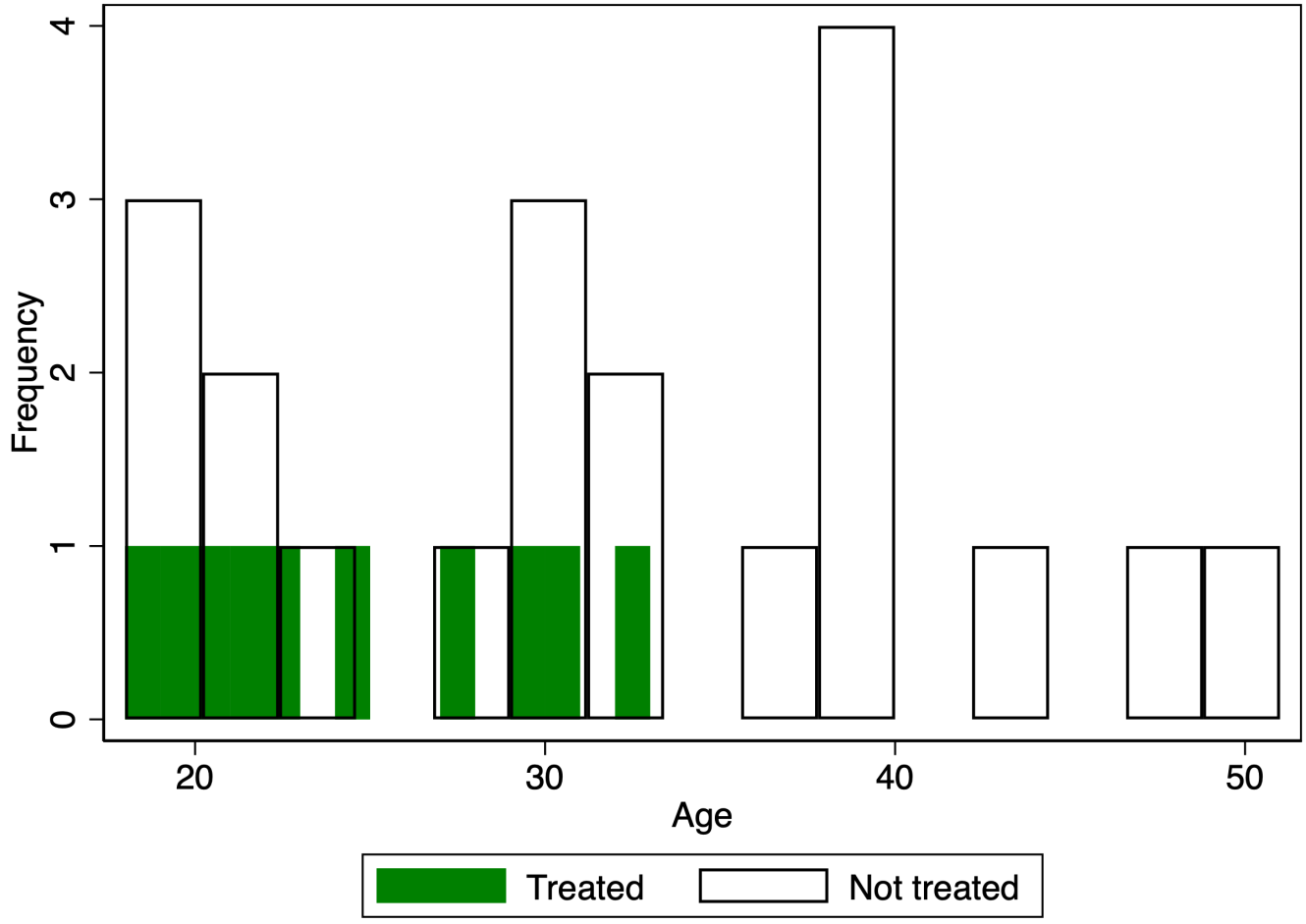
If we can't find one, it means common support failed, the estimate would need to use nearest neighbors (with matching bias), which we will discuss

Training example (unmatched)

Trainee sample			Comparison group		
unit1	age1	earnings1	unit2	age2	earnings2
1	18	$9,500.00	1	20	$8,500.00
2	29	$12,250.00	2	27	$10,075.00
3	24	$11,000.00	3	21	$8,725.00
4	27	$11,750.00	4	39	$12,775.00
5	33	$13,250.00	5	38	$12,550.00
6	22	$10,500.00	6	29	$10,525.00
7	19	$9,750.00	7	39	$12,775.00
8	20	$10,000.00	8	33	$11,425.00
9	21	$10,250.00	9	24	$9,400.00
10	30	$12,500.00	10	30	$10,750.00
Mean	24.3	$11,075.00	11	33	$11,425.00
			12	36	$12,100.00
			13	22	$8,950.00
SDO	-$26.25		14	18	$8,050.00
			15	43	$13,675.00
			16	39	$12,775.00
			17	19	$8,275.00
			18	30	$9,000.00
			19	51	$15,475.00
			20	48	$14,800.00
			Mean	31.95	$11,101.25

Age Imbalance

Figure: Age distribution of a job training program's trainees (green) versus a sample of workers who were not enrolled in the trainee program (white).



Simple example

Let's review together `nnmatch.do` to see how we do this using a program called `teffects nnmatch` in Stata. Focus only on the first ten lines and pay attention to who got matched to who. Calculate the answers by hand.

Exact matching

- Exact matching finds a person in the control group whose value of X_j is *exactly* equal to each person in the treatment group i
- Will not work if the conditioning set includes a continuous variable
- Will also not work if K gets large (curse of dimensionality we discuss later)
- Numerically identical to subclassification

ATT estimator

We will focus on the ATT for the rest of today and the equation is:

$$\widehat{ATT} = \frac{1}{N_T} \sum_{D_i=1} (Y_i - Y_{j(i)}) \tag{2}$$

where $Y_{j(i)}$ is the jth unit matched to the ith unit based on the jth being "exactly equal to" the ith unit with respect to the X conditioning set

Number of matches

What if I find two or more M units with the identical X value? Then what?

$$\widehat{ATT} = \frac{1}{N_T} \sum_{D_i=1} \left(Y_i - \left[\frac{1}{M} \sum_{m=1}^{M} Y_{j_m(1)} \right] \right) \tag{3}$$

Notice that we are only dealing with Y_i^0 by matching; The Y_i^1 is fine as is.

Matching algorithm

1. For each unit i in the treatment group with known and quantified confounder $X = x_i$, find all units j in the donor pool for whom $x_i = x_j$. These j units are our M matches and M can be one or it can be greater than one if you want it to be.
2. For each unit i, replace its missing potential outcome, Y_i^0, with the matched j units' realized outcomes, $\frac{1}{M}\sum Y_{j(i)}$, from Step 1. Do this for all i units in the treatment group.
3. For each unit i, calculate the difference between realized earnings and matched earnings, $\widehat{\delta_i} = Y_i - \frac{1}{M}\sum Y_{j(i)}$.
4. Finally, estimate the sample ATT by averaging over all i differences in earnings from Step 3 as $\frac{1}{N_T}\sum \widehat{\delta_i}$, where N_T is the number of treatment units.

Matched sample

Trainee sample			Comparison group			Matched Trainee sample		
unit1	age1	earnings1	unit2	age2	earnings2	unit2	age2	earnings2
1	18	$9,500.00	1	20	$8,500.00	14	18	$8,050.00
2	29	$12,250.00	2	27	$10,075.00	6	29	$10,525.00
3	24	$11,000.00	3	21	$8,725.00	9	24	$9,400.00
4	27	$11,750.00	4	39	$12,775.00	2	27	$10,075.00
5	33	$13,250.00	5	38	$12,550.00	8,11	33	$11,425.00
6	22	$10,500.00	6	29	$10,525.00	13	22	$8,950.00
7	19	$9,750.00	7	39	$12,775.00	17	19	$8,275.00
8	20	$10,000.00	8	33	$11,425.00	1	20	$8,500.00
9	21	$10,250.00	9	24	$9,400.00	3	21	$10,075.00
10	30	$12,500.00	10	30	$10,750.00	10,18	30	$9,875.00
Mean	24.3	$11,075.00	11	33	$11,425.00	**Mean**	24.3	$9,515.00
			12	36	$12,100.00			
			13	22	$8,950.00			
SDO	-$26.25		14	18	$8,050.00			
ATT	$1,560.00		15	43	$13,675.00			
			16	39	$12,775.00			
			17	19	$8,275.00			
			18	30	$9,000.00			
			19	51	$15,475.00			
			20	48	$14,800.00			
			Mean	31.95	$11,101.25			

Matched sample

- Notice how you don’t use all the data with matching (only the yellow units)
- That’s because not all of the data is relevant or helpful
- This is an important point – not all of the data in causal inference has the variation you need for estimating causal effects
- You’re imputing missing counterfactuals, not predicting outcomes – it’s a different task

Inexact matching



Curse of Dimensionality

- If no matches can be found, it means many cells may contain either only treatment units or only control units but not both (”empty cells”) which violates our common support assumption
- Sometimes people want to ”bin” or ”coarsen” the covariates, and you can do that, but then it’s no longer exact matching and any differences in potential outcomes across the units still create biases
- So here we look at inexact matches and then we look at ways to reduce the matching biases when common support fails

Nearest Neighbor Matching

- Estimate $\widehat{ATT}$ by *imputing* the missing potential outcome of each treatment unit i using the observed outcome from that outcome's "nearest" neighbor j in the control set using X for the matching

$$\widehat{ATT} = \frac{1}{N_T} \sum_{D_i=1} (Y_i - Y_{j(i)})$$

where $Y_{j(i)}$ is the observed outcome of a control unit such that $X_{j(i)}$ is the **closest** value to X_i among all of the control observations (eg match on X)

Matching

- We could also use the average observed outcome over M closest matches:

$$\widehat{ATT} = \frac{1}{N_T} \sum_{D_i=1} \left(Y_i - \left[\frac{1}{M} \sum_{m=1}^{M} Y_{j_m(1)} \right] \right)$$

- Works well when we can find good matches for each treatment group unit, so M is usually defined to be small (i.e., $M = 1$ or $M = 2$)

Matching example with single covariate

i	Y_i^1	Y_i^0	D_i	X_i
1	6	?	1	3
2	1	?	1	1
3	0	?	1	10
4		0	0	2
5		9	0	3
6		1	0	-2
7		1	0	-4

Question: What is $\widehat{ATT} = \frac{1}{N_T} \sum_{D_i=1} (Y_i - Y_{j(i)})$?

Matching example with single covariate

i	Y_i^1	Y_i^0	D_i	X_i
1	6	?	1	3
2	1	?	1	1
3	0	?	1	10
4		0	0	2
5		9	0	3
6		1	0	-2
7		1	0	-4

Question: What is $\widehat{ATT} = \frac{1}{N_T} \sum_{D_i=1} (Y_i - Y_{j(i)})$?

Match and plug in!

Matching example with single covariate

i	Y_i^1	Y_i^0	D_I	X_i
1	6	9	1	3
2	1	0	1	1
3	0	9	1	10
4		0	0	2
5		9	0	3
6		1	0	-2
7		1	0	-4

Question: What is $\widehat{\delta_{ATT}} = \frac{1}{N_T} \sum_{D_i=1} (Y_i - Y_{j(i)})$?

$$\widehat{ATT} = \frac{1}{3} \cdot (6 - 9) + \frac{1}{3} \cdot (1 - 0) + \frac{1}{3} \cdot (0 - 9) = -3.7$$

Measuring the matching discrepancy

- What does it mean to be close when I am working with a large number of covariates?
- What if we had a way of measuring a match in terms of how "close" each unit's X_i value was to the matched X_j
- Let's do that and use the square root of the sum of all squared differences in each unit's $X_i - X_{j(i)}$ as a measure of how bad the match is
- This is called the Euclidean distance

Euclidean distance

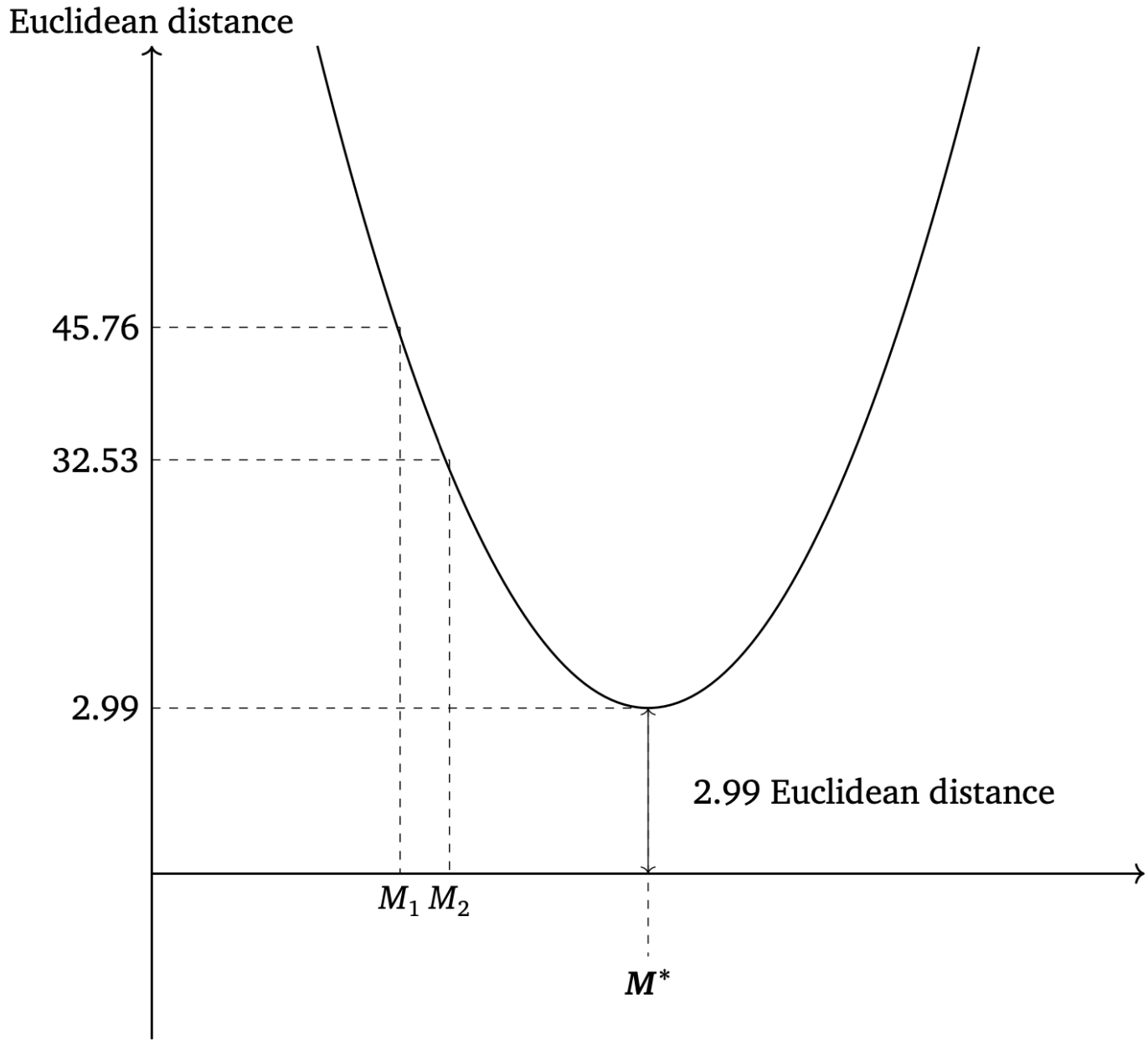
Definition: Euclidean distance

$$
\begin{aligned}
||X_i - X_j|| &= \sqrt{(X_i - X_j)'(X_i - X_j)} \\
&= \sqrt{\sum_{n=1}^{k} (X_{ni} - X_{nj})^2}
\end{aligned}
$$

Let's do this together – sometimes it helps to manually calculate this

https://docs.google.com/spreadsheets/d/1iro1Qzrr1eLDY_LJVzOYvnQZWmxY8JyTcDf6YcdhkwQ/edit?usp=sharing

Visualization of Optimal Match



Minimizing the Euclidean distance

- We are minimizing Euclidean distance, and if that is not zero, then there is matching discreprancy
- And since it is a minimum, any other match will have more matching discreprancies
- `Matching` in R and `teffects` in Stata (not sure in python)
- Let's look at `nnmatch1.do` together and see how simple this is

Other distance metrics

- Our example treated a one unit difference in age and one unit difference in GPA as the same, but those scales are different and matter a lot
- The Euclidean distance is not invariant to changes in the scale of the X's.
- Alternative distance metrics that *are* invariant to changes in scale are more commonly used
- Normalized Euclidean distance and Mahalanobis distance both try to normalize it so that scale doesn't matter

Normalized Euclidean distance

Definition: Normalized Euclidean distance

A commonly used distance is the normalized Euclidean distance:

$$||X_i - X_j|| = \sqrt{(X_i - X_j)' \widehat{V}^{-1} (X_i - X_j)}$$

where

$$\widehat{V}^{-1} = \text{diag}(\widehat{\sigma}_1^2, \widehat{\sigma}_2^2, \ldots, \widehat{\sigma}_k^2)$$

Normalized Euclidean distance

- Notice that the normalized Euclidean distance is equal to:

$$||X_i - X_j|| = \sqrt{\sum_{n=1}^{k} \frac{(X_{ni} - X_{nj})^2}{\widehat{\sigma}_n^2}}$$

- Thus, if there are changes in the scale of X_{ni}, these changes also affect $\widehat{\sigma}_n^2$, and the normalized Euclidean distance does not change

Mahalanobis distance

Definition: Mahalanobis distance

The Mahalanobis distance is the scale-invariant distance metric:

$$||X_i - X_j|| = \sqrt{(X_i - X_j)' \widehat{\Sigma}_X^{-1} (X_i - X_j)}$$

where $\widehat{\Sigma}_X$ is the sample variance-covariance matrix of X.

Let's review them using `nnmatch2.do`

Matching and the Curse of Dimensionality

- The larger the dimensions of the conditioning set, the less likely common support holds, and you can't not do it because you need these covariate dimensions to satisfy weak unconfoundedness!
- This problem is caused by the finite dataset, and it introduces a particular type of selection bias
- Abadie and Imbens (2011) derived a bias adjustment based on the unconfoundedness assumption and an outcome regression model

Deriving the matching bias

$$\widehat{\delta}_{ATT} = \frac{1}{N_T} \sum_{D_i=1} (Y_i - Y_{j(i)}),$$

where each i and $j(i)$ units are matched, $X_i \approx X_{j(i)}$ and $D_{j(i)} = 0$.

Define potential outcomes and switching eq.

$$\begin{aligned}
\mu^0(x) &= E[Y|X = x, D = 0] = E[Y^0|X = x], \\
\mu^1(x) &= E[Y|X = x, D = 1] = E[Y^1|X = x], \\
Y_i &= \mu^{D_i}(X_i) + \varepsilon_i
\end{aligned}$$

Deriving the matching bias

Substitute and distribute terms

$$
\begin{aligned}
\widehat{\delta}_{ATT} &= \frac{1}{N_T} \sum_{D_i=1} (Y_i - Y_{j(i)}) \\
&= \frac{1}{N_T} \sum_{D_i=1} \left[(\mu^1(X_i) + \varepsilon_i) - (\mu^0(X_{j(i)}) + \varepsilon_{j(i)}) \right] \\
&= \frac{1}{N_T} \sum_{D_i=1} (\mu^1(X_i) - \mu^0(X_{j(i)})) + \frac{1}{N_T} \sum_{D_i=1} (\varepsilon_i - \varepsilon_{j(i)})
\end{aligned}
$$

Deriving the matching bias

Difference between sample estimate and population parameter is:

$$\begin{aligned}\widehat{\delta}_{ATT} - \delta_{ATT} &= \frac{1}{N_T}\sum_{D_i=1}\left(\mu^1(X_i) - \mu^0(X_{j(i)}) - \delta_{ATT}\right) \\ &+ \frac{1}{N_T}\sum_{D_i=1}\left(\varepsilon_i - \varepsilon_{j(i)}\right)\end{aligned}$$

Add a zero, algebraic manipulation and simplification:

$$\begin{aligned}\widehat{\delta}_{ATT} - \delta_{ATT} &= \frac{1}{N_T}\sum_{D_i=1}\left(\mu^1(X_i) - \mu^0(X_i) - \delta_{ATT}\right) \\ &+ \frac{1}{N_T}\sum_{D_i=1}\left(\varepsilon_i - \varepsilon_{j(i)}\right) \\ &+ \frac{1}{N_T}\sum_{D_i=1}\left(\mu^0(X_i) - \mu^0(X_{j(i)})\right).\end{aligned}$$

Deriving the matching bias

- Matching under unconfoundedness is consistent which means this:

$$\widehat{\delta}_{ATT} - \delta_{ATT} \rightarrow 0 \text{ as N } \rightarrow \infty$$

.

- But in a finite sample, we may never reach the necessary sample size for unbiaseness

$$E[\sqrt{\frac{1}{N}}(\widehat{\delta}_{ATT} - \delta_{ATT})] = E[\sqrt{\frac{1}{N}}(\mu^0(X_i) - \mu^0(X_{j(i)}))|D = 1]$$

- When might this happen? When the dimensions, k, are very large (lots of covariates and/or strata dimensions)

Bias is often an issue when we match in many dimensions

- The difference between X_i and $X_{j(i)}$ converges to zero very slowly
- The selection bias term, $\mu^0(X_i) - \mu^0(X_{j(i)})$, converges to zero very slowly
- $E[\sqrt{\frac{1}{N}}(\mu^0(X_i) - \mu^0(X_{j(i)}))|D = 1]$ may not converge to zero and an be very large
- And so, $E[\sqrt{\frac{1}{N}}(\widehat{\delta}_{ATT} - \delta_{ATT})]$ may not converge to zero because the bias of the matching discrepancy is dominating the matching estimator

Solutions to matching bias problem

The bias of the matching estimator is caused by large matching discrepancies $||X_i - X_{j(i)}||$ which is virtually guaranteed by the curse of dimensionality. However:

1. But the matching discrepancies are observed. We can always check in the data how well we're matching the covariates.
2. For $\widehat{\delta}_{ATT}$ we can sometimes make the matching discrepancies small by using a large reservoir of untreated units to select the matches (that is, by making N_C large).
3. If the matching discrepancies are large, so we are worried about potential biases, we can apply bias correction techniques

Matching with bias correction

- Each treated observation contributes

$$\mu^0(X_i) - \mu^0(X_{j(i)})$$

  to the bias.
- Bias-corrected (BC) matching:

$$\widehat{\delta}^{BC}_{ATT} = \frac{1}{N_T} \sum_{D_i=1} \left[(Y_i - Y_{j(i)}) - (\widehat{\mu^0}(X_i) - \widehat{\mu^0}(X_{j(i)})) \right]$$

  where $\widehat{\mu^0}(X)$ is an estimate of $E[Y|X = x, D = 0]$. For example using OLS but other maybe too (neural nets?).
- Under some conditions, the bias correction eliminates the bias of the matching estimator without affecting the estimator's variance.

Intuition

- Under unconfoundedness, we know that

$$E[Y^0|D=1, X=x] = E[Y^0|D=0, X=x]$$

- This implies that whatever the relationship is between Y^0 and X in the control group, it's the same in the treatment group, even though we cannot observe Y^0 in the treatment group
- We will therefore fit a regression of Y^0 onto X in the $D=0$ group only and use that to predict $\widehat{Y^0}$ in both groups

Bias correction steps

Unit	*Worker*	Y^1	Y^0	*D*	*Age*
1	Andy	$200	.	1	23
2	Betty	$250	.	1	27
3	Chad	$150	.	1	29
4	Doug	$300	.	1	22
5	Edith	.	$225	0	27
6	Fred	.	$500	0	32
7	Gina	.	$200	0	26
8	Hank	.	$190	0	26
9	Inez	.	$180	0	28
10	Janet	.	$140	0	29

Table 35: Ten workers eligible, conditional on age, for a job training program. The first four enrolled and the second six did not. We have their age and earnings, but not their counterfactuals.

Bias correction steps

$Unit$	$Worker$	Y^1	$\widehat{Y^0}, Y^0$	$\widehat{\delta_i}$	D	Age
1	Andy	\$200	\$195	\$5	1	23
2	Betty	\$250	\$225	\$25	1	27
3	Chad	\$150	\$140	\$10	1	29
4	Doug	\$300	\$195	\$105	1	22
5	Edith	.	\$225		0	27
6	Fred	.	\$500		0	32
7	Gina	.	\$200		0	26
8	Hank	.	\$190		0	26
9	Inez	.	\$180		0	28
10	Janet	.	\$140		0	29

Table 36: Matched sample with imputed counterfactuals and estimated individual treatment effects for the treatment group using nearest neighbor matching.

Steps

1. Regress Y on X with OLS except only use the *matched* sample of control group units:

$$Y_j^0 = \alpha + \beta X_j + \varepsilon_j$$

where j are the units for which $D_j = 0$. Examine the code for this in `nnmatch_bc1.do` and `nnmatch_bc2.do`.

Steps

2. Use the fitted values $\widehat{\alpha}$ and $\widehat{\beta}$ to predict $\widehat{\mu}^0(X)$ for both the i and the matched $j(i)$ units:

$$\begin{aligned} \widehat{\mu}_i^0 &= \widehat{\alpha} + \widehat{\beta} X_i \\ \widehat{\mu}_{j(i)}^0 &= \widehat{\alpha} + \widehat{\beta} X_{j(i)} \end{aligned}$$

Steps

3. Subtract $\widehat{\mu}_i^0(X_i) - \widehat{\mu}_{j(i)}^0(X_{j(i)})$, our estimate of the selection bias caused by matching discrepancies, from the sample estimate of the ATT:

$$\widehat{\delta}_{ATT}^{BC} = \frac{1}{N_T} \sum_{D_i=1} \left[(Y_i - Y_{j(i)}) - \left(\widehat{\mu}^0(X_i) - \widehat{\mu}^0(X_{j(i)}) \right) \right]$$

Steps

4. Estimate Abadie-Imbens robust standard error (Abadie and Imbens 2006; 2008; 2011)

Let's review all of this `nnmatch_bc1.do` and `nnmatch_bc2.do`

Matching bias: Implications for practice

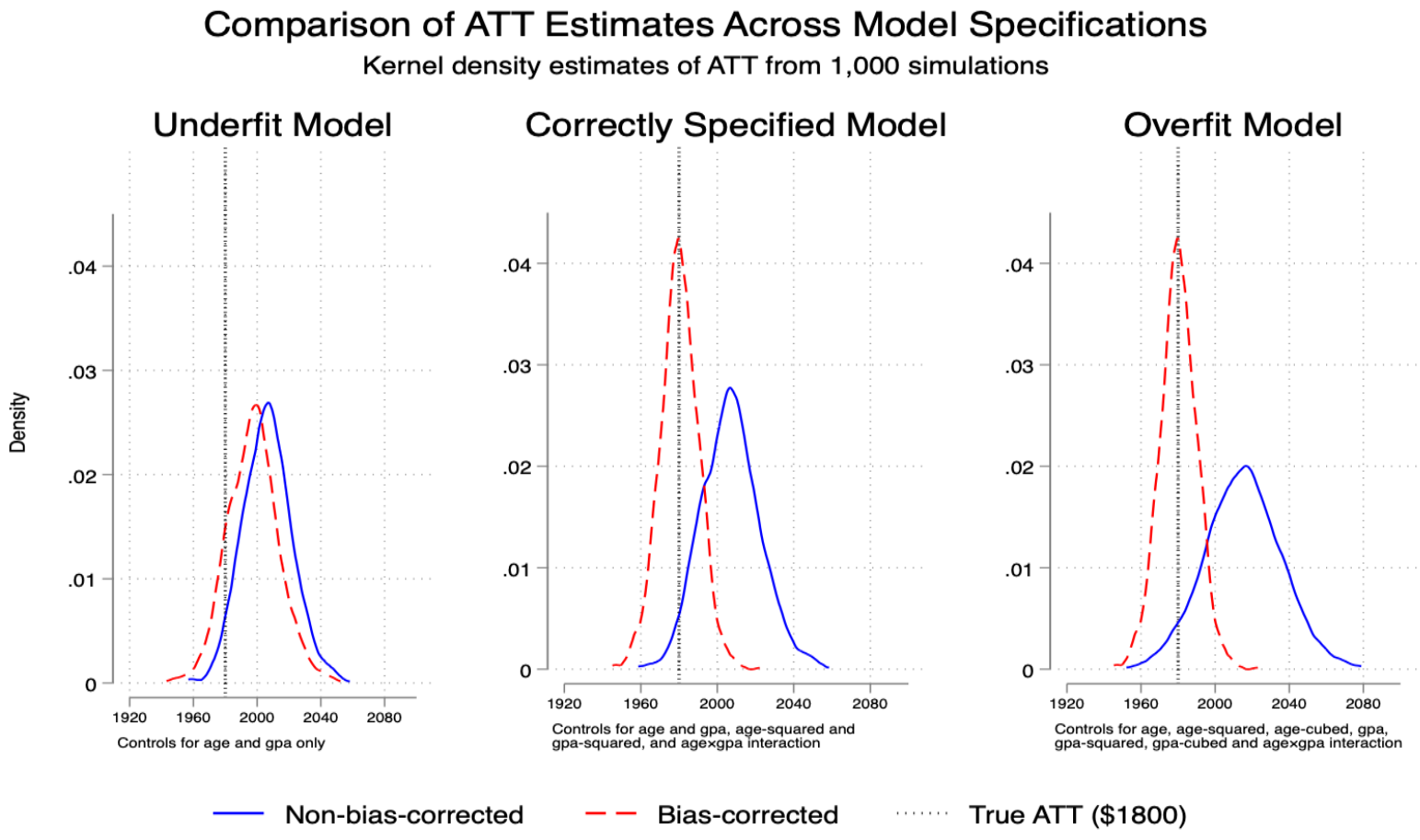
Matching bias arises because of the effect of large matching discrepancies on $\mu^0(X_i) - \mu^0(X_{j(i)})$ due to a lack of common support. To minimize matching discrepancies:

1. Use a small M (e.g., $M = 1$). Larger values of M produce large matching discrepancies.
2. Use matching with replacement. Because matching with replacement can use untreated units as a match more than once, matching with replacement produces smaller matching discrepancies than matching without replacement.
3. Try to match covariates with a large effect on $\mu^0(\cdot)$ particularly well.

Bias Correction

- Consider `matching_biasadj.do` that examines three models:
  1. Underfit model with and without bias correction
  2. Correctly specified model with and without bias correction
  3. Overfit model with and without bias correction
- Overfit model included cubic even though correct model was quadratic

Simulation


Comparison of ATT Estimates Across Model Specifications
Kernel density estimates of ATT from 1,000 simulations
Underfit Model
Correctly Specified Model
Overfit Model
Density
.04
.03
.02
.01
0
1920
1960
2000
2040
2080
Controls for age and gpa only
Controls for age and gpa, age-squared and gpa-squared, and age×gpa interaction
Controls for age, age-squared, age-cubed, gpa, gpa-squared, gpa-cubed and age×gpa interaction
Non-bias-corrected
Bias-corrected
True ATT ($1800)

Large sample distribution for matching estimators

- Cannot use the bootstrap, so Abadie and Imbens derived the variance (Abadie and Imbens 2008)
- Matching estimators have a Normal distribution in large samples (provided the bias is small):

$$\sqrt{N_T}(\widehat{\delta}_{ATT} - \delta_{ATT}) \xrightarrow{d} N(0, \sigma^2_{ATT})$$

- For matching without replacement, the "usual" variance estimator:

$$\widehat{\sigma}^2_{ATT} = \frac{1}{N_T} \sum_{D_i=1} \left(Y_i - \frac{1}{M} \sum_{m=1}^{M} Y_{j_m(i)} - \widehat{\delta}_{ATT} \right)^2,$$

  is valid.

Large sample distribution for matching estimators

- For matching with replacement:

$$\begin{aligned}\widehat{\sigma}^2_{ATT} &= \frac{1}{N_T}\sum_{D_i=1}\left(Y_i - \frac{1}{M}\sum_{m=1}^{M} Y_{j_m(i)} - \widehat{\delta}_{ATT}\right)^2 \\ &+ \frac{1}{N_T}\sum_{D_i=0}\left(\frac{K_i(K_i-1)}{M^2}\right)\widehat{var}(\varepsilon|X_i, D_i=0)\end{aligned}$$

  where K_i is the number of times observation i is used as a match.

- $\widehat{var}(Y_i|X_i, D_i=0)$ can be estimated also by matching. For example, take two observations with $D_i = D_j = 0$ and $X_i \approx X_j$, then

$$\widehat{var}(Y_i|X_i, D_i=0) = \frac{(Y_i - Y_j)^2}{2}$$

  is an unbiased estimator of $\widehat{var}(\varepsilon_i|X_i, D_i=0))$

Example

Let's now review the syntax in Stata. It's called `matching.do` and is in /Labs/Matching

https://github.com/Mixtape-Sessions/Causal-Inference-1/tree/main/Lab/Matching

Propensity scores

- Alternative to the distance minimization method is to use something called the ”propensity score” to summarize all of the covariates into a single scalar and then weight or match on it
- Propensity score was introduced by Rosenbaum and Rubin (1983) and has become extremely influential in causal inference

Propensity score theorem

“We are interested in estimating the average effect of a binary treatment on a scaler outcome. If assignment to the treatment is exogenous or unconfounded, that is, independent of the potential outcomes given covariates, biases associated with simple treatment-control average comparisons can be removed by adjusted for differences in the covariates. Rosenbaum and Rubin (1983) show that adjusting solely for differences between treated and control units in the propensity score removes all biases associated with differences in covariates.” – Hirano, Imbens and Ridder (2003, Econometrica)

Formal Definition

Definition of Propensity score

A propensity score is a person (or ”unit”) specific probability bounded between 0 and 1 measuring the probability of treatment assignment given a collection of variables: $p(X) = Pr(D = 1|X)$

Propensity score theorem

Propensity score theorem

If $(Y^1, Y^0) \perp\!\!\!\perp D|X$ (full unconfoundedness), then $(Y^1, Y^0) \perp\!\!\!\perp D|\rho(X)$ where $\rho(X) = Pr(D = 1|X)$, the propensity score

- Conditioning on the propensity score is enough to have independence between D and (Y^1, Y^0) (Rosenbaum and Rubin 1983)
- With full unconfoundedness, you can estimate the ATE, but you can estimate the ATT with weaker assumptions (independence with respect to $(Y^0) \perp\!\!\!\perp D|\rho(X))$
- But remember – propensity scores don't solve unconfoundedness; if you don't have it with respect to K dimensions of X, you won't have it when you collapse it to the propensity score

True vs Estimating Propensity Score

- Outside the randomized experiment, we don't know the "true propensity score"
  - → We use DAGs or hunches to select covariates, careful not to include outcomes or colliders
  - → But we still won't know the functional form (i.e., logit, probit, polynomials, interactions)
- Often then people will use higher order terms and interactions to provide flexibility but it does technically have misspecification biases
- This is an area most likely where machine learning could greatly enhance the estimation (take your pick)

Step 1: Pick your parameter ATE vs ATT vs ATU

- Which population are you studying? Only those who were discriminated against? That's the ATT
- Do you want to imagine "what if blacks and whites were both discriminated against?" That's the ATE
- The more you can focus on one particular causal parameter, the easier and more justified it gets as it weakens both assumptions

Step 2: Estimate the propensity score

- Estimate the conditional probability of treatment using probit or logit model (or ML)

$$Pr(D_i = 1|X_i) = F(\beta X_i)$$

- Note: don't use OLS because while it will get the mean right, it will not get correct values in the tails because of its linear projections
- OLS will give propensity scores outside the [0,1] bounds and probabilities cannot be negative or greater then one

Step 2: Estimate the propensity score

- Use the estimated coefficients to predict the propensity score for each unit i

$$\widehat{\rho}_i(X_i) = \widehat{\beta} X_i$$

- Note that each unit i now has a predicted probability of treatment given the values of their covariates relative to everyone else's

Step 2: Estimate the propensity score

- Think of the propensity score as a frequentist concept of probability
- “If I drew someone from the sample with these characteristics, then how many of those are in the treatment group divided by the total with those characteristics”
- Or for each dimension of X, a ratio of $\frac{N_T}{(N_T+N_C)}$

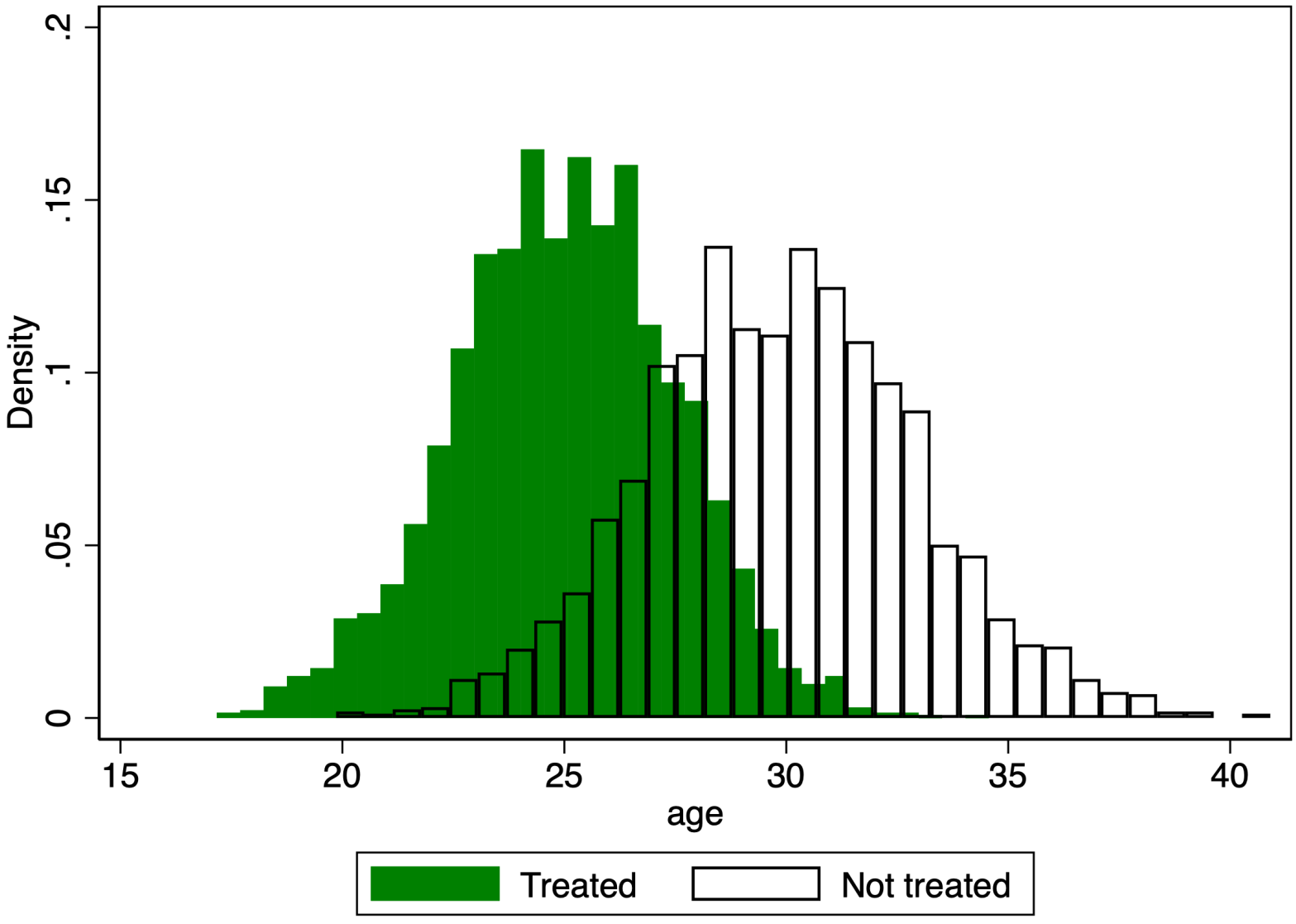
Propensity score as dimension reduction

- Rubin (1977) and Rosenbaum and Rubin (1983) developed the propensity score method
- They show that if treatment is independent of potential outcomes conditional on K covariates, then it will be independent of potential outcomes conditional on propensity score
- Main value of the propensity score is dimension reduction to reduce K covariates into a single scalar without loss of information
- Variety of ways to incorporate the propensity score – stratification, weighting and matching

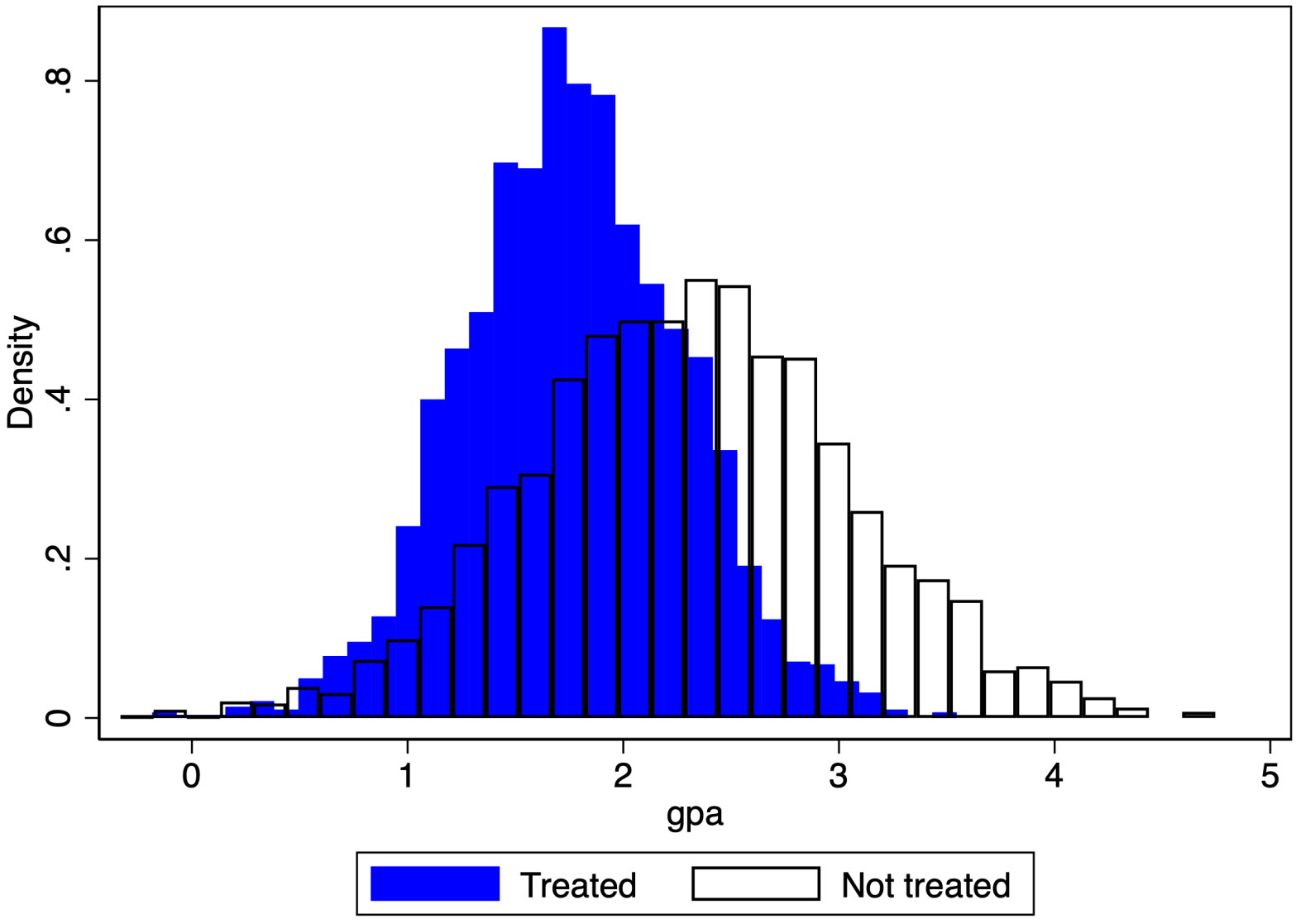
Investigating overlap

- Once you obtain the propensity score, you can use it for estimation, but you can also use it for evaluating covariate balance
- It's an easy way to do it with histograms, and since the propensity score theorem holds for the dimensions of X, there's no loss of generality in investigating overlap that way versus one by one
- Remember: you need common support, not on X individual covariates alone, but within the K dimensions, so investigating with the propensity score is easier to do

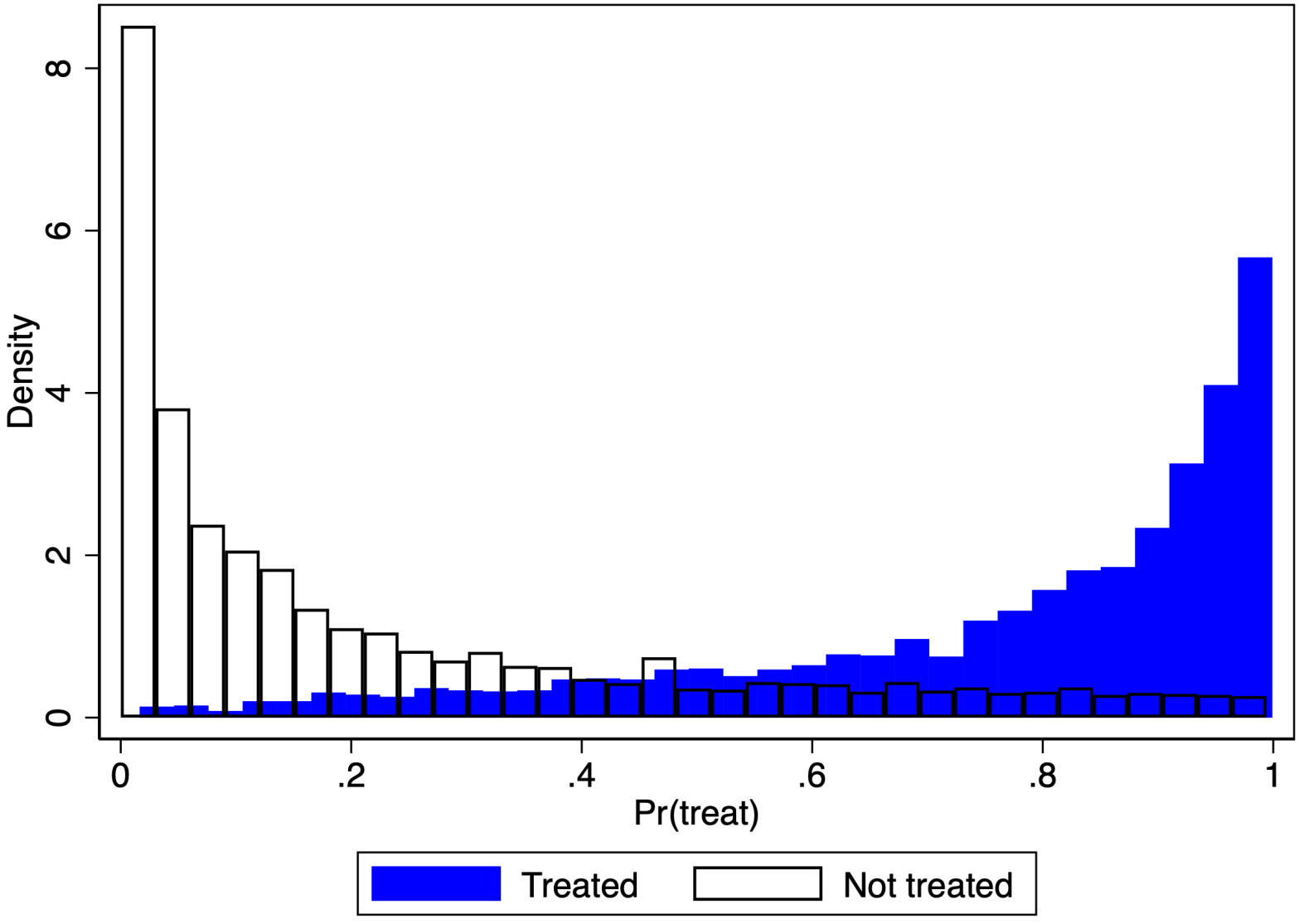
Covariate 1 histograms



Covariate 2 histograms



Propensity scores



Navigating the vastness of estimation

- Estimators abound and can be a little bewildering so to summarize them:
  1. Match units from one group to another using the propensity score (with various rules for finding how close to be)
  2. Weighting by the inverse propensity score
- Variety of techniques to derive standard errors from parametric methods to bootstrapping
- Can even introduce “doubly robust” methods to deal with matching bias like we did with nearest neighbor bias correction
- But all these estimators assume unconfoundedness, so really it’s all about addressing the lack of overlap; that’s the only bias that exists when you have unconfoundedness remember

Step 3a: Estimation with matching

- Most common method is to use matching
- Matching finds a unit in the comparison group with a similar $\widehat{\rho}_i(X)$ to service as counterfactual for the unit
- For the ATE, you'll need matches on both side; for the ATT, you'll need matches for the treatment group among controls
- Lack of overlap creates issues for matching which we'll note later

Step 3a: Estimation with stratification

- Rare to see this done anymore, though it was one of the methods that Dehejia and Wahba (2002) tried
- Stratification is a kind of weighting method similar to Cochran's subclassification method where weights are group shares within certain ranges of the propensity score
- Four steps to doing this; I'll review again later, but let me briefly do it now

Step 3a: Estimation with stratification

1. First, split the sample by ranges on the propensity score until you find covariate balance within each range
2. Next calculate weights for each region depend on the causal parameter you're seeking:
   - → ATE: number of units in that region divided by total units in sample ;
   - → ATT: number of treated units in that region divided by the number of treated units in the sample
   - → ATU: number of control units in that region divided by the number of control units in the sample
3. Calculate simple difference in mean outcomes (i.e., without controls) for each propensity score region (Step 1)
4. Then take weighted average using weights from step 2 and difference in means from step 3

Step 3b: Early weighting methods

- Heckman, Ichimura, and Todd (1997, 1998) and Heckman, Ichimura, Smith, and Todd (1998) focus on the ATT
- Estimators based on local linear regressions of the outcome on treatment status and either covariates or the propensity score.
- They conclude that in general there is no clear ranking of their estimators
- Under some conditions the estimator based on adjustment for all covariates is superior to the estimator based on adjustment for the propensity score,
- Under other conditions the second estimator is to be preferred and lack of knowledge of the propensity score does not alter this conclusion

Step 3b: IPW by Hirano, Imbens and Ridder

- Hirano, Imbens and Ridder (2003, Econometrica) considers estimation of the ATT
- Focus is on the efficient estimation and they show that while weighting on the inverse of the true propensity score is not an efficient estimator, paradoxically weighting each observation by the *inverse* of a nonparametric estimator of the propensity score is

Step 3b: Estimation with inverse probability weighting

- IPW uses the estimated propensity score to reweight the outcomes for which there are several historical methods for doing so
- IPW is non-parametric – you are just taking averages and multiplying by the inverse of the propensity score weights depending on which parameter you want to estimate
- There are fewer implementation choices than in matching (i.e., no choice over distance, number of neighbors)
- There are bias adjustment methods called double robust where you combine imputing counterfactuals with weighting by the propensity score

Step 3b: Estimation of ATT with IPW

Estimating ATT with IPW

Given $Y^0 \perp\!\!\!\perp D|X$ and weak common support, then

$$\begin{aligned} \delta_{ATT} &= E[Y^1 - Y^0|D=1] \\ &= \frac{1}{Pr(D=1)} \cdot E\left[Y \cdot \frac{D-\rho(X)}{1-\rho(X)}\right] \end{aligned}$$

Notice that when $D=1$, the outcome is not weighted, but when $D=0$ it is. You're missing the Y^0 for the treatment, not Y^1 so you the treatment group Y values alone and weight "up" or "down" the comparison groups by their propensity scores

Step 4: Standard Errors

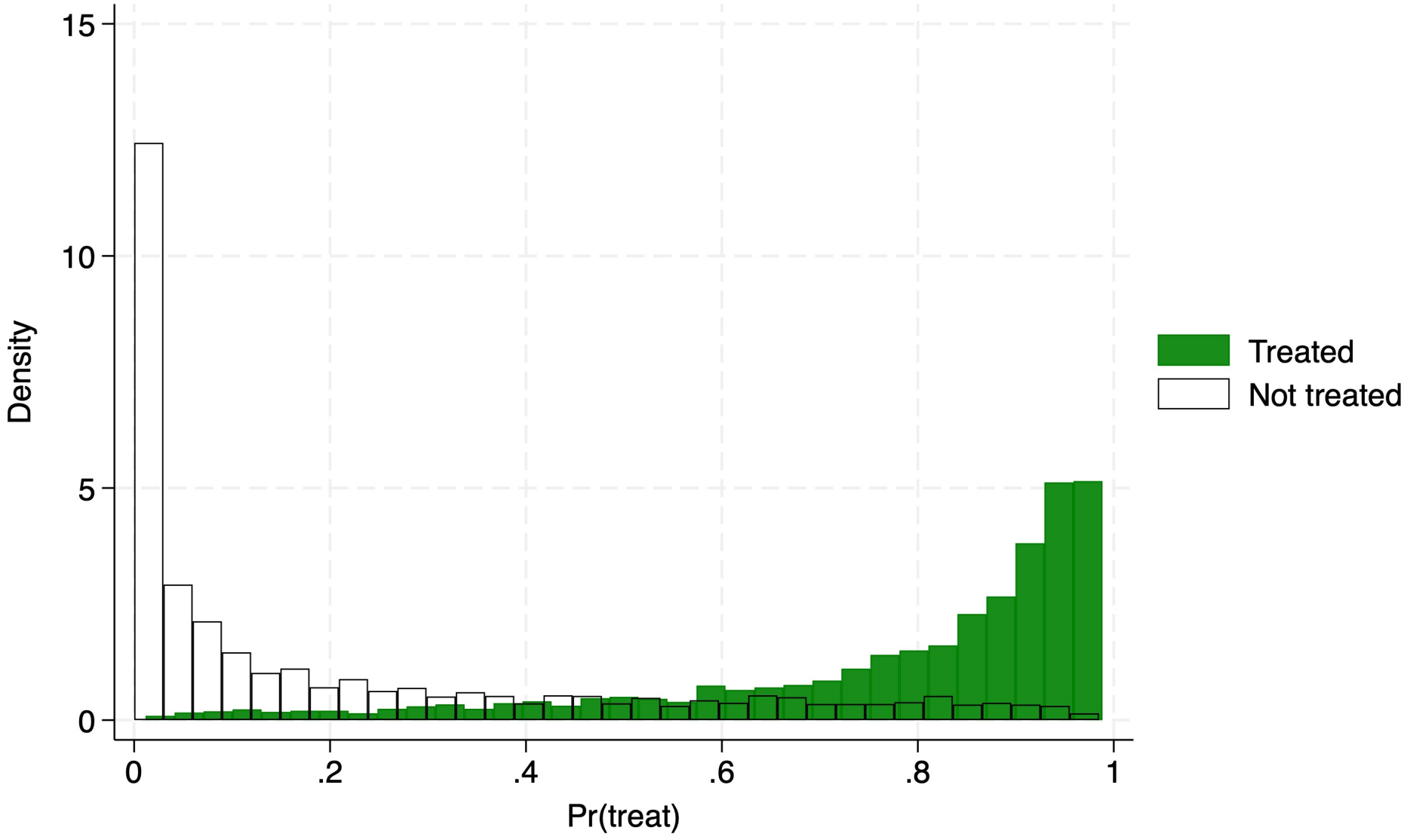
Standard errors can be constructed a few different ways:

- We need to adjust the standard errors for first-step estimation of $\rho(X)$
  - → Parameteric first step: Newey and McFadden (1994)
  - → Non-parametric first step: Newey (1994)
- IPW is a smooth estimator which means the bootstrap is valid for inference (Adudumilli 2018 and Bodory et al. 2020)

Check for common support using histograms

- Assessing whether there are units it both groups for whichever parameter you're focused on is simple with propensity score as shown earlier using histograms of the propensity score for treated and control
- Crump, et al. (2009) suggest keeping propensity scores within the interval [0.1,0.9] ("trimming") but any trimming will drop units and dropping units means we are not estimating the ATT
- We are rather estimating the average treatment effect for the remaining units satisfying overlap (ATTO or something)
- Let's look at a picture again just to remind ourselves (and we can look at `matching.do` again)

Assessing overlap



Estimating ATE with IPW

- *Any* of the parameters including the ATE are possible with propensity scores
- Assumptions are strongest for ATE: full unconfoundedness and full support versus weak versions of both
- You should only focus on the ATE if you care about the entire population's treatment effects, but remember that requires more assumptions than if you just focused on the ATT

Estimating ATE with IPW

Estimating ATE with IPW

Given $Y^1, Y^0 \perp\!\!\!\perp D|X$ and common support, then

$$\begin{aligned} \delta_{ATE} &= E[Y^1 - Y^0] \\ &= E\left[Y \cdot \frac{D - \rho(X)}{\rho(X) \cdot (1 - \rho(X))}\right] \end{aligned}$$

Notice that since treated units are missing counterfactuals, but so are controls, all of the data is weighted for the ATE (not just the controls for ATT)

Inverse Probability Weighting

Proof.

$$
\begin{aligned}
E\left[Y \cdot \frac{D-\rho(X)}{\rho(X)(1-\rho(X))}\middle|X\right] &= E\left[\frac{Y}{\rho(X)}\middle|X, D=1\right]\rho(X) \\
&\quad + E\left[\frac{-Y}{1-\rho(X)}\middle|X, D=0\right](1-\rho(X)) \\
&= E[Y|X, D=1] - E[Y|X, D=0]
\end{aligned}
$$

and the results follow from integrating over $P(X)$ and $P(X|D=1)$. □

Other comments about propensity scores

- Only other comments to make is that you can get outlier weights
- If the propensity score is very high for a control group in the ATT, the weight can explode
- Certain normalizations of the way the IPW is constructed to try and minimize those influences
- Stata's `teffects` or R's `ipw` both let you estimate parameters using IPW and get standard errors
- But you can do it manually using OLS and use the propensity scores as analytical weights

Simulation

Lets look at the `matching.do` again

Double robust estimators

- You can have the right covariates but the wrong model and unbiasedness requires the correct model
- What if you had a way to control for the covariates using propensity scores and something else like regression?
- Buys you some insurance against model misspecification if such a thing existed

Double robust estimators

- Question: Is it possible to combine the virtues of regression and propensity score?
- Answer: Double robust estimators
- DR address the model misspecification problem by combining propensity scores with other methods
- Basic idea in all of them was to control for covariates twice *at the same time* without paying for it (two for one)

Double robust estimators

- We say that estimators combining regression with IPW are double robust so long as
- The regression for the outcome is properly specified, or
- The propensity score is properly specified
- We give ourselves two chances to get it right (either/or not both/and) but if neither is properly specified, then you didn't really gain much
- Three strikes in baseball

Doubly Robust Estimator

Easy to verify: with true $m_1(X)$ (an outcome regression or OR model) and $\rho(X)$

$$ATE = \mathbb{E}\left[\frac{DY}{\rho(X)} - \frac{D-\rho(X)}{\rho(X)}m_1(X)\right] - \mathbb{E}\left[\frac{(1-D)Y}{1-\rho(X)} + \frac{D-\rho(X)}{1-\rho(X)}m_0(X)\right]$$

$$= \mathbb{E}\left[m_1(X_i) + \frac{D_i\{Y_i - m_1(X_i)\}}{\rho(X_i)}\right] - \mathbb{E}\left[m_0(X_i) + \frac{(1-D_i)\{Y_i - m_0(X_i)\}}{1-\rho(X_i)}\right]$$

$$= \mu_1 - \mu_0 = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$$

Doubly Robust Estimator

In the previous formula, replace the true PS $\rho(X)$ and outcome $m_1(X)$ by the estimated ones from postulated models $\hat{\rho}(X)$ and $\hat{m}_1(X)$, we obtain two augmented estimators:

$$\widehat{\tau}_{dr} = \hat{\mu}_{1,dr} - \hat{\mu}_{0,dr}$$

$$= \frac{1}{N}\sum_{i=1}^{N}\left[\frac{D_i Y_i - D_i - \hat{\rho}(X_i)}{\hat{\rho}(X_i)}\hat{m}_1(X_i)\right] - \frac{1}{N}\sum_{i=1}^{N}\left[\frac{(1-D_i)Y_i - D_i - \hat{\rho}(X_i)}{1-\hat{\rho}(X_i)}\hat{m}_0(X_i)\right]$$

Double Robust Estimator

- The two estimators are mathematically the same, but they have different statistical implications:
  - → the first estimator augments an IPW estimator by outcome regression (OR);
  - → the second augments an OR estimator by IPW.
- The first estimator is usually referred to as the “doubly-robust (DR) estimator” and we saw it in Sant’Anna and Zhou (2020) actually for diff-in-diff

Doubly Robust Estimator

Now focus on the DR estimator

Notice that $\hat{\mu}_{1,dr}$ and $\hat{\mu}_{0,dr}$ have the same structure

$$\hat{\mu}_{1,dr} = N^{-1} \sum_{i=1}^{N} \left[\frac{D_i Y_i}{\hat{\rho}_i} - \frac{(D_i - \hat{\rho}_i)\hat{m}_1}{\hat{\rho}_i} \right]$$

$$\hat{\mu}_{0,dr} = N^{-1} \sum_{i=1}^{N} \left[\frac{(1 - D_i) Y_i}{1 - \hat{\rho}_i} + \frac{(D_i - \hat{\rho}_i)\hat{m}_0}{1 - \hat{\rho}_i} \right]$$

We will study the properties of $\hat{\mu}_{1,dr}$ as an estimator for $\mu_1 = \mathbb{E}\{Y(1)\}$; $\hat{\mu}_{0,dr}$ is symmetric

Doubly Robust Estimator

What does $\hat{\mu}_{1,dr}$ estimate? Simple algebra shows that $\mu_{1,dr}$ converges to

$$\mathbb{E}\left[\frac{DY}{\rho(X)} - \frac{D-\rho(X)}{\rho(X)}m(X)\right]$$

$$= \mathbb{E}\{Y(1)\} + \mathbb{E}\left[\frac{D-\rho(X)}{\rho(X)}\{Y(1) - m(X)\}\right]$$

- Thus, in order for $\hat{\mu}_{1,dr}$ to estimate $\mathbb{E}\{Y(1)\}$, the second term in (4) must be 0
- Regression estimator augmented by weighted residuals
- When does the second term $= 0$?

Doubly Robust Estimator

When does $R = \mathbb{E}\left[\frac{D-\rho(X)}{\rho(X)}\{Y(1) - m_1(X)\}\right] = 0$?

- **Scenario 1:** Postulated propensity score model $\rho(X;\beta)$ is incorrect, but postulated regression model $m_1(X;\alpha)$ is correct

$$R = \mathbb{E}\left[\mathbb{E}\left[\frac{D-\rho(X)}{\rho(X)}\{Y(1) - m_1(X)\}\middle| X\right]\right]$$

$$= \mathbb{E}\left[\frac{\rho_{true}(X)-\rho(X)}{\rho(X)}\mathbb{E}\{Y(1) - m_1(X)\middle| X\}\right]$$

- Easy to see that $\mathbb{E}\{Y(1) - m_1(X)|X\} = \mathbb{E}\{Y(1)|X\} - m_1(X) = m_{1,true}(X) - m_1(X) = 0$

Doubly Robust Estimator

When does $\mathbb{E}\left[\frac{D-\rho(X)}{\rho(X)}\{Y(1)-m_1(X)\}\right]=0$?

- **Scenario 2:** Postulated propensity score model $\rho(X;\beta)$ is correct, but postulated regression model $m_1(X;\alpha)$ is incorrect

$$R=\mathbb{E}\left[\mathbb{E}\left[\frac{D-\rho(X)}{\rho(X)}\{Y(1)-m_1(X)\}\middle|X\right]\right]$$

$$=\mathbb{E}\left[\frac{\rho_{true}(X)-\rho(X)}{\rho(X)}\mathbb{E}\{Y(1)-m_1(X)\middle|X\}\right]$$

- Equals to zero because $\rho_{true}(X)-\rho(X)=0$

Doubly Robust Estimator

- In both cases, the second term goes to 0 in large samples, and thus $\hat{\mu}_{1,dr}$ is consistent (asymptotically unbiased) for $E(Y(1))$.
- Similarly, $\hat{\mu}_{0,dr}$ is consistent for $E(Y(0))$, and hence $\hat{\tau}_{dr}$ is consistent for the ATE.
- Obviously, if both models are correct, $\hat{\tau}_{dr}$ is consistent for estimating the ATE.

Doubly Robust Estimator

Double Robustness: $\hat{\tau}_{dr}$ is a consistent estimator of the ATE if either the propensity score model or the potential outcome model is, but not necessarily both are, correctly specified

- DR is a large sample property
- Offers protection against model mis-specification: gives you two chances to get it right (and wrong)!
- If $\rho(X)$ and $m_1(X)$ are modeled correctly, $\hat{\tau}_{dr}$ will have smaller variance than the IPW estimator (in large samples)
- If the outcome model $m_1(X)$ is correct, $\hat{\tau}_{dr}$ has larger variance (in large samples) than the direct regression estimator
- ... but gives protection in the event it is not

Propensity score matching

- Matching, or “imputation”, is another way that utilizes the $\widehat{\rho_i}(X_i)$
- Matching estimation based on the propensity score has the same first step as IPW, but not the second and third steps
- Common support starts to be more complex with imputation methods because you will need to decide how far away from a unit’s own propensity score is a tolerable distance to be considered a “neighbor”

Standard matching strategy

- Pair each treatment unit i with one or more *comparable* control group unit j, where comparability is in terms of proximity, or distance, to the estimated propensity score
- Impute the unit's missing counterfactual outcome $Y_{i(j)}$ based on the unit or units chosen in the previous step
- If more than one are "nearest neighbors", then use the neighbors' weighted outcomes

$$Y_{i(j)} = \sum_{j \in C(i)} w_{ij} Y_j$$

where $C(i)$ is the set of neighbors with $W = 0$ of the treatment unit $i$and w_{ij} is the weight of control group units j with $\sum_{j \in C(i)} w_{ij} = 1$

Imputing the counterfactuals

Let the ATT be our parameter of interest:

$$E[Y_i^1|D_i = 1] - E[Y_i^0|D_i = 1]$$

We estimate it as follows

$$\widehat{ATT} = \frac{1}{N_T} \sum_{i:D_i=1} \left[Y_i - Y_{i(j)} \right]$$

where N_T is the number of matched treatment units in the sample. Note the difference between *imputation* and IPW – the only weight here is $\frac{1}{N_T}$

Matching methods

- The probability of observing two units with *exactly* the same propensity score is in principle zero if $Pr(X = x)$ is continuous
- Several matching methods have been proposed in the literature, but the most widely used are:
  - → Stratification matching
  - → Nearest-neighbor matching (with or without caliper)
  - → Radius matching
  - → Kernel matching
- Typically, one treatment unit i is matched to several control units j, but sometimes one-to-one matching is used

Stratification

- Stratification based on the propensity score is a multi step process that bears resemblance to the stratification/subclassification method proposed by Cochran (1968)
- Method uses brute force to achieve the balancing property discussed earlier, which is then used with weighted differences in means within propensity score “strata”
- Dehejia and Wahba (2002) used stratification matching in their seminal paper

Stratification: Achieving Balance

First create "propensity score strata" inside which you have balanced covariates

1. Sort the data by propensity score and divide into groups of observations with similar propensity scores (e.g., percentiles)
2. Within each strata, test (e.g., t-test) whether the means of the k covariates are equal between treatment and control
3. If so, then stop. If not, it means the covariates aren't balanced *within that propensity score strata* so then divide that strata in half and repeat step 2
4. If a particular covariate is unbalanced for multiple groups, modify the initial logit or probit equation by including higher order terms and/or interactions with that covariate and repeat

Propensity score matching

- Next we review explicit imputation based on the propensity score or what is sometimes called propensity score matching
- King and Nielsen (2019) is a critique of using propensity scores *for matching* (i.e., imputation)
- But not a critique of the propensity score itself or to stratification, regression adjustment, or IPW
- Issues raised have to do with forced balance through trimming and a myriad of other common choices made by the researcher

Ad hoc user choices introduce bias

“[The] more balanced the data, or the more balance it becomes by [trimming] some of the observations through matching, the more likely propensity score matching will degrade inferences.” – King and Nielsen (2019)

Nearest Neighbor

Pretty similar to covariate matching. Formula is

$$\widehat{ATT} = \frac{1}{N_T} \sum_{i:D_i=1} \left[Y_i - \sum_{j \in C(i)_M} w_{ij} Y_j \right]$$

- N_T is the number of treated units i and N_C is number of control units j
- w_{ij} is equal to $\frac{1}{N_C}$ if j is a control unit and zero otherwise
- And unit j is chosen as a control for i if it's propensity score is nearest to that of i

NN Matching: Bias vs. Variance

How far away on the propensity score will you use is what makes some of the different types of matching proposed differ

- Matching just one nearest neighbor minimizes bias at the cost of larger variance
- Matching using additional nearest neighbors increases the bias but decreases the variance

NN Matching: Bias vs. Variance

Matching with or without replacement

- with replacement keeps bias low at the cost of larger variance
- without replacement keeps variance low but at the cost of potential bias

Distance between treatment and control units

- What was historically done was limiting “distance” through various *ad hoc* choices
- Imagine these choices as creating like a cowboy rope lasso that matches to everything inside that circle
- There were two common ways for creating the circle – caliper matching and radius matching.

Caliper matching

- Caliper matching is a variation on NN matching that tries to build brakes into the algorithm as to avoid "bad neighbors" by imposing a tolerable maximum distance (e.g., 0.2 units in the propensity score away from a treatment unit i's propensity score)
- Note – this is a one-to-one imputation, and if there doesn't exist anybody in the control group unit j within that "caliper", then treatment unit i is discarded which as with all trimming changes the parameter we are estimating
- It's difficult to know what this caliper should be *ex ante*, hence why I said it is somewhat *ad hoc*

Radius matching

- Each treatment unit i is matched with the control group units whose propensity score are in a "predefined neighborhood" of the propensity score of the treatment unit.
- **All** the control units with $\widehat{\rho_j}(X_j)$ falling within a radius r from $\widehat{\rho_i}(X_i)$ are matched to the treatment unit i – this is what distinguishes it from calipers, and makes it more similar to covariate matching (Abadie and Imbens 2006, 2008)
- The smaller the radius, the better the quality of the matches, but the higher the possibility some treatment units are not matched because the neighborhood does not contain control group units j

Software

- You can use -`teffects, psmatch`- to get at these two nearest neighbor approaches by setting the number of matches
- You can use -`pscore2`- for stratification
- You can use the `MatchIt` package in R

Failure of econometric estimators (LaLonde 1986)

- Evaluation of the Job Trainings Program (NSW) has a rich history in causal inference
- Bob LaLonde (passed away November 2015) was a Card and Ashenfelter student at Princeton whose job market paper evaluated, not NSW itself, but econometric methods one would use in something like NSW
- Dehejia and Wahba (1999; 2002) used LaLonde's data with propensity score matching and found they could recover known effects
- Critiques by Petra Todd, Jeff Smith and others followed which I'll summarize

Summarizing LaLonde (1986)

- Very clever study that combined experimental and non-experimental data to ascertain whether popular econometric methods could recover unbiased effects when those effects were already known
- Damning conclusion – 1986 AER (it was LaLonde's JMP) found econometric methods failed to get the number right, and worse, failed to get the sign right
- Was a critical paper in the emerging "credibility crisis" within labor and helped fuel the type of work we now broadly consider to be design based causal inference

LaLonde, Robert J. (1986). "Evaluating the Econometric Evaluations of Training Programs with Experimental Data". *American Economic Review*.

LaLonde's study was **not** an evaluation of the NSW program, as that had been done, but rather an evaluation of econometric models done by:

- replacing the experimental NSW control group with non-experimental control group drawn from two nationally representative survey datasets: Current Population Survey (CPS) and Panel Study of Income Dynamics (PSID)
- estimating the average effect using non-experimental workers as controls for the NSW trainees
- comparing his non-experimental estimates to the experimental estimates of $900

LaLonde (1986)

- LaLonde's conclusion: available econometric approaches were biased and inconsistent
  - → His estimates were way off and usually the wrong sign
  - → Conclusion was influential in policy circles and led to greater push for more experimental evaluations

Description of NSW Job Trainings Program

The National Supported Work Demonstration (NSW), operated by Manpower Demonstration Research Corp in the mid-1970s:

- was a temporary employment program designed to help disadvantaged workers lacking basic job skills move into the labor market by giving them work experience and counseling in a sheltered environment
- was also unique in that it **randomly assigned** qualified applicants to training positions:
  - → **Treatment group**: received all the benefits of NSW program
  - → **Control group**: left to fend for themselves
- admitted AFDC females, ex-drug addicts, ex-criminal offenders, and high school dropouts of both sexes

NSW Program

- Treatment group members were:
  - → guaranteed a job for 9-18 months depending on the target group and site
  - → divided into crews of 3-5 participants who worked together and met frequently with an NSW counselor to discuss grievances and performance
  - → paid for their work
- Control group members were randomized so the same
- Note: the randomization balanced observables and unobservables across the two arms, thus enabling the estimation of an ATE for the people who self-selected into the program

NSW Program

- Other details about the NSW program:
  - → Wages: NSW offered the trainees lower wage rates than they would've received on a regular job, but allowed their earnings to increase for satisfactory performance and attendance
  - → Post-treatment: after their term expired, they were forced to find regular employment
  - → Job types: varied within sites – gas station attendant, working at a printer shop – and males and females were frequently performing different kinds of work

NSW Data

- NSW data collection:
  - → MDRC collected earnings and demographic information from both treatment and control at baseline and every 9 months thereafter
  - → Conducted up to 4 post-baseline interviews
  - → Different sample sizes from study to study can be confusing, but has simple explanations

NSW Data

- Estimation:
  - → NSW was a randomized job trainings program; therefore estimating the average treatment effect is straightforward:

$$SDO = \frac{1}{N_t} \sum_{D_i=1} Y_i - \frac{1}{N_c} \sum_{D_i=0} Y_i \approx E[Y^1 - Y^0]$$

    in large samples assuming treatment selection is independent of potential outcomes (randomization) – i.e., $(Y^0, Y^1) \perp\!\!\!\perp D$.

- NSW worked: Treatment group participants' real earnings post-treatment (1978) was positive and economically meaningful – $\approx$ \$900 (LaLonde 1986) to \$1,800 (Dehejia and Wahba 2002) depending on the sample used

TABLE 5—EARNINGS COMPARISONS AND ESTIMATED TRAINING EFFECTS FOR THE NSW MALE PARTICIPANTS USING COMPARISON GROUPS FROM THE *PSID* AND THE *CPS-SSA*[a,b]

Name of Comparison Group[d]	Comparison Group Earnings Growth 1975–78 (1)	NSW Treatment Earnings Less Comparison Group Earnings: Pre-Training Year, 1975 Unadjusted (2)	NSW Treatment Earnings Less Comparison Group Earnings: Pre-Training Year, 1975 Adjusted[c] (3)	NSW Treatment Earnings Less Comparison Group Earnings: Post-Training Year, 1978 Unadjusted (4)	NSW Treatment Earnings Less Comparison Group Earnings: Post-Training Year, 1978 Adjusted[c] (5)	Difference in Differences: Difference in Earnings Growth 1975–78 Treatments Less Comparisons: Without Age (6)	Difference in Differences: Difference in Earnings Growth 1975–78 Treatments Less Comparisons: With Age (7)	Unrestricted Difference in Differences: Quasi Difference in Earnings Growth 1975–78: Unadjusted (8)	Unrestricted Difference in Differences: Quasi Difference in Earnings Growth 1975–78: Adjusted[c] (9)	Controlling for All Observed Variables and Pre-Training Earnings (10)
Controls	$2,063	$39	$−21	$886	$798	$847	$856	$897	$802	$662
	(325)	(383)	(378)	(476)	(472)	(560)	(558)	(467)	(467)	(506)
PSID-1	$2,043	−$15,997	−$7,624	−$15,578	−$8,067	$425	−$749	−$2,380	−$2,119	−$1,228
	(237)	(795)	(851)	(913)	(990)	(650)	(692)	(680)	(746)	(896)
PSID-2	$6,071	−$4,503	−$3,669	−$4,020	−$3,482	$484	−$650	−$1,364	−$1,694	−$792
	(637)	(608)	(757)	(781)	(935)	(738)	(850)	(729)	(878)	(1024)
PSID-3	($3,322	($455	$455	$697	−$509	$242	−$1,325	$629	−$552	$397
	(780)	(539)	(704)	(760)	(967)	(884)	(1078)	(757)	(967)	(1103)
CPS-SSA-1	$1,196	−$10,585	−$4,654	−$8,870	−$4,416	$1,714	$195	−$1,543	−$1,102	−$805
	(61)	(539)	(509)	(562)	(557)	(452)	(441)	(426)	(450)	(484)
CPS-SSA-2	$2,684	−$4,321	−$1,824	−$4,095	−$1,675	$226	−$488	−$1,850	−$782	−$319
	(229)	(450)	(535)	(537)	(672)	(539)	(530)	(497)	(621)	(761)
CPS-SSA-3	$4,548	$337	$878	−$1,300	$224	−$1,637	−$1,388	−$1,396	$17	$1,466
	(409)	(343)	(447)	(590)	(766)	(631)	(655)	(582)	(761)	(984)

[a] The columns above present the estimated training effect for each econometric model and comparison group. The dependent variable is earnings in 1978. Based on the experimental data an unbiased estimate of the impact of training presented in col. 4 is $886. The first three columns present the difference between each comparison group's 1975 and 1978 earnings and the difference between the pre-training earnings of each comparison group and the NSW treatments.

[b] Estimates are in 1982 dollars. The numbers in parentheses are the standard errors.

[c] The exogenous variables used in the regression adjusted equations are age, age squared, years of schooling, high school dropout status, and race.

[d] See Table 3 for definitions of the comparison groups.

Switching out the control group

- Think of $800 to $900 as the “ground truth” since row 1 was using the RCT
- LaLonde “drops” the experimental controls (which satisfied independence) and “replaces” it with six different draws from two nationally representative surveys (PSID and CPS)
- Now the dataset contains a negatively selected treatment group compared to a nationally representative control group
- Will selection on observable methods “work”?

TABLE 5—EARNINGS COMPARISONS AND ESTIMATED TRAINING EFFECTS FOR THE NSW MALE PARTICIPANTS USING COMPARISON GROUPS FROM THE *PSID* AND THE *CPS-SSA*[a,b]

Name of Comparison Group[d]	Comparison Group Earnings Growth 1975–78 (1)	NSW Treatment Earnings Less Comparison Group Earnings: Pre-Training Year, 1975 Unadjusted (2)	NSW Treatment Earnings Less Comparison Group Earnings: Pre-Training Year, 1975 Adjusted[c] (3)	NSW Treatment Earnings Less Comparison Group Earnings: Post-Training Year, 1978 Unadjusted (4)	NSW Treatment Earnings Less Comparison Group Earnings: Post-Training Year, 1978 Adjusted[c] (5)	Difference in Differences: Difference in Earnings Growth 1975–78 Treatments Less Comparisons: Without Age (6)	Difference in Differences: Difference in Earnings Growth 1975–78 Treatments Less Comparisons: With Age (7)	Unrestricted Difference in Differences: Quasi Difference in Earnings Growth 1975–78: Unadjusted (8)	Unrestricted Difference in Differences: Quasi Difference in Earnings Growth 1975–78: Adjusted[c] (9)	Controlling for All Observed Variables and Pre-Training Earnings (10)
Controls	$2,063	$39	$−21	$886	$798	$847	$856	$897	$802	$662
	(325)	(383)	(378)	(476)	(472)	(560)	(558)	(467)	(467)	(506)
PSID-1	$2,043	−$15,997	−$7,624	−$15,578	−$8,067	$425	−$749	−$2,380	−$2,119	−$1,228
	(237)	(795)	(851)	(913)	(990)	(650)	(692)	(680)	(746)	(896)
PSID-2	$6,071	−$4,503	−$3,669	−$4,020	−$3,482	$484	−$650	−$1,364	−$1,694	−$792
	(637)	(608)	(757)	(781)	(935)	(738)	(850)	(729)	(878)	(1024)
PSID-3	($3,322	($455	$455	$697	−$509	$242	−$1,325	$629	−$552	$397
	(780)	(539)	(704)	(760)	(967)	(884)	(1078)	(757)	(967)	(1103)
CPS-SSA-1	$1,196	−$10,585	−$4,654	−$8,870	−$4,416	$1,714	$195	−$1,543	−$1,102	−$805
	(61)	(539)	(509)	(562)	(557)	(452)	(441)	(426)	(450)	(484)
CPS-SSA-2	$2,684	−$4,321	−$1,824	−$4,095	−$1,675	$226	−$488	−$1,850	−$782	−$319
	(229)	(450)	(535)	(537)	(672)	(539)	(530)	(497)	(621)	(761)
CPS-SSA-3	$4,548	$337	$878	−$1,300	$224	−$1,637	−$1,388	−$1,396	$17	$1,466
	(409)	(343)	(447)	(590)	(766)	(631)	(655)	(582)	(761)	(984)

[a] The columns above present the estimated training effect for each econometric model and comparison group. The dependent variable is earnings in 1978. Based on the experimental data an unbiased estimate of the impact of training presented in col. 4 is $886. The first three columns present the difference between each comparison group's 1975 and 1978 earnings and the difference between the pre-training earnings of each comparison group and the NSW treatments.

[b] Estimates are in 1982 dollars. The numbers in parentheses are the standard errors.

[c] The exogenous variables used in the regression adjusted equations are age, age squared, years of schooling, high school dropout status, and race.

[d] See Table 3 for definitions of the comparison groups.

Imbalanced covariates for experimental and non-experimental samples

	All		CPS Controls $N_c = 15,992$	NSW Trainees $N_t = 297$		
covariate	mean	(s.d.)	mean	mean	t-stat	diff
Black	0.09	0.28	0.07	0.80	47.04	-0.73
Hispanic	0.07	0.26	0.07	0.94	1.47	-0.02
Age	33.07	11.04	33.2	24.63	13.37	8.6
Married	0.70	0.46	0.71	0.17	20.54	0.54
No degree	0.30	0.46	0.30	0.73	16.27	-0.43
Education	12.0	2.86	12.03	10.38	9.85	1.65
1975 Earnings	13.51	9.31	13.65	3.1	19.63	10.6
1975 Unemp	0.11	0.32	0.11	0.37	14.29	-0.26

Dehija and Wahba (1999)

- Dehejia and Wahba (DW) update LaLonde's original study using propensity score matching
    1. Dehejia, Rajeev H. and Sadek Wahba (1999). "Causal Effects in Nonexperimental Studies: Reevaluating the Evaluation of Training Programs". Journal of the American Statistical Association, vol. 94(448): 1053-1062 (pdf)
- Can propensity score matching improve over the estimators that LaLonde examined?

Table 1. Sample Means of Characteristics for NSW and Comparison Samples

	No. of observations	*Age*	*Education*	*Black*	*Hispanic*	*No degree*	*Married*	*RE74 (U.S. $)*	*RE75 (U.S. $)*
NSW/Lalonde:[a]									
Treated	297	24.63	10.38	.80	.09	.73	.17		3,066
		(.32)	(.09)	(.02)	(.01)	(.02)	(.02)		(236)
Control	425	24.45	10.19	.80	.11	.81	.16		3,026
		(.32)	(.08)	(.02)	(.02)	(.02)	(.02)		(252)
RE74 subset:[b]									
Treated	185	25.81	10.35	.84	.059	.71	.19	2,096	1,532
		(.35)	(.10)	(.02)	(.01)	(.02)	(.02)	(237)	(156)
Control	260	25.05	10.09	.83	.1	.83	.15	2,107	1,267
		(.34)	(.08)	(.02)	(.02)	(.02)	(.02)	(276)	(151)
Comparison groups:[c]									
PSID-1	2,490	34.85	12.11	.25	.032	.31	.87	19,429	19,063
		[.78]	[.23]	[.03]	[.01]	[.04]	[.03]	[991]	[1,002]
PSID-2	253	36.10	10.77	.39	.067	.49	.74	11,027	7,569
		[1.00]	[.27]	[.04]	[.02]	[.05]	[.04]	[853]	[695]
PSID-3	128	38.25	10.30	.45	.18	.51	.70	5,566	2,611
		[1.17]	[.29]	[.05]	[.03]	[.05]	[.05]	(686)	[499]
CPS-1	15,992	33.22	12.02	.07	.07	.29	.71	14,016	13,650
		[.81]	[.21]	[.02]	[.02]	[.03]	[.03]	[705]	[682]
CPS-2	2,369	28.25	11.24	.11	.08	.45	.46	8,728	7,397
		[.87]	[.19]	[.02]	[.02]	[.04]	[.04]	[667]	[600]
CPS-3	429	28.03	10.23	.21	.14	.60	.51	5,619	2,467
		[.87]	[.23]	[.03]	[.03]	[.04]	[.04]	[552]	[288]

NOTE: Standard errors are in parentheses. Standard error on difference in means with RE74 subset/treated is given in brackets. Age = age in years; Education = number of years of schooling; Black = 1 if black, 0 otherwise; Hispanic = 1 if Hispanic, 0 otherwise; No degree = 1 if no high school degree, 0 otherwise; Married = 1 if married, 0 otherwise; RE*x* = earnings in calendar year 19*x*.

[a] NSW sample as constructed by Lalonde (1986).

[b] The subset of the Lalonde sample for which RE74 is available.

[c] Definition of comparison groups (Lalonde 1986):

PSID-1: All male household heads under age 55 who did not classify themselves as retired in 1975.

PSID-2: Selects from PSID-1 all men who were not working when surveyed in the spring of 1976.

PSID-3: Selects from PSID-2 all men who were not working in 1975.

CPS-1: All CPS males under age 55.

CPS-2: Selects from CPS-1 all males who were not working when surveyed in March 1976.

CPS-3: Selects from CPS-2 all the unemployed males in 1976 whose income in 1975 was below the poverty level.

PSID1–3 and CPS-1 are identical to those used by Lalonde. CPS2–3 are similar to those used by Lalonde, but Lalonde's original subset could not be recreated.

Table 2. Lalonde's Earnings Comparisons and Estimated Training Effects for the NSW Male Participants Using Comparison Groups From the PSID and the CPS[a]

	A. Lalonde's original sample					B. RE74 subsample (results do not use RE74)					C. RE74 subsample (results use RE74)				
	NSW treatment earnings less comparison group earnings 1978		Unrestricted differences in differences: Quasi-difference in earnings growth 1975−1978		Controlling for all	NSW treatment earnings less comparison group earnings 1978		Unrestricted differences in differences: Quasi-difference in earnings growth 1975−1978		Controlling for all	NSW treatment earnings less comparison group earnings 1978		Unrestricted differences in differences: Quasi-difference in earnings growth 1975−1978		Controlling for all
Comparison group	Unadjusted[b] (1)	Adjusted[c] (2)	Unadjusted[d] (3)	Adjusted[e] (4)	variables[f] (5)	Unadjusted[b] (1)	Adjusted[c] (2)	Unadjusted[d] (3)	Adjusted[e] (4)	variables[f] (5)	Unadjusted[b] (1)	Adjusted[c] (2)	Unadjusted[d] (3)	Adjusted[e] (4)	variables[f] (5)
NSW	886	798	879	802	820	1,794	1,672	1,750	1,631	1,612	1,794	1,688	1,750	1,672	1,655
	(472)	(472)	(467)	(468)	(468)	(633)	(637)	(632)	(637)	(639)	(633)	(636)	(632)	(638)	(640)
PSID-1	−15,578	−8,067	−2,380	−2,119	−1,844	−15,205	−7,741	−582	−265	186	−15,205	−879	−582	218	731
	(913)	(990)	(680)	(746)	(762)	(1155)	(1175)	(841)	(881)	(901)	(1155)	(931)	(841)	(866)	(886)
PSID-2	−4,020	−3,482	−1,364	−1,694	−1,876	−3,647	−2,810	721	298	111	−3,647	94	721	907	683
	(781)	(935)	(729)	(878)	(885)	(960)	(1082)	(886)	(1004)	(1032)	(960)	(1042)	(886)	(1004)	(1028)
PSID-3	697	−509	629	−552	−576	1,070	35	1,370	243	298	1,070	821	1,370	822	825
	(760)	(967)	(757)	(967)	(968)	(900)	(1101)	(897)	(1101)	(1105)	(900)	(1100)	(897)	(1101)	(1104)
CPS-1	−8,870	−4,416	−1,543	−1,102	−987	−8,498	−4,417	−78	525	709	−8,498	−8	−78	739	972
	(562)	(577)	(426)	(450)	(452)	(712)	(714)	(537)	(557)	(560)	(712)	(572)	(537)	(547)	(550)
CPS-2	−4,195	−2,341	−1,649	−1,129	−1,149	−3,822	−2,208	−263	371	305	−3,822	615	−263	879	790
	(533)	(620)	(459)	(551)	(551)	(671)	(746)	(574)	(662)	(666)	(671)	(672)	(574)	(654)	(658)
CPS-3	−1,008	−1	−1,204	−263	−234	−635	375	−91	844	875	−635	1,270	−91	1,326	1,326
	(539)	(681)	(532)	(677)	(675)	(657)	(821)	(641)	(808)	(810)	(657)	(798)	(641)	(796)	(798)

NOTES: Panel A replicates the sample of Lalonde (1986, table 5). The estimates for columns (1)–(4) for NSW, PSID1–3, and CPS-1 are identical to Lalonde's. CPS-2 and CPS-3 are similar but not identical, because we could not exactly recreate his subset. Column (5) differs because the data file that we obtained did not contain all of the covariates used in column (10) of Lalonde's Table 5.

[a] Estimated effect of training on RE78. Standard errors are in parentheses. The estimates are in 1982 dollars.

[b] The estimates based on the NSW control groups are unbiased estimates of the treatment impacts for the original sample ($886) and for the RE74 sample ($1,794).

[c] The exogenous variables used in the regressions-adjusted equations are age, age squared, years of schooling, high school dropout status, and race (and RE74 in Panel C).

[d] Regresses RE78 on a treatment indicator and RE75.

[e] The same as (d), but controls for the additional variables listed under (c).

[f] Controls for all pretreatment covariates.

Covariate imbalance

- Conditional on the propensity score, the covariates are independent of the treatment, suggesting that the distribution of covariate values should be the same for both treatment and control groups
- This can be checked as we have data on all three once we've estimated the propensity score
- DW note that the two samples have *severe* imbalance on *observables* – a huge number of non-experimental controls have propensity scores almost exactly equal to 0
- Their analysis will "trim" (which will ultimately have implications for interpretation)

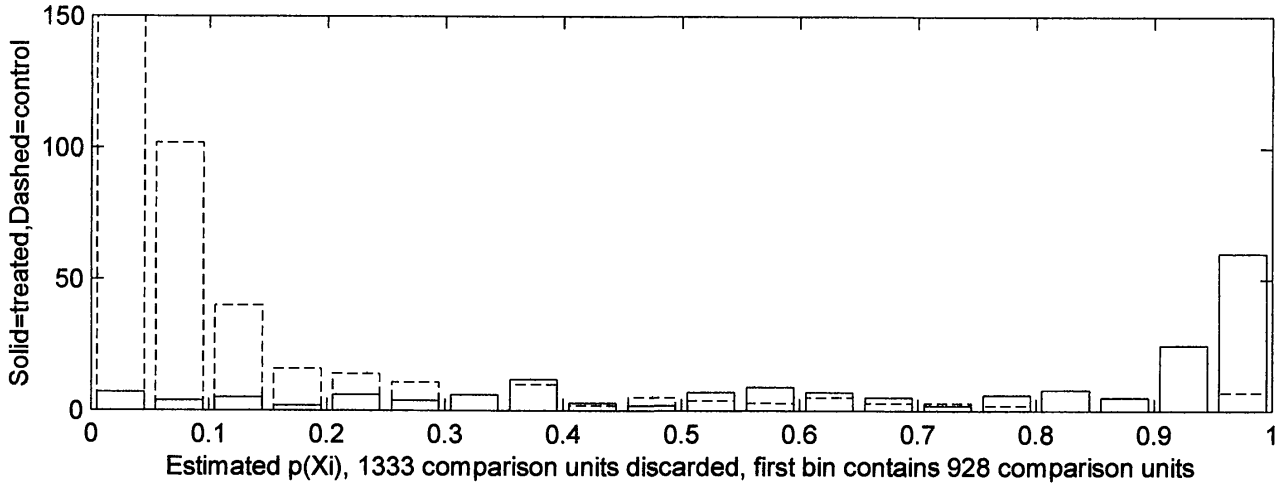


Figure 1. Histogram of the Estimated Propensity Score for NSW Treated Units and PSID Comparison Units. The 1,333 PSID units whose estimated propensity score is less than the minimum estimated propensity score for the treatment group are discarded. The first bin contains 928 PSID units. There is minimal overlap between the two groups. Three bins (.8–.85, .85–.9, and .9–.95) contain no comparison units. There are 97 treated units with an estimated propensity score greater than .8 and only 7 comparison units.

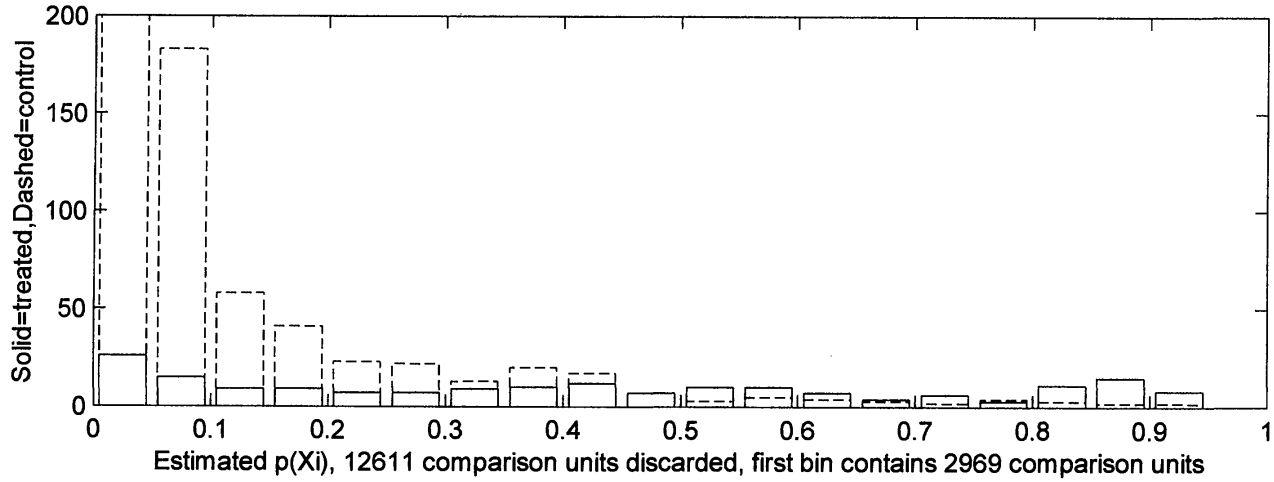


Figure 2. Histogram of the Estimated Propensity Score for NSW Treated Units and CPS Comparison Units. The 12,611 CPS units whose estimated propensity score is less than the minimum estimated propensity score for the treatment group are discarded. The first bin contains 2,969 CPS units. There is minimal overlap between the two groups, but the overlap is greater than in Figure 1; only one bin (.45–.5) contains no comparison units, and there are 35 treated and 7 comparison units with an estimated propensity score greater than .8.

Table 3. Estimated Training Effects for the NSW Male Participants Using Comparison Groups From PSID and CPS

	NSW earnings less comparison group earnings		*NSW treatment earnings less comparison group earnings, conditional on the estimated propensity score*					
			Quadratic in score[b]	*Stratifying on the score*			*Matching on the score*	
	(1) Unadjusted	*(2) Adjusted*[a]	*(3)*	*(4) Unadjusted*	*(5) Adjusted*	*(6) Observations*[c]	*(7) Unadjusted*	*(8) Adjusted*[d]
NSW	1,794	1,672						
	(633)	(638)						
PSID-1[e]	−15,205	731	294	1,608	1,494	1,255	1,691	1,473
	(1,154)	(886)	(1,389)	(1,571)	(1,581)		(2,209)	(809)
PSID-2[f]	−3,647	683	496	2,220	2,235	389	1,455	1,480
	(959)	(1,028)	(1,193)	(1,768)	(1,793)		(2,303)	(808)
PSID-3[f]	1,069	825	647	2,321	1,870	247	2,120	1,549
	(899)	(1,104)	(1,383)	(1,994)	(2,002)		(2,335)	(826)
CPS-1[g]	−8,498	972	1,117	1,713	1,774	4,117	1,582	1,616
	(712)	(550)	(747)	(1,115)	(1,152)		(1,069)	(751)
CPS-2[g]	−3,822	790	505	1,543	1,622	1,493	1,788	1,563
	(670)	(658)	(847)	(1,461)	(1,346)		(1,205)	(753)
CPS-3[g]	−635	1,326	556	1,252	2,219	514	587	662
	(657)	(798)	(951)	(1,617)	(2,082)		(1,496)	(776)

[a] Least squares regression: RE78 on a constant, a treatment indicator, age, age^2, education, no degree, black, Hispanic, RE74, RE75.

[b] Least squares regression of RE78 on a quadratic on the estimated propensity score and a treatment indicator, for observations used under stratification; see note (g).

[c] Number of observations refers to the actual number of comparison and treatment units used for (3)–(5); namely, all treatment units and those comparison units whose estimated propensity score is greater than the minimum, and less than the maximum, estimated propensity score for the treatment group.

[d] Weighted least squares: treatment observations weighted as 1, and control observations weighted by the number of times they are matched to a treatment observation [same covariates as (a)]. Propensity scores are estimated using the logistic model, with specifications as follows:

[e] PSID-1: Prob (T_i = 1) = F(age, age^2, education, education^2, married, no degree, black, Hispanic, RE74, RE75, RE74^2, RE75^2, u74*black).

[f] PSID-2 and PSID-3: Prob (T_i = 1) = F(age, age^2, education, education^2, no degree, married, black, Hispanic, RE74, RE74^2, RE75, RE75^2, u74, u75).

[g] CPS-1, CPS-2, and CPS-3: Prob (T_i = 1) = F(age, age^2, education, education^2, no degree, married, black, Hispanic, RE74, RE75, u74, u75, education*RE74, age^3).

Table 4. Sample Means of Characteristics for Matched Control Samples

Matched samples	*No. of observations*	*Age*	*Education*	*Black*	*Hispanic*	*No degree*	*Married*	*RE74 (U.S. $)*	*RE75 (U.S. $)*
NSW	185	25.81	10.35	.84	.06	.71	.19	2,096	1,532
MPSID-1	56	26.39	10.62	.86	.02	.55	.15	1,794	1,126
		[2.56]	[.63]	[.13]	[.06]	[.13]	[.12]	[1,406]	[1,146]
MPSID-2	49	25.32	11.10	.89	.02	.57	.19	1,599	2,225
		[2.63]	[.83]	[.14]	[.08]	[.16]	[.16]	[1,905]	[1,228]
MPSID-3	30	26.86	10.96	.91	.01	.52	.25	1,386	1,863
		[2.97]	[.84]	[.13]	[.08]	[.16]	[.16]	[1,680]	[1,494]
MCPS-1	119	26.91	10.52	.86	.04	.64	.19	2,110	1,396
		[1.25]	[.32]	[.06]	[.04]	[.07]	[.06]	[841]	[563]
MCPS-2	87	26.21	10.21	.85	.04	.68	.20	1,758	1,204
		[1.43]	[.37]	[.08]	[.05]	[.09]	.08	[896]	[661]
MCPS-3	63	25.94	10.69	.87	.06	.53	.13	2,709	1,587
		[1.68]	[.48]	[.09]	[.06]	[.10]	[.09]	[1,285]	[760]

NOTE: Standard error on the difference in means with NSW sample is given in brackets.
MPSID1–3 and MCPS1–3 are the subsamples of PSID1–3 and CPS1–3 that are matched to the treatment group.

Replies by econometricians to DW

- Heckman, Smith and Todd concluded from their own work that in order for matching estimators to have low bias, you need the following:
  1. A rich set of variables related to program participation and predictive of Y^0 labor market outcomes,
  2. Nonexperimental comparison group be drawn from the same local labor markets as the participants and
  3. Dependent variable (e.g., earnings) be measured in the same way for participants and nonparticipants
- All three of these conditions fail to hold in DW (1999, 2002) according to Smith and Todd (2005)
- DW also note the importance of conditioning on pre-treatment lagged outcomes (e.g., real earnings in $t-1$, $t-2$, etc.) as well as *trimming*

Smith and Todd, diff-in-diff, doubly robust

- Difference-in-differences with propensity scores tended to work well in Smith and Todd (2005) though the effect sizes are much larger
- In my Causal Inference II workshop, we use Sant'anna And Zhao's double robust DiD and get nearly the exact same parameter estimate as the experimental finding

Coding together

- Let’s spend some time together trying to replicate the Lalonde study ourselves
- I use replicate lightly – our goal is syntax only
- It’s in the Lalonde lab on githbub for this workshop

Interpreting OLS coefficients

- Most common causal model is OLS with covariate controls

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- OLS is the best linear predictor (Angrist and Pischke 2009)
- But the best linear predictor of the *realized* outcome is not the same as being the best linear predictor of the *missing potential outcome*

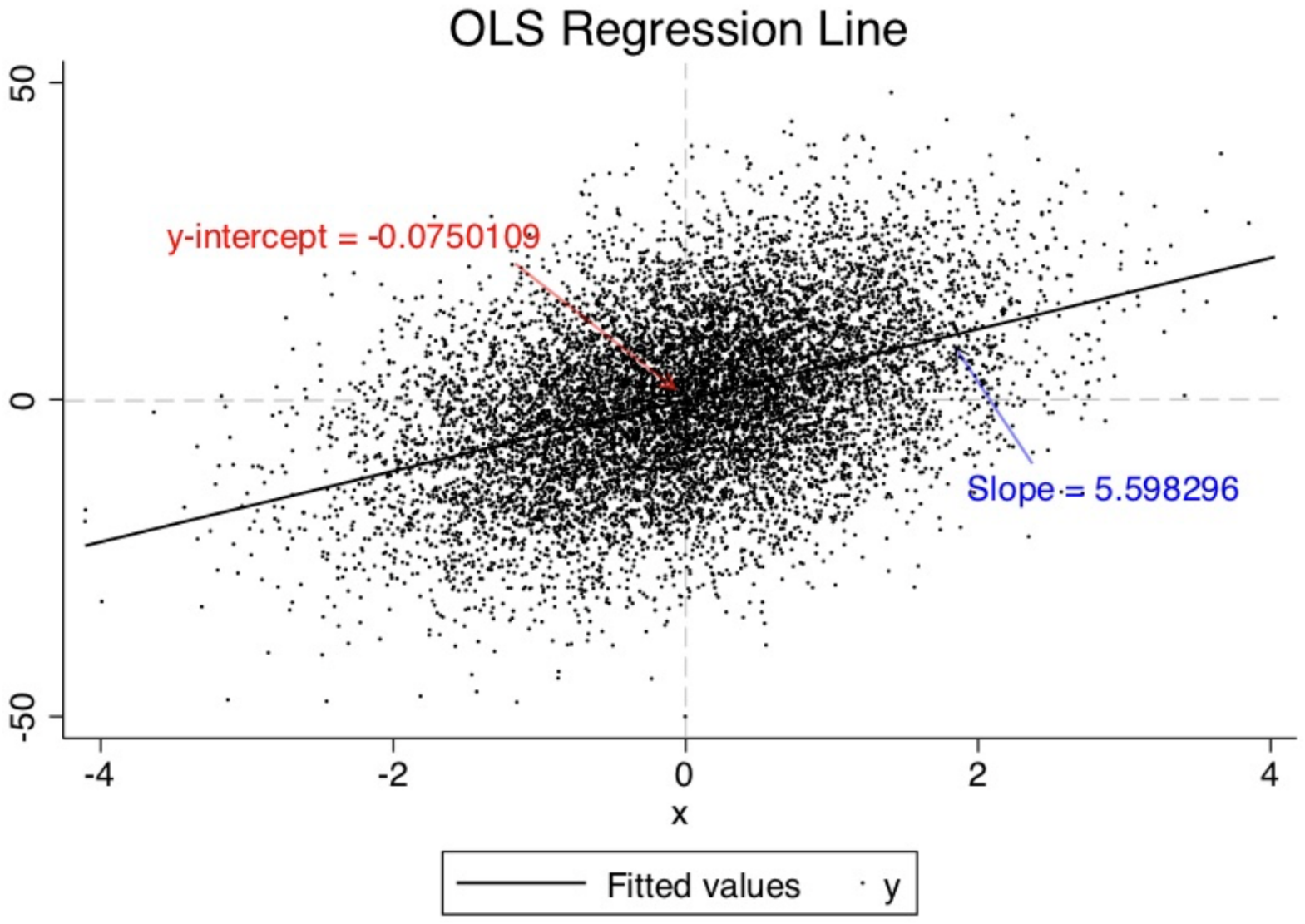
OLS Simulated Code

Stata Code | R Code | Python Code

▼ Code

```
set seed 1
clear
set obs 10000
gen x = rnormal()
gen u  = rnormal()
gen y  = 5.5*x + 12*u
reg y x
predict yhat1
gen yhat2 = -0.0750109  + 5.598296*x // Compare yhat1 and yhat2
sum yhat*
predict uhat1, residual
gen uhat2=y-yhat2
sum uhat*
twoway (lfit y x, lcolor(black) lwidth(medium)) (scatter y x, mcolor(bla
msize(tiny) msymbol(point)), title(OLS Regression Line)
rvfplot, yline(0)
```

OLS Extrapolates Using Functional Form (e.g., Lines)



Bivariate vs multivariate regressions

- Interpreting the OLS coefficient in bivariate regression is written out as a scaled covariance:

$$\widehat{\beta_1} = \frac{Cov(Y_i, X_i)}{Var(X_i)}$$

- But when we are looking at a multivariate regression, what is it?

Applying FWL theorem

- Frisch-Waugh-Lovell also helps us interpret $\widehat{\beta}_1$ when there are covariates
- Use the FWL theorem to turn the multivariate regression coefficient into the simple bivariate one
- Angrist and Pischke (2009) call FWL the “regression anatomy theorem” and Filoso (2013) has an excellent and original proof

Applying FWL theorem

- Can we estimate the causal effect of family size on labor supply by regressing labor supply (`Y`) on family size (`X`)?

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- If family size is random, then we can interpret $\widehat{\beta_1}$ as the ATE
- But how do we interpret $\widehat{\beta_1}$ if `family size` is only conditionally random?

Applying FWL theorem

- Assume the longer model is:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + \gamma_1 \texttt{White}_i + \gamma_2 \texttt{Married}_i \\ &\quad + \gamma_3 \texttt{Age}_i + \gamma_4 \texttt{Employed}_i + u_i \end{aligned}$$

- Assume unconfoundedness and constant treatment effects so that family size is independent conditional on all controls
- FWL shows that in a multivariate regression, any one coefficient fitted can be reconceived as a simple scaled covariance of the outcome and residualized treatment variable

FWL Theorem

FWL Theorem

Assume your main multiple regression model of interest:

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \cdots + \beta_K x_{Ki} + e_i$$

and an auxiliary regression in which the variable x_{1i} is regressed on all the remaining independent variables

$$x_{1i} = \gamma_0 + \gamma_{k-1} x_{k-1i} + \gamma_{k+1} x_{k+1i} + \cdots + \gamma_K x_{Ki} + f_i$$

and $\tilde{x}_{1i} = x_{1i} - \widehat{x}_{1i}$ being the residual from the auxiliary regression. The parameter β_1 can be rewritten as:

$$\beta_1 = \frac{Cov(y_i, \tilde{x}_{1i})}{Var(\tilde{x}_{1i})}$$

FWL Proof (Proof by Filoso 2013)

To prove the theorem, note $E[\tilde{x}_{ki}] = E[x_{ki}] - E[\widehat{x}_{ki}] = E[f_i]$, and plug y_i and residual $\tilde{x}_{ki}$ from x_{ki} auxiliary regression into the covariance $cov(y_i, \tilde{x}_{ki})$

$$\begin{aligned}
\beta_k &= \frac{cov(y_i, \tilde{x}_{ki})}{var(\tilde{x}_{ki})} \\
&= \frac{cov(\beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \cdots + \beta_K x_{Ki} + e_i, \tilde{x}_{ki})}{var(\tilde{x}_{ki})} \\
&= \frac{cov(\beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \cdots + \beta_K x_{Ki} + e_i, f_i)}{var(f_i)}
\end{aligned}$$

1. Since by construction $E[f_i] = 0$, it follows that the term $\beta_0 E[f_i] = 0$.
2. Since f_i is a linear combination of all the independent variables with the exception of x_{ki}, it must be that

$$\beta_1 E[f_i x_{1i}] = \cdots = \beta_{k-1} E[f_i x_{k-1i}] = \beta_{k+1} E[f_i x_{k+1i}] = \cdots = \beta_K E[f_i x_{KI}] = 0$$

FWL Proof (Proof by Filoso 2013)

3. Consider now the term $E[e_i f_i]$. This can be written as:

$$\begin{aligned} E[e_i f_i] &= E[e_i f_i] \\ &= E[e_i \tilde{x}_{ki}] \\ &= E[e_i (x_{ki} - \widehat{x}_{ki})] \\ &= E[e_i x_{ki}] - E[e_i \tilde{x}_{ki}] \end{aligned}$$

Since e_i is uncorrelated with any independent variable, it is also uncorrelated with x_{ki}: accordingly, we have $E[e_i x_{ki}] = 0$. With regard to the second term of the subtraction, substituting the predicted value from the x_{ki} auxiliary regression, we get

$$E[e_i \tilde{x}_{ki}] = E[e_i (\widehat{\gamma}_0 + \widehat{\gamma}_1 x_{1i} + \cdots + \widehat{\gamma}_{k-1} x_{k-1} i + \widehat{\gamma}_{k+1} x_{k+1i} + \cdots + \widehat{\gamma}_K x_{Ki})]$$

Once again, since e_i is uncorrelated with any independent variable, the expected value of the terms is equal to zero. Then, it follows $E[e_i f_i] = 0$.

FWL Proof (Proof by Filoso 2013)

4. The only remaining term is $E[\beta_k x_{ki} f_i]$ which equals $E[\beta_k x_{ki} \tilde{x}_{ki}]$ since $f_i = \tilde{x}_{ki}$. The term x_{ki} can be substituted using a rewriting of the auxiliary regression model, x_{ki}, such that

$$x_{ki} = E[x_{ki}|X_{-k}] + \tilde{x}_{ki}$$

This gives

$$\begin{aligned} E[\beta_k x_{ki} \tilde{x}_{ki}] &= E[\beta_k E[\tilde{x}_{ki}(E[x_{ki}|X_{-k}] + \tilde{x}_{ki})]] \\ &= \beta_k E[\tilde{x}_{ki}(E[x_{ki}|X_{-k}] + \tilde{x}_{ki})] \\ &= \beta_k \{E[\tilde{x}_{ki}^2] + E[(E[x_{ki}|X_{-k}]\tilde{x}_{ki})]\} \\ &= \beta_k var(\tilde{x}_{ki}) \end{aligned}$$

which follows directly from the orthogonoality between $E[x_{ki}|X_{-k}]$ and $\tilde{x}_{ki}$. From previous derivations we finally get

$$cov(y_i, \tilde{x}_{ki}) = \beta_k var(\tilde{x}_{ki})$$

which completes the proof.

FWL

- Let's review the FWL "partialing out" interpretation of OLS estimated $\widehat{\beta_1}$ in code
- We will use the fwl.do at /Labs/Matching for this

Core OLS assumptions

- Return to OLS model with additive covariates controls:

$$Y_i = \alpha + \delta D_i + \beta X_i + \gamma Z_i + \varepsilon_i$$

- Which average treatment effect parameter does $\widehat{\delta}$ estimate? What assumptions are imposed by exogeneity?
- This hopefully is where I'll be able to convince you of how important it is that you identify ahead of time *which* causal effect
- This is **not** a criticism of OLS but rather of that specification above

OLS Assumptions

- Typically we assume that the mean error is zero conditional on all covariates, called exogeneity
- This is a pregnant assumption as it turns out
- Imbens and Rubin discussed it in their book on causal inference from 2015
- I'll pull out their quote and proof but we will then discuss some other materials

OLS Assumptions

"In many empirical studies in social sciences, causal effects are estimated through linear regression, where, typically it is implicitly assumed that in the super-population,

$$E[Y_i^D | X_i] = \alpha + \delta_{sp} \cdot D + X_i \beta$$

for some values of the three unknown parameters, α, δ_{sp} and β where $\delta_{sp} = E_{sp}[Y_i^1 - Y_i^0]$."

What about OLS? (Imbens and Rubin 2015)

"Defining $\varepsilon_i = Y_i - \delta_{sp} \cdot D_i - X_i\beta$ *so that we can write*

$$Y_i = \alpha + \delta_{sp} \cdot D_i + X_i\beta + \varepsilon_i$$

it is then assumed that

$$\varepsilon_i \perp\!\!\!\perp D_i, X_i$$

This assumption is often referred to as ***exogeneity*** *of the treatment (and the pre-treatment variables) in the econometrics literature."*

OLS Assumptions

"The regression function is interpreted as a causal relation, in our sense of the term "causal", namely that if we manipulate the treatment D_i, then the outcome would change in expectation by an amount δ_{sp}. Hence in the potential outcomes formulation, we have

$$
\begin{aligned}
Y_i^0 &= \alpha + X_i\beta + \varepsilon_i \\
Y_i^1 &= Y_i^0 + \delta_{sp}
\end{aligned}
$$

OLS Assumptions

"Then, because ε_i is a function of Y_i^0 and X_i given the parameters,

$$Pr(D_i = 1|Y_i^0, Y_i^1 X_i) = Pr(D_i|\varepsilon_i, X_i),$$

and by exogeneity of the treatment indicator, we have

$$Pr(D_i|\varepsilon_i, X_i) = Pr(D_i|X_i)$$

and thus [conditional independence] holds."

OLS Assumptions

"However, the exogeneity assumption combines unconfoundedness with functional form and constant treatment effect assumptions that are quite strong, and arguably unnecessary." – Imbens and Rubin (2015)

Regression with correct functional form



Imbens and Rubin are saying regression can identify an aggregate causal parameter under unconfoundedness because it assumes constant treatment effects and the correct functional form which is like an army without a bridge shooting arrows and using catapults to the target. If the functional form is right, then they'll hit the target everytime.

Constant Treatment Effects and Linearity

Most commonly used method is OLS where the outcome is an additive model of the observed outcome, Y, on the treatment, D, and covariates, X like:

$$Y_i = \alpha + \delta D_i + \beta_1 X_i + \varepsilon_i$$

Take conditional expectations

$$\begin{aligned} E[Y_i|D_i = 1, X_i] &= \alpha + \delta E[D_i|D_i = 1, X_i] + \beta_1 E[X_i|D_i = 1, X_i] \\ E[Y_i|D_i = 0, X_i] &= \alpha + \delta E[D_i|D_i = 0, X_i] + \beta_1 E[X_i|D_i = 0, X_i] \end{aligned}$$

Constant Treatment Effects and Linearity

Replace realized variables with potential notation (both outcomes and covariates):

$$\begin{aligned} E[Y_i^1|D_i = 1, X_i] &= \alpha + \delta + \beta_{11} E[X_i^1|D_i = 1, X_i] \\ E[Y_i^0|D_i = 0, X_i] &= \alpha + \beta_{01} E[X_i^0|D_i = 0, X_i] \end{aligned}$$

X^1 is the effect of X on Y^1 when treated and X^0 is effect on Y^0 when not treated; note if treatment changes X then $X^1 \neq X^0$ and it will be a problem

Constant Treatment Effects and Linearity

With a binary treatment variable, the OLS estimator is equivalent to simple difference in conditional means:

$$
\begin{aligned}
\widehat{\delta} &= E[Y_i^1 | D_i = 1, X_i] - E[Y_i^0 | D_i = 0, X_i] \\
&= \left(\alpha + \delta E[D_i | D_i = 1, X_i] + \beta_{11} E[X_i^1 | D_i = 1, X_i] \right) \\
&\quad - \left(\alpha + \delta E[D_i | D_i = 0, X_i] + \beta_{01} E[X_i^0 | D_i = 0, X_i] \right) \\
&= \delta + \beta_{11} E[X_i^1 | D_i = 1, X_i] - \beta_{01} E[X_i^0 | D_i = 0, X_i]
\end{aligned}
$$

OLS model requires: (1) linearity, (2) treatment cannot change covariate values (e.g., bad controls), (3) $\beta_{11} = \beta_{01}$ (homogenous treatment effects with respect to X).

Simulation

- Simulation will have linear DGP, heterogeneity with respect to covariates and common support violation (1000 trials)
- Will show a variety of estimators and specifications so that we see how to recover causal parameters with regression and matching
- Focus is on estimated ATE and estimated ATT under a variety of specifications

Heterogenous Treatment Effects wrt X

```
* Simulation with heterogenous treatment effects, unconfoundedness and OLS estimation
clear all
program define het_te, rclass
version 14.2
syntax [, obs(integer 1) mu(real 0) sigma(real 1) ]

    clear
    drop _all
    set obs 5000
    gen     treat = 0
    replace treat = 1 in 2501/5000

    * Poor pre-treatment fit
    gen     age = rnormal(25,2.5)       if treat==1
    replace age = rnormal(30,3)         if treat==0
    gen     gpa = rnormal(2.3,0.75)     if treat==0
    replace gpa = rnormal(1.76,0.5)     if treat==1

    su age
    replace age = age - `r(mean)'

    su gpa
    replace gpa = gpa - `r(mean)'

    gen age_sq = age^2
    gen gpa_sq = gpa^2
    gen interaction=gpa*age

    gen y0 = 15000 + 10.25*age + -10.5*age_sq + 1000*gpa + -10.5*gpa_sq + 500*
interaction + rnormal(0,5)
    gen y1 = y0 + 2500 + 100 * age + 1000*gpa
    gen delta = y1 - y0

    su delta // ATE = 2500
    su delta if treat==1 // ATT = 1980
    local att = r(mean)
    scalar att = `att'
    gen att = `att'

    gen earnings = treat*y1 + (1-treat)*y0
```

Parameters

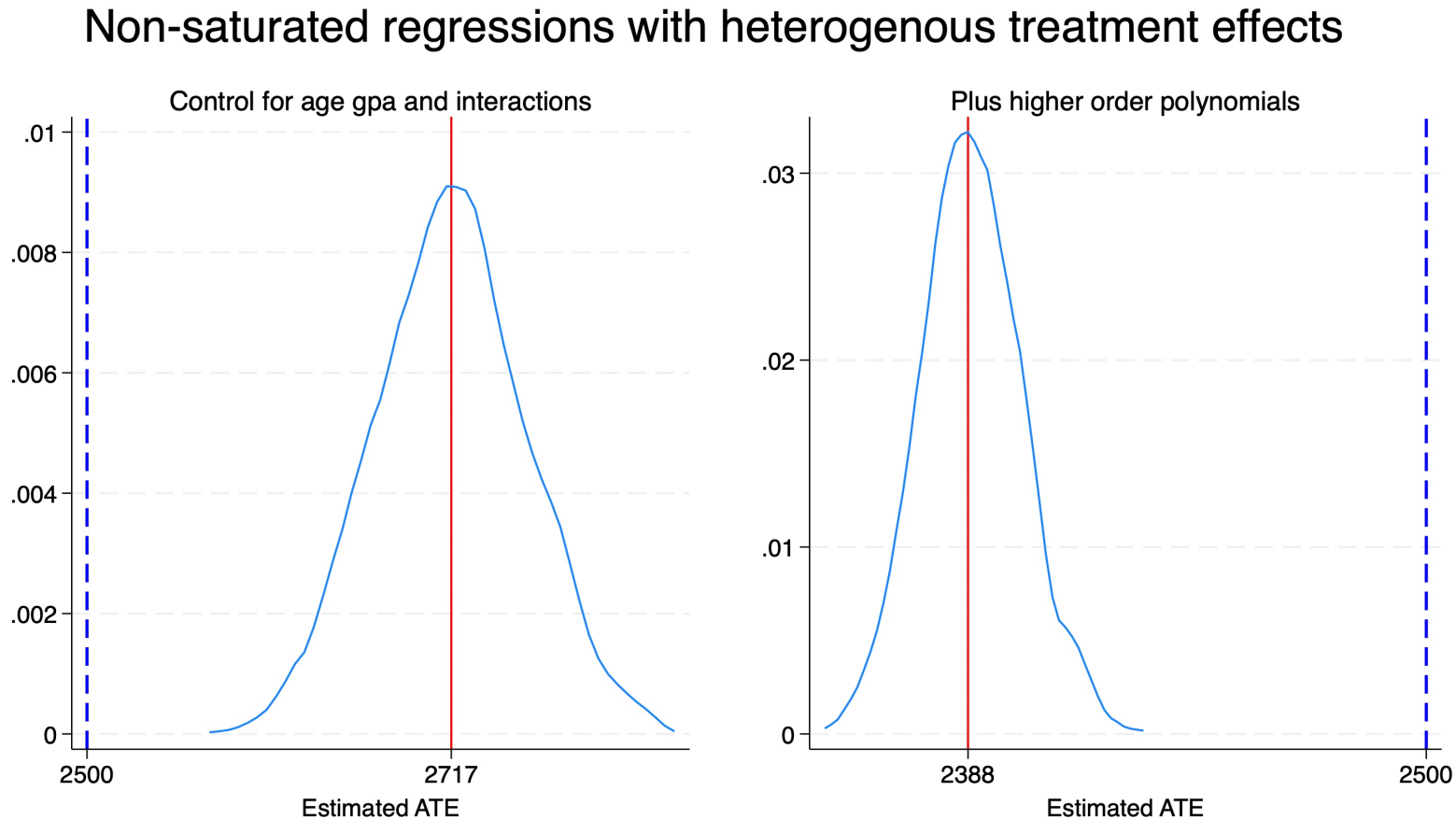
- We have two parameters: the ATE is $2500 but the ATT is $1980
- What is the specification for each of them?
- Let's look at what people usually do
- 1,000 simulations of DGP with regression estimates plotting coefficient on treatment dummy: first with just age and GPA, second with the precise model used for Y^0 (but not Y^1)

Constant Treatment Effects and Linearity

```
* Regression 1: constant treatment effects, no quadratics
reg earnings treat age gpa, robust
local treat1=_b[treat]
scalar treat1 = `treat1'
gen treat1=`treat1'

* Regression 2: constant treatment effects, quadratics and interaction
reg earnings treat age age_sq gpa gpa_sq c.gpa#c.age, robust
local treat2=_b[treat]
scalar treat2 = `treat2'
gen treat2=`treat2'
```

Coefficient on Treatment Dummy is Wrong



ATE is 2500 and ATT is 1980

Commentary

- Three ifs and a then:
  - → *If* unconfoundedness held, and
  - → *if* the potential outcome model was linear, and
  - → *if* the treatment effect had been homogenous with respect to age and GPA,
  - → *then* the coefficient on the treatment variable would have been the ATE
- But it wasn't because homogeneity with respect to X was not true (recall $Y(0)$ coefficients were not the same as $Y(1)$ coefficients)
- Why were these models biased? Didn't we have the correct specification in the second graph?

Heterogenous treatment effects

Write down a simplified version of the DGP from the code:

$$
\begin{aligned}
Y_i^0 &= \alpha + \beta_{01} X_i + \varepsilon_i \\
Y_i^1 &= \alpha + \beta_{01} X_i + \delta D_i + \beta_{11} X_i \times D_i + \varepsilon_i
\end{aligned}
$$

Notice that the setup before, X_i had a different effect on Y^0 than it did on Y_i^1 – that's because of heterogenous treatment effects with respect to conditioning set.

Heterogenous treatment effects

Take conditional expectations of the *potential* outcomes:

$$
\begin{aligned}
E[Y_i^0|D_i = 0, X_i] &= \alpha + \beta_{01} E[X_i|D_i = 0, X_i] \\
E[Y_i^1|D_i = 1, X_i] &= \alpha + \beta_{01} E[X_i|D_i = 0, X_i] \\
&\quad + \delta + \beta_{11} E[X_i|D_i = 1, X_i]
\end{aligned}
$$

Average treatment effect is: $E[Y_i^1|D_i, X_i^1] - E[Y_i^0|D_i, X_i]$

$$
\begin{aligned}
&= \left(\alpha + \beta_{01} E[X_i|D_i = 0, X_i] + \delta + \beta_{11} E[X_i \times D_i|D_i = 1, X_i]\right) \\
&\quad - \left(\alpha + \beta_{01} E[X_i|D_i = 0, X_i]\right) \\
&= \delta + \beta_{11} E[X_i \times D_i|D_i = 1]
\end{aligned}
$$

Regression adjustment

This implies an interacted OLS model or what Wooldridge (2010) calls regression adjustment:

$$Y_i = \alpha + \delta D_i + \beta_{01} X_i + \beta_{11} D_i \times X_i + \varepsilon_i$$

$\widehat{\delta}$ is the ATE but the ATT is equal to $\widehat{\delta} + \widehat{\beta_{11}} E[X_i | D_i = 1]$ where $E[X_i | D_i = 1]$ is the sample average of X_i for the treatment group

We will estimate two models: (1) once with simplified but incorrectly specified saturated and (2) another with the correctly specified saturated model – warning, it's a huge pain and you can easily mess it up even with just a few variables

Misspecified Saturated OLS Regression

```
* Regression 3: Heterogenous treatment effects, partial saturation
regress earnings i.treat##c.age##c.gpa, robust
local ate1=_b[1.treat]
scalar ate1 = `ate1'
gen ate1=`ate1'

* Obtain the coefficients
local treat_coef = _b[1.treat]
local age_treat_coef = _b[1.treat#c.age]
local gpa_treat_coef = _b[1.treat#c.gpa]
local age_gpa_treat_coef = _b[1.treat#c.age#c.gpa]

* Save the coefficients as scalars and generate variables
scalar treat_coef = `treat_coef'
gen treat_coef_var = `treat_coef'

scalar age_treat_coef = `age_treat_coef'
gen age_treat_coef_var = `age_treat_coef'

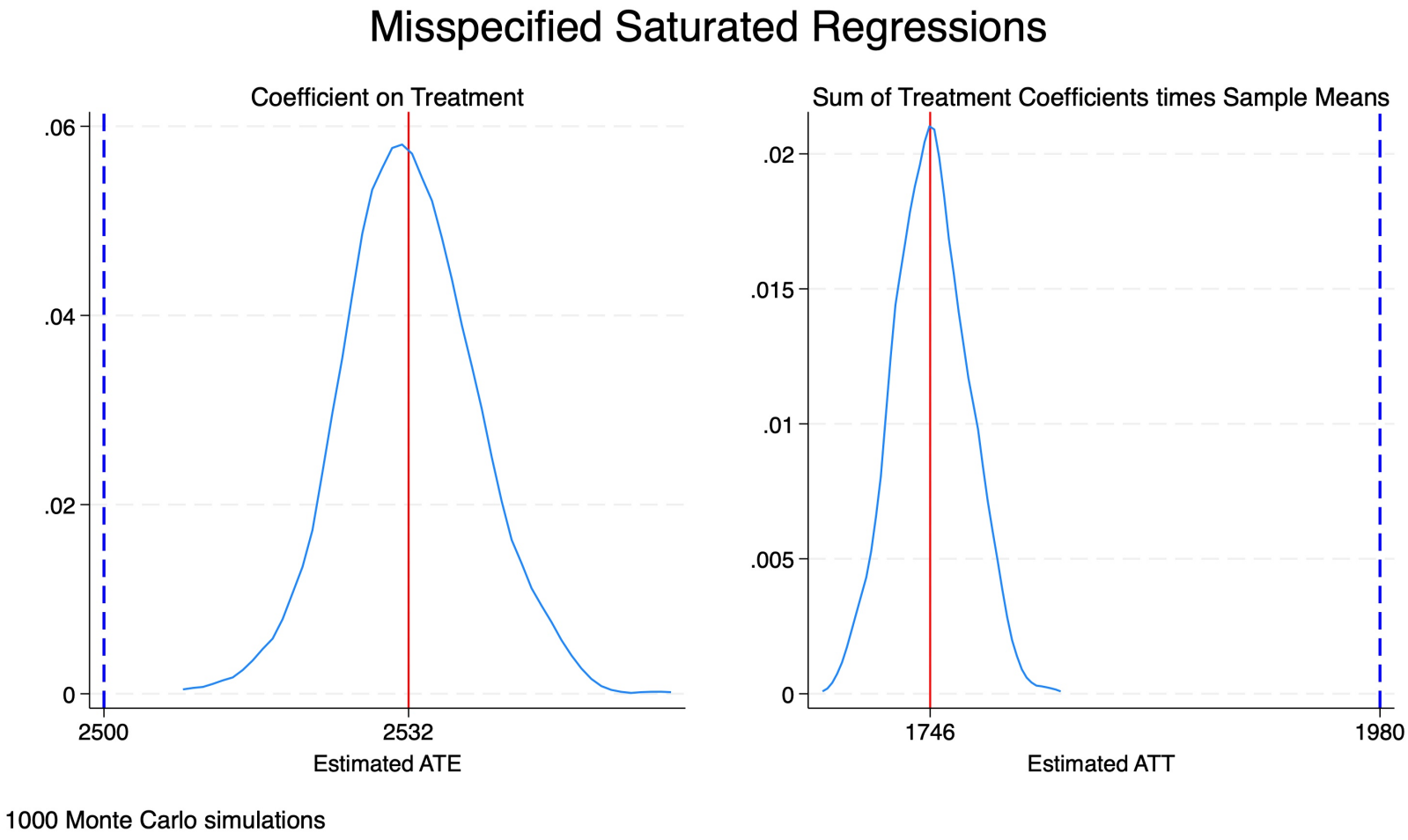
scalar gpa_treat_coef = `gpa_treat_coef'
gen gpa_treat_coef_var = `gpa_treat_coef'

scalar age_gpa_treat_coef = `age_gpa_treat_coef'
gen age_gpa_treat_coef_var = `age_gpa_treat_coef'

* Calculate the mean of the covariates
egen mean_age = mean(age), by(treat)
egen mean_gpa = mean(gpa), by(treat)

* Calculate the ATT
gen treat3 = treat_coef_var + ///
            age_treat_coef_var * mean_age + ///
            gpa_treat_coef_var * mean_gpa + ///
            age_gpa_treat_coef_var * mean_age * mean_gpa if treat == 1
```

Misspecified Saturated OLS Regression



Comically Long Saturated OLS Regression

```
    * Regression 4: Fully saturated regression model
#delimit ;
    regress earnings    i.treat##c.age
                        i.treat##c.age_sq
                        i.treat##c.gpa
                        i.treat##c.gpa_sq
                        i.treat##c.age##c.gpa;
#delimit cr

    local ate2=_b[1.treat]
    scalar ate2 = `ate2'
    gen ate2=`ate2'

    * Obtain the coefficients
    local treat_coef        = _b[1.treat] // 0
    local age_treat_coef    = _b[1.treat#c.age] // 1
    local agesq_treat_coef  = _b[1.treat#c.age_sq] // 2
    local gpa_treat_coef    = _b[1.treat#c.gpa] // 3
    local gpasq_treat_coef  = _b[1.treat#c.gpa_sq] // 4
    local age_gpa_coef      = _b[1.treat#c.age#c.gpa] // 5


    * Save the coefficients as scalars and generate variables
    scalar  treat_coef = `treat_coef'
    gen     treat_coef_var = `treat_coef' // 0

    scalar  age_treat_coef = `age_treat_coef'
    gen     age_treat_coef_var = `age_treat_coef' // 1

    scalar  agesq_treat_coef = `agesq_treat_coef'
    gen     agesq_treat_coeff_var = `agesq_treat_coef' // 2

    scalar  gpa_treat_coef = `gpa_treat_coef'
    gen     gpa_treat_coef_var = `gpa_treat_coef' // 3

    scalar  gpasq_treat_coef = `gpasq_treat_coef'
    gen     gpasq_treat_coef_var = `gpasq_treat_coef' // 4

    scalar  age_gpa_coef = `age_gpa_coef'
    gen     age_gpa_coef_var = `age_gpa_coef' // 5


    * Calculate the mean of the covariates
    su  age if treat==1
    local mean_age = `r(mean)'
    gen mean_age = `mean_age'

    su age_sq if treat==1
    local mean_agesq = `r(mean)'
    gen mean_agesq = `mean_agesq'

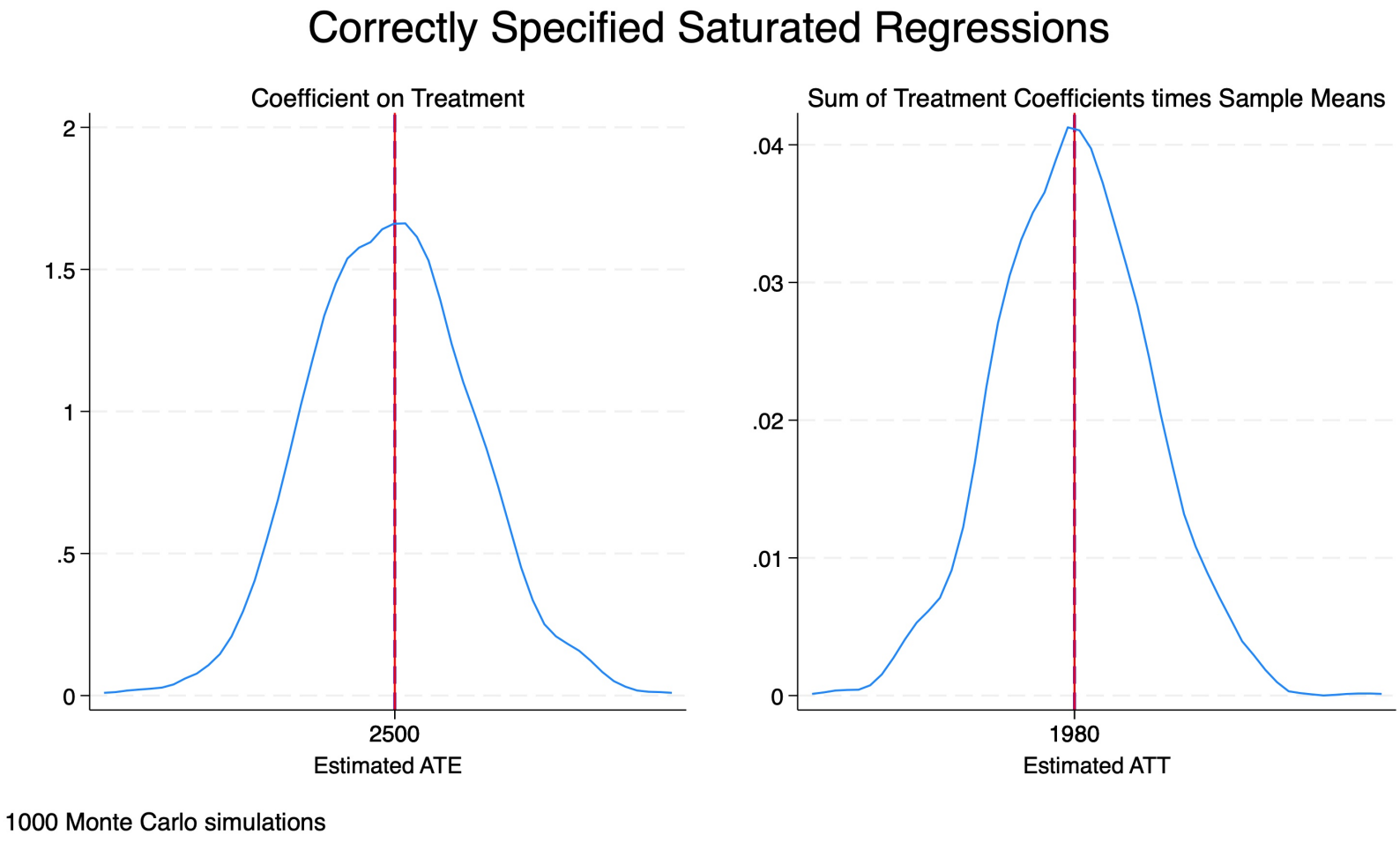
    su gpa if treat==1
    local mean_gpa = `r(mean)'
    gen mean_gpa = `mean_gpa'

    su gpa_sq if treat==1
    local mean_gpasq = `r(mean)'
    gen mean_gpasq = `mean_gpasq'

    su agegpa if treat==1
    local mean_agegpa = `r(mean)'
    gen mean_agegpa = `mean_agegpa'


* Calculate the ATT
gen treat4 =    treat_coef_var + /// 0
                age_treat_coef_var * mean_age + /// 1
                agesq_treat_coeff_var * mean_agesq + /// 2
                gpa_treat_coef_var * mean_gpa + /// 3
                gpasq_treat_coef_var * mean_gpasq + /// 4
                age_gpa_coef_var * mean_agegpa
```

Correctly Saturated OLS Regression



Regression adjustment

- Notice how the same regression (fully interacted) led to *both* the ATE and the ATT?
- That means if you don't state ahead of time which parameter you want, how are you going to know how to set it up, and how will you know how to recover it?
- Thankfully software exists that does this for you called "regression adjustment" by Wooldridge (2010) or Oaxaca-Blinder (Kline 2011; Graham and Pinto 2022)
- In Stata teffects, you can just the `ra` to get it – see `comparison.do`

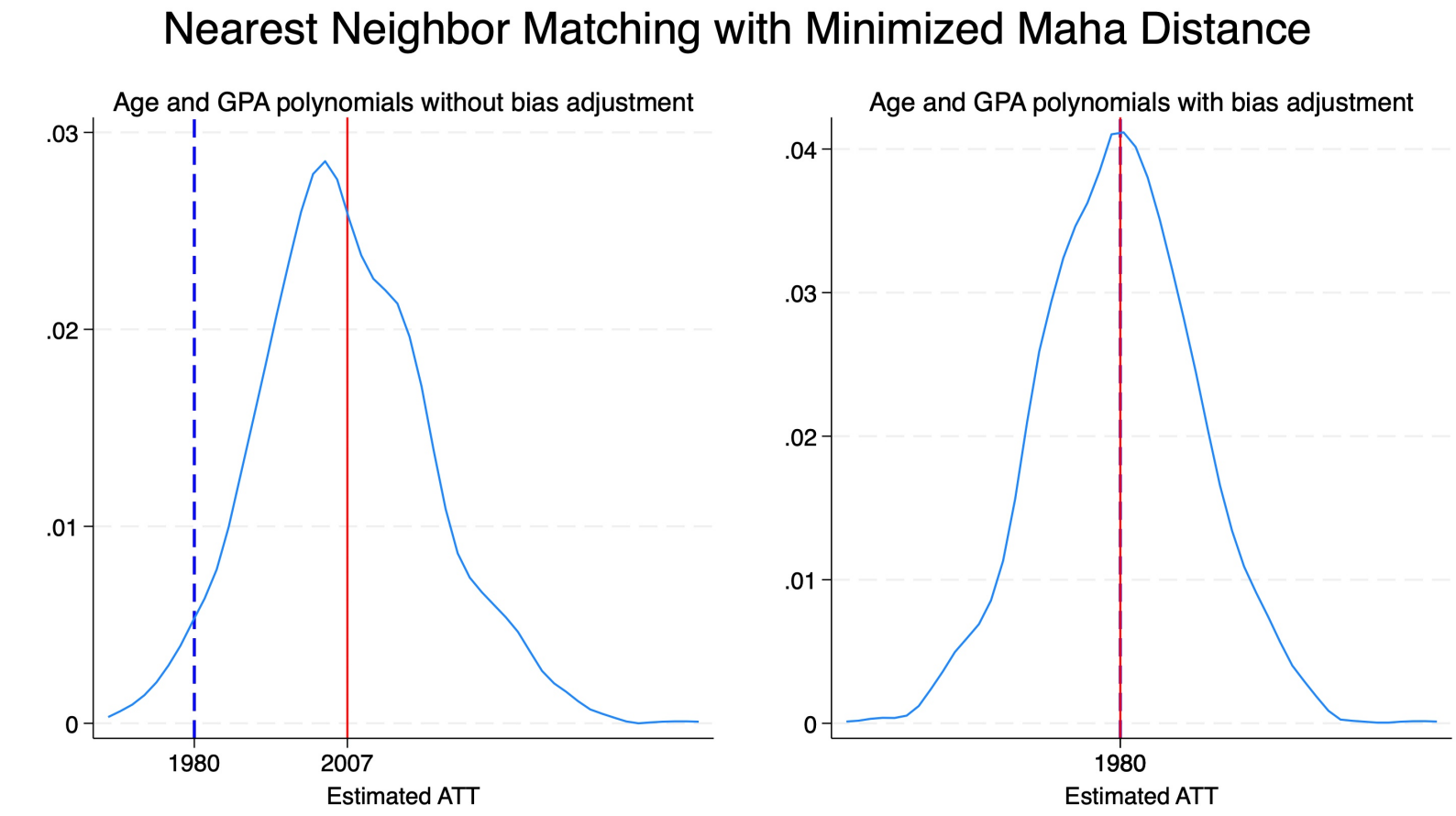
Great quote

"OLS is . . . the linear projection of Y on D and X [and] provides the best linear predictor of Y given D and X (Angrist and Pishcke 2009). However, if our goal is to conduct causal inference, then this is not, in fact, a good reason to use this method. OLS is 'best' in predicting actual outcomes, but causal inference is about predicting [fictional potential] outcomes. . . . In other words, OLS . . . is optimal for predicting 'what is'. Instead we are interested in predicting 'what would be' if treatment were assigned differently." - Tymon Sloczynski, Restat 2022

Matching

- Now let's estimate the ATT ($1980) using nearest neighbor matching by minimizing Mahalanobis distance on age, GPA, polynomials and interaction
- One line in Stata using `teffects` and only 1 match (variance is simple to estimate until we use matches multiple times, then the variance grows)
- In R, the package is `Matching`, not sure in python

Matching Estimation



Estimated ATT from 1000 simulations using nearest neighbor matching

Commentary

- Technically I was only “fully interacting” – full saturation would be to interact the treatment dummy with every value of the covariates yielding a huge number of parameters likely that cannot be estimated
- Regression adjustment within -teffects- will do this for you
- But note, with heterogeneity you have to use the fully saturated model (I just hadn’t dummied every value of the covariates) to get both the ATE and the ATT
- Otherwise you are imposing strong and unnecessary assumptions on the data that the treatment effects are the same for all values of X and constant treatment effects so that the single coefficient is the ATE and the ATT

Misspecified functional form regression visual



Using our medieval castle example, if you have the wrong functional form, then regression won’t hit.

Nonlinear DGP

- We've seen what happens if there's heterogenous treatment effects with respect to covariates
- But this was resolvable with "full interaction" or "regression adjustment"
- But Imbens and Rubin (2015) noted that exogeneity also implied functional forms, namely linearity
- Let's examine this now using a very unusual DGP using `nonlinear_matching.do`

Nonlinear DGP

```
clear
drop _all
set obs 5000
gen     treat = 0
replace treat = 1 in 2501/5000
gen     age = rnormal(35,2.5)       if treat==0
replace age = rnormal(30,2)         if treat==1
gen     gpa = rnormal(2.3,0.75)     if treat==0
replace gpa = rnormal(1.76,0.25)    if treat==1

su age
replace age = age - `r(mean)'

su gpa
replace gpa = gpa - `r(mean)'

su age, detail
replace age = age - `r(mean)'

gen gpa_age = age*gpa

su age, detail

* Admittedly weird nonlinear data generating process: Y0
gen     y0 = rnormal()
replace y0 = 200 + rnormal() if age < `r(p25)'
replace y0 = 0 +   runiform() if age>= `r(p25)' & age<=`r(p75)'
replace y0 = 150 + rnormal() if age>`r(p75)'

su gpa_age, detail

* Admittedly weird nonlinear data generating process: Y1
gen     y1 = 0
replace y1 = y0 + 2000 * (0.25)*gpa_age + rnormal(1,25) if gpa_age >= `r(p5)'  & gpa_age < `r(p25)'
replace y1 = -10*y0 + 25*age + (0.1)*gpa if age >= `r(p25)' & gpa_age < `r(p50)'
replace y1 = 10*y0 + (5)*gpa_age + gpa + rnormal(5,5) if gpa_age >= `r(p50)' & gpa_age<`r(p75)'
replace y1 = y0 + (0.05) * gpa_age + 2*age if gpa_age>=`r(p75)'

gen delta = y1-y0

su delta // ATE = approximately 15
local ate = r(mean)
scalar ate = `ate'
gen ate = `ate'

su delta if treat==1 // ATT = approximately 272
local att = r(mean)
scalar att = `att'
```

Nonlinear DGP, OLS with RA

```
    * Regression: Heterogenous treatment effects with age
    regress earnings i.treat##c.age i.treat##c.gpa i.treat##c.gpa_age, robust

    ** ATE
    local reg_ate2=_b[1.treat]
    scalar reg_ate2 = `reg_ate2'
    gen reg_ate2=`reg_ate2'

    ** ATT
    * Obtain the coefficients
    local treat_coef = _b[1.treat]
    local age_treat_coef = _b[1.treat#c.age]
    local gpa_treat_coef = _b[1.treat#c.gpa]
    local gpaage_treat_coef = _b[1.treat#c.gpa_age]


    * Save the coefficients as scalars and generate variables
    scalar treat_coef = `treat_coef'
    gen treat_coef_var = `treat_coef'

    scalar age_treat_coef = `age_treat_coef'
    gen age_treat_coef_var = `age_treat_coef'

    scalar gpa_treat_coef = `gpa_treat_coef'
    gen gpa_treat_coef_var = `gpa_treat_coef'

    scalar gpaage_treat_coef = `gpaage_treat_coef'
    gen gpaage_treat_coef_var = `gpaage_treat_coef'

    * Calculate the mean of the age covariate for treatment group only
    egen mean_age = mean(age) if treat==1
    egen max_age = max(mean_age)
    replace mean_age = max_age if treat==0

    egen mean_gpa = mean(gpa) if treat==1
    egen max_gpa = max(mean_gpa)
    replace mean_gpa = max_gpa if treat==0

    egen mean_gpaage = mean(gpaage) if treat==1
    egen max_gpaage = max(mean_gpaage)
    replace mean_gpaage = max_gpaage if treat==0

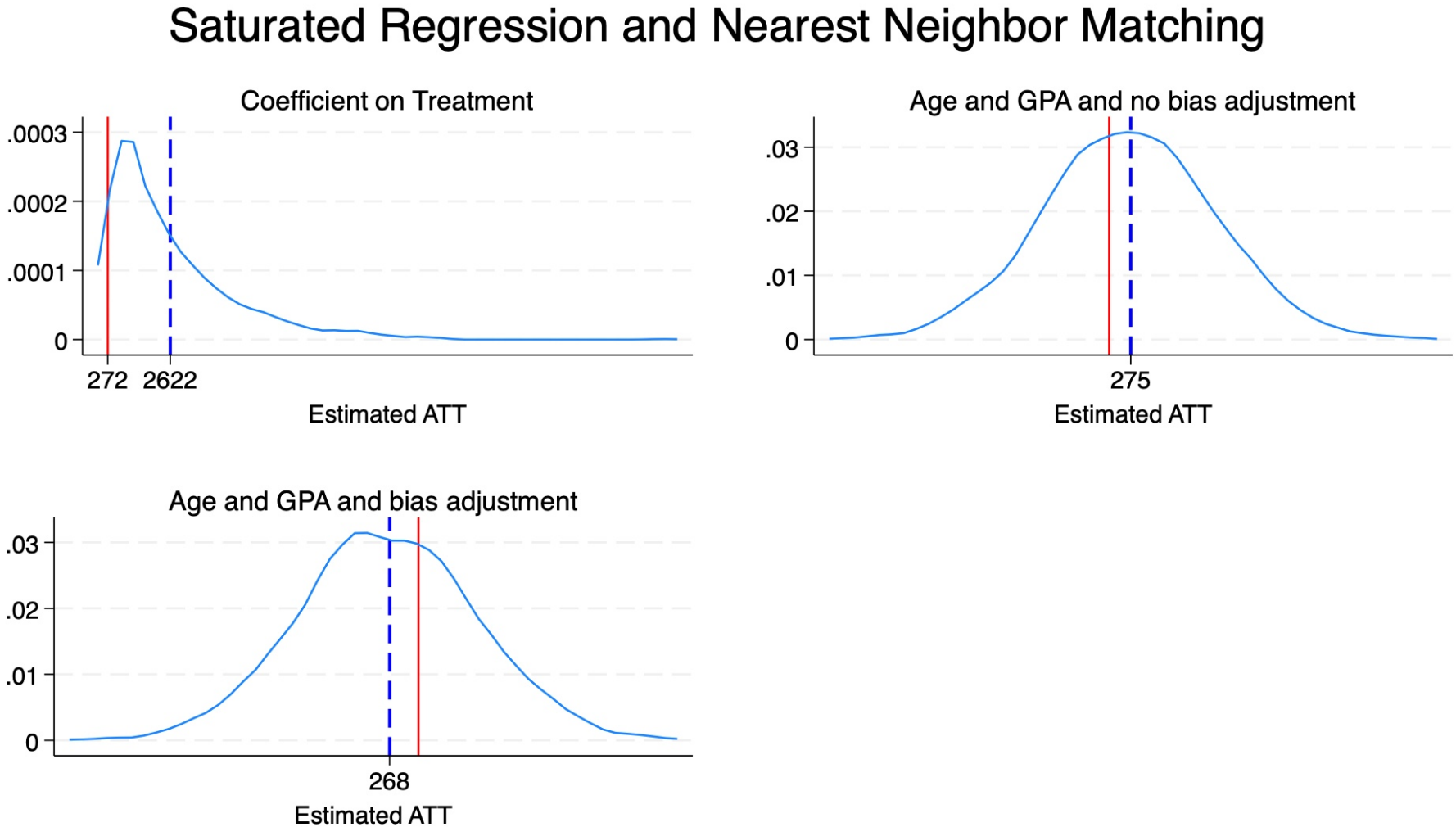
    * Calculate the ATT
gen reg_att =   treat_coef_var + /// 0
                age_treat_coef_var * mean_age + /// 1
                gpa_treat_coef_var * mean_gpa + /// 2
                gpaage_treat_coef_var * mean_gpaage
```

Nonlinear DGP and Matching

```
** Distance minimization matching method
teffects nnmatch (earnings age gpa) (treat), atet nn(1) metric(maha)
mat b=e(b)
local nn_att1 = b[1,1]
scalar nn_att1=`nn_att1'
gen nn_att1=`nn_att1'

** Distance minimization matching method with bias adjustment
teffects nnmatch (earnings age gpa) (treat), atet nn(1) metric(maha) biasadj(age gpa)
mat b=e(b)
local nn_att2 = b[1,1]
scalar nn_att2=`nn_att2'
gen nn_att2=`nn_att2'
end
```

Nonlinear DGP, OLS and Matching



The dashed blue line is estimated ATT from 1000 simulations using saturated OLS and nearest neighbor matching.

Back up. What *is* that coefficient?

You ran this and read $\widehat{\delta}$ off the output:

$$Y_i = \alpha + \delta D_i + \beta X_i + \varepsilon_i$$

- With heterogeneous effects, $\widehat{\delta}$ is an average of *something*
- The question is what, and with what weights
- Two papers answer it. Twenty-four years apart

Angrist (1998): the variance weights

Binary D, discrete controls G. Apply FWL and the algebra falls out:

$$\beta_1 = \sum_g w_g \, \tau_g, \qquad w_g = \frac{\Pr(G = g)\, p_g(1 - p_g)}{\sum_{g'} \Pr(G = g')\, p_{g'}(1 - p_{g'})}$$

- τ_g is the within-stratum effect — each one correctly estimated
- But the weights are *not* the population shares $\Pr(G = g)$
- They're population share $\times$ **treatment variance** $p_g(1 - p_g)$

Why $p_g(1-p_g)$? Because OLS likes a coin flip

Stratum size $\Pr(G=g)$ $\times$ **Treatment variance** $p_g(1-p_g)$ $\propto$ **Weight** w_g

- $p_g(1-p_g)$ peaks at $p_g = 0.5$ and vanishes at 0 or 1
- Strata where treatment looks like a coin flip get *more* weight
- Strata where almost everyone is treated (or nobody) get almost none
- Makes sense as *precision*. Makes no sense as a *population*

A job training program run at two sites

- **Site A** had plenty of slots. They randomized with a *coin flip*
- **Site B** was budget-rationed. You got a slot only if you *rolled a six*

Site	N	p_g	Treated / Control	τ_g
A	300	$1/2$	150 / 150	$-\$9{,}000$
B	600	$1/6$	100 / 500	$+\$6{,}000$

- Site A pulled people out of jobs they already had — lock-in, and it cost them
- Site B genuinely helped
- $ATE = \frac{1}{3}(-9{,}000) + \frac{2}{3}(6{,}000) =$ **+$1,000**

Now run the regression

Site	$\Pr(G)$	$p_g(1-p_g)$	w_g	τ_g
A	$1/3$	0.250	**0.474**	$-\$9{,}000$
B	$2/3$	0.139	**0.526**	$+\$6{,}000$

$$\beta_{OLS} = 0.474(-9{,}000) + 0.526(6{,}000) = \textbf{-\$1,105}$$

- Site A is a third of the population and gets *half* the weight
- Because a coin flip has more treatment variance than a die
- The program raises earnings by \$1,000. OLS says it lowers them by \$1,100

The gap needs *both* kinds of variation

- τ_g constant $\Rightarrow \beta = \sum_g w_g \tau = \tau = ATE$. No gap
- p_g constant $\Rightarrow w_g = \Pr(G = g)$. No gap

- You need heterogeneous effects *and* uneven take-up
- An RCT with $p_g = 0.5$ everywhere: no gap
- Observational data with selection: gap essentially always
- Same structure as our balance slide — imbalance has to interact with something

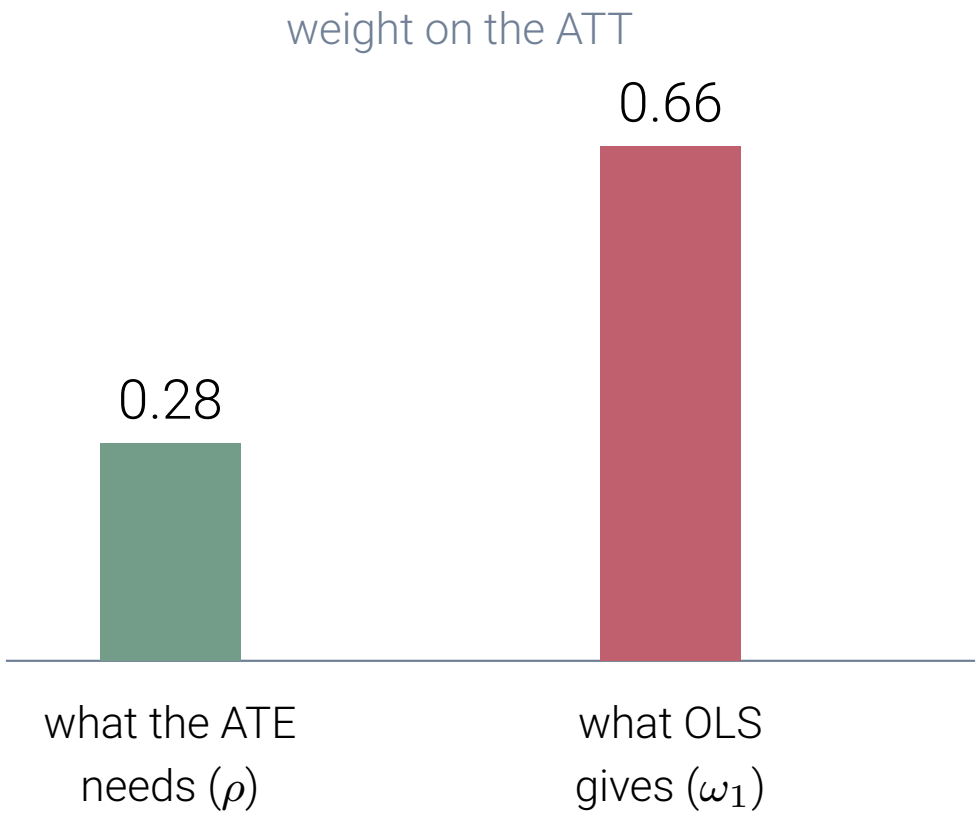
Sloczynski (2022): say it in ATT and ATU

Angrist decomposes into *strata*. Tymon decomposes into the two parameters we name:

$$\tau = \omega_1 \cdot ATT + \omega_0 \cdot ATU$$

- Same coefficient. Different bookkeeping
- Our two sites: $ATT = -\$3{,}000$, $ATU = +\$2{,}538$, $\omega_1 = 0.658$
- $0.658(-3{,}000) + 0.342(2{,}538) =$ **-$1,105** — the same number as Angrist

And now look at the weights



- $ATE = \rho \cdot ATT + (1 - \rho) \cdot ATU$ with $\rho = 0.28$
- OLS uses $\omega_1 = 0.66 \approx 1 - \rho$
- **The weights are flipped** — and here that flips the sign too

So why did we need Tymon if we had Angrist?

	Angrist (1998)	**Sloczynski (2022)**
Controls	discrete, saturated	any X
Decomposes into	stratum effects τ_g	ATT and ATU
Weights	$\Pr(g)p_g(1-p_g)$	$\omega_1 \approx 1-\rho$
Need to know	the whole joint dist.	the treated share
Misspecified?	no — it's saturated	yes, generally

- They agreed on our example *because* two sites *is* a saturated model
- That's the special case where the two coincide
- **Angrist says it's a weighted average. Tymon says the weights are backwards**

You'll see this exact move again next week

- Goodman-Bacon (2021): twoway fixed effects with staggered adoption
- Same result, different setting — the estimate is a weighted average of every possible 2×2 DiD
- Weights again $\propto$ group size $\times$ treatment variance
- And again the pathology is the weights, not the comparisons

Once you've seen it here, Bacon's decomposition is the same story wearing a panel-data coat.

Two diagnostics you can run before lunch

Both need only ρ, the treated share:

- Going for the **ATT**? The damage is ρ
- Going for the **ATE**? The damage is $|2\rho - 1|$ — twice your distance from 50/50

- Each is the fraction of the $ATU - ATT$ gap that leaks into your estimate
- So OLS is fine for the ATE when the arms are similar in size
- And fine for the ATT only when very few are treated
- `hettreatreg` in R and Stata reports these for you

Where the gap comes from

Sloczynski splits the distance between your coefficient and your target in two:

$$\beta_{OLS} - ATE = \underbrace{\text{bias from nonlinearity}}_{\text{wrong functional form}} + \underbrace{\text{bias from heterogeneity}}_{\text{wrong weights}}$$

- Saturate the model and the first term goes away
- The second one does *not* — that's the reversal, and no amount of flexibility fixes it
- You fix it by changing the *estimator*, not the specification

Stata exercises

We will now look at two simulations to help you understand both of these

Visualizing the problem one way

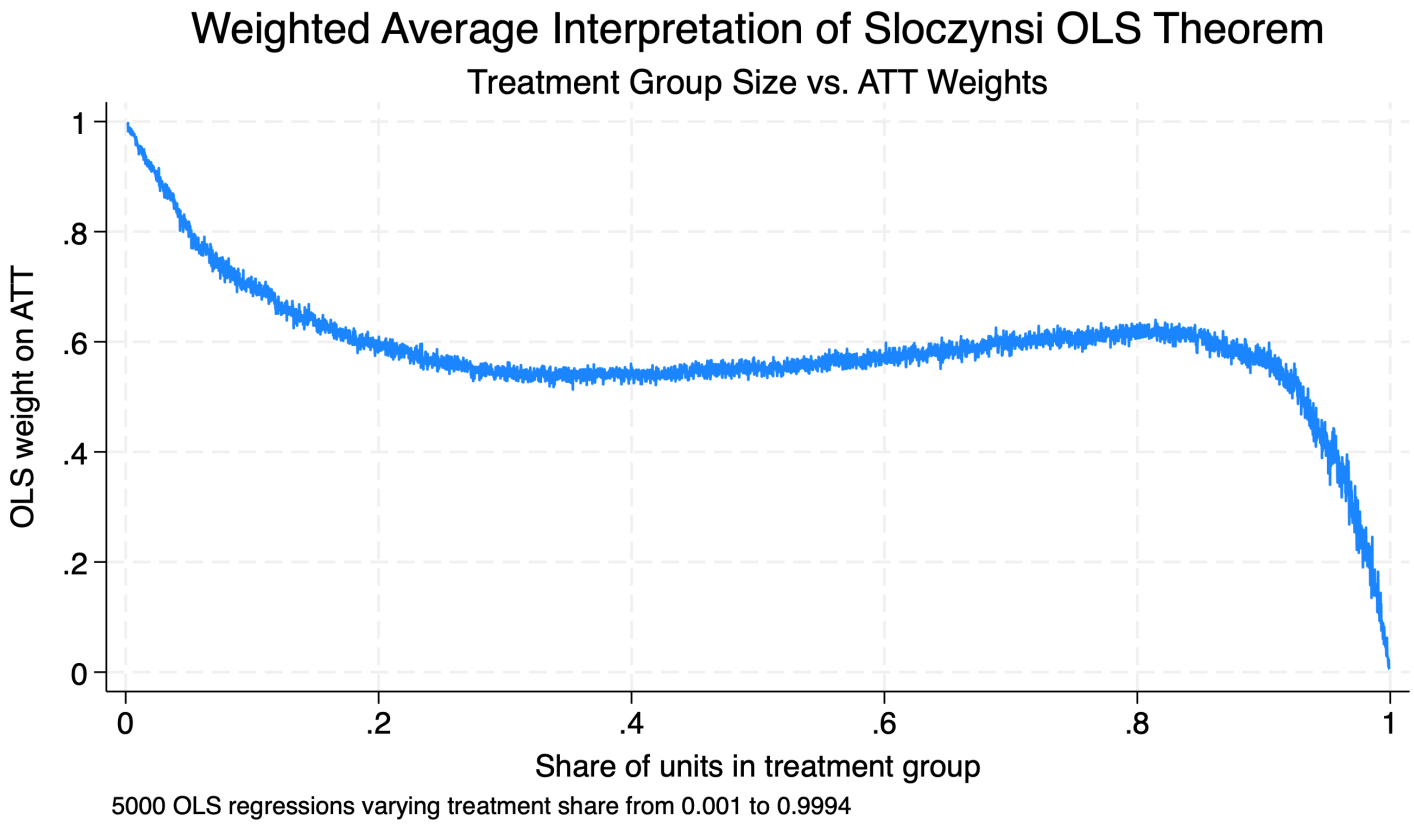


Figure: As more units are placed in treatment, the weight OLS puts on the ATT *falls*.

The coefficient sweeps with the treated share

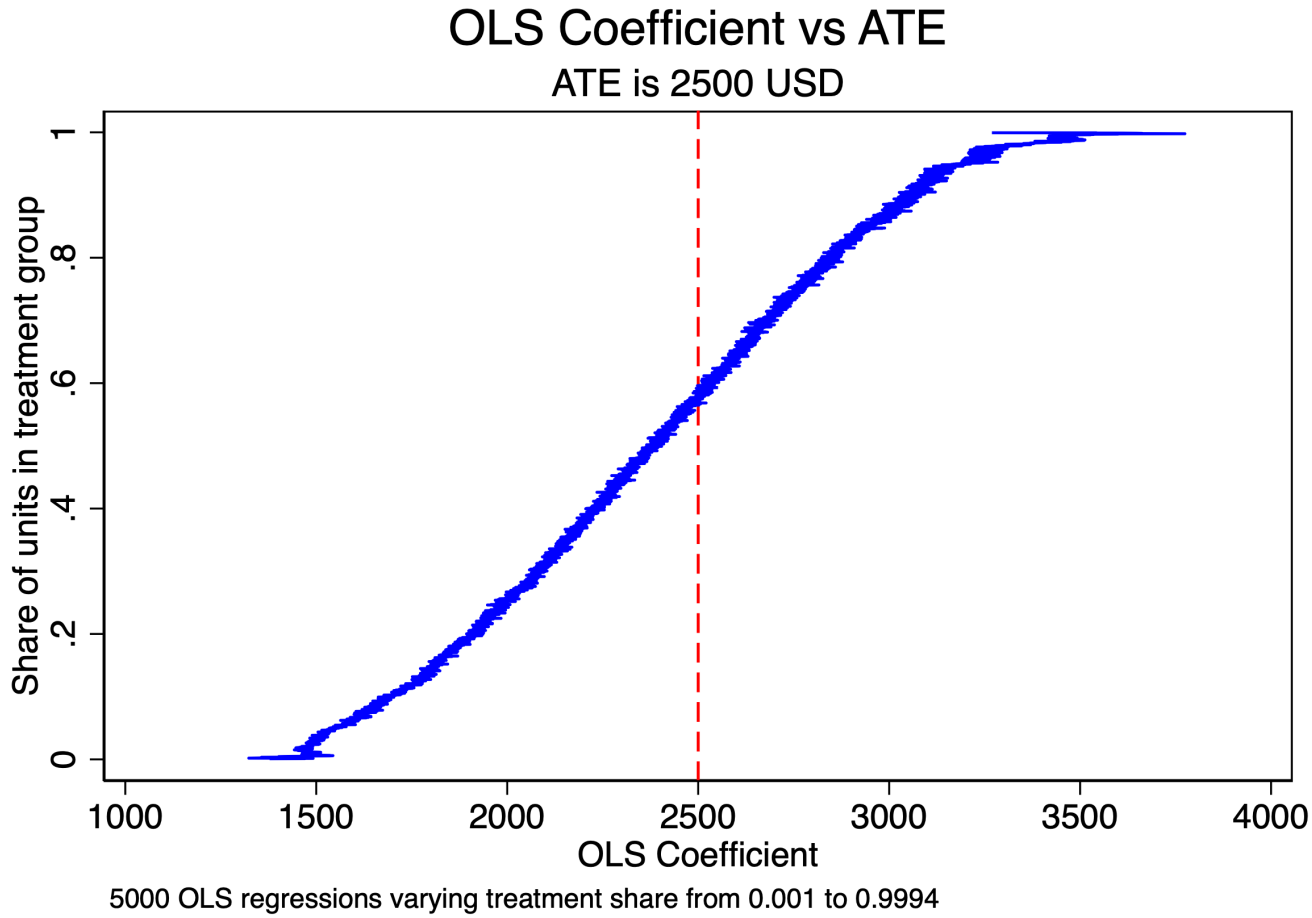


Figure: Few treated (bottom left): OLS leans toward the ATT ($1,962). Many treated (top right): toward the ATU ($3,037). It hits the true ATE of $2,500 only in between.

Prevalence of OLS Bias in Evaluating Labor Market Programs

- OLS may frequently show substantial bias for ATE and ATT in published studies
- In studies like Card, Kluve, and Weber (2018), with a mean treatment rate of 17.7%, biases are common:
  - → The expected bias in OLS for ATE is 64.6% of the ATU-ATT difference.
  - → For ATT, the expected OLS bias is 17.7% of the ATU-ATT difference.
- These findings indicate that biases in OLS estimates could often be significant.
  - → LaLonde's (1986) training program effects aligning with ATT.
  - → Aizer et al.'s (2016) cash transfer effects resembling ATU.
- Implementation of results is available in R and Stata through the `hettreatreg` packages.

Tymon's replication of Angrist and Pischke

TABLE 1.—THE EFFECTS OF A TRAINING PROGRAM ON EARNINGS

	(1)	(2)	(3)	(4)
	Original estimates			
OLS	−3,437***	−78	623	794
	(612)	(596)	(610)	(619)
	Diagnostics			
$\hat{w}_0$	0.019	0.001	0.017	0.017
$\hat{w}_0^* = \hat{\rho}$	0.011	0.011	0.011	0.011
$\hat{\delta}$	−0.970	−0.987	−0.971	−0.971
$\hat{\delta}^* = 2\hat{\rho} - 1$	−0.977	−0.977	−0.977	−0.977
	Decomposition			
$\widehat{\text{ATT}}$	−3,373***	−69	754	928
	(620)	(595)	(619)	(630)
$\hat{w}_1$	0.981	0.999	0.983	0.983
$\widehat{\text{ATU}}$	−6,753***	−6,289**	−6,841***	−6,840***
	(1,219)	(2,807)	(1,294)	(1,319)
$\hat{w}_0$	0.019	0.001	0.017	0.017
$\widehat{\text{ATE}}$	−6,714***	−6,218**	−6,754***	−6,751***
	(1,206)	(2,777)	(1,281)	(1,305)
Demographic controls	√		√	√
Earnings in 1974				√
Earnings in 1975		√	√	√
$\hat{\rho} = \hat{\text{P}}(d = 1)$	0.011	0.011	0.011	0.011
Observations	16,177	16,177	16,177	16,177

The estimates in the top panel correspond to column 2 in table 3.3.3 in Angrist and Pischke (2009, p. 89). The dependent variable is earnings in 1978. Demographic controls include age, age squared, years of schooling, and indicators for married, high school dropout, Black, and Hispanic. For treated individuals, earnings in 1974 correspond to real earnings in months 13 to 24 prior to randomization, which overlaps with calendar year 1974 for a number of individuals. Formulas for w_0, w_1, and δ are given in theorem 1 and corollary 2. Following these results, OLS = $\hat{w}_1 \times \widehat{\text{ATT}} + \hat{w}_0 \times \widehat{\text{ATU}}$. Estimates of ATE, ATT, and ATU are sample analogs of τ_{APLE}, $\tau_{APLE,1}$, and $\tau_{APLE,0}$, respectively. Also, $\widehat{\text{ATE}} = \hat{\rho} \times \widehat{\text{ATT}} + (1 - \hat{\rho}) \times \widehat{\text{ATU}}$. Huber–White standard errors (OLS) and bootstrap standard errors ($\widehat{\text{ATE}}$, $\widehat{\text{ATT}}$, and $\widehat{\text{ATU}}$) are in parentheses. Statistically significant at *10%, **5%, and ***1%.

Stata simulation

Let’s do this ourselves using lalonde.do in the labs/matching on github repo

Replications of Lalonde data using Dehejia and Wahba sample with CPS control

	Dependent variable: Real earnings in 1978				
	(1)	(2)	(3)	(4)	(5)
Controls:	No controls	Demographics	RE75 Only	Demo and RE75	Demo, RE74 and RE75
Treatment	-8497.516***	-3436.795***	-77.705	622.547	**793.587**
	(581.916)	(612.056)	(596.215)	(609.582)	(618.609)

Each column corresponds to a set of observable controls below:

- Demographics: Age, Age-squared, Years of schooling, Marriage dummy, High school dropout dummy, Black and Hispanic dummies
- Real earnings in 1975 (RE75)
- Real earnings in 1974 (RE74)

Our focus will be on the last estimate. Question is what is this?

What that $794 actually was

Interact treatment with every covariate and all three parameters fall out:

$\widehat{ATE}$	$-\$4{,}930$
$\widehat{ATT}$	$+\$796$
$\widehat{ATU}$	$-\$4{,}996$
OLS coefficient	$+\$794$

- Only 1.1% of the sample is treated — so OLS put **99.96%** of its weight on the ATT
- Which is why it looked "close to the experimental benchmark"
- The OLS estimate and the ATE have opposite signs

Diagnosis is not a method

- Everything so far tells you what goes wrong. None of it tells you what to run
- But the fix has been sitting in that last table the whole time
- We got the ATE, the ATT *and* the ATU out of one regression
- By interacting the treatment with every covariate

That estimator has a name, and it is older than the problem it solves.

Regression adjustment

Fit the two potential-outcome surfaces *separately*:

$$\widehat{\mu}^1(X) \text{ on the treated,} \qquad \widehat{\mu}^0(X) \text{ on the controls}$$

Then impute everyone's missing side and average over whichever population your estimand names:

- Average over *everyone* $\rightarrow$ ATE
- Average over the *treated* $\rightarrow$ ATT
- Average over the *untreated* $\rightarrow$ ATU

You choose the weights. OLS chose them for you.

This is Oaxaca-Blinder

- Oaxaca (1973) and Blinder (1973) built it to decompose the black-white and male-female wage gap
- Separate wage equations by group, then ask how much of the gap is composition and how much is returns
- Structurally identical to what we just did
- Kline (2011) showed the "unexplained component" *is* a treatment effect estimator — and a doubly robust one
- Also called regression adjustment (Wooldridge 2010; Graham and Pinto 2022)

A footnote you'll enjoy

- Ron Oaxaca was Orley Ashenfelter's student in the Princeton Industrial Relations Section
- The same room we started the day in
- The tool built there in 1973 to measure discrimination turns out to be the repair for a problem economists wouldn't name until 2022

Two rivers, one estimator.

In practice

- Stata: `teffects ra (y X) (d)` — and `atet` if you want the ATT
- R: `estimatr::lm_lin`, or the `hettreatreg` package for the diagnostics
- Do it by hand once so you believe it, then let the software do it
- Full interaction is easy to botch — we saw the "comically long" version earlier

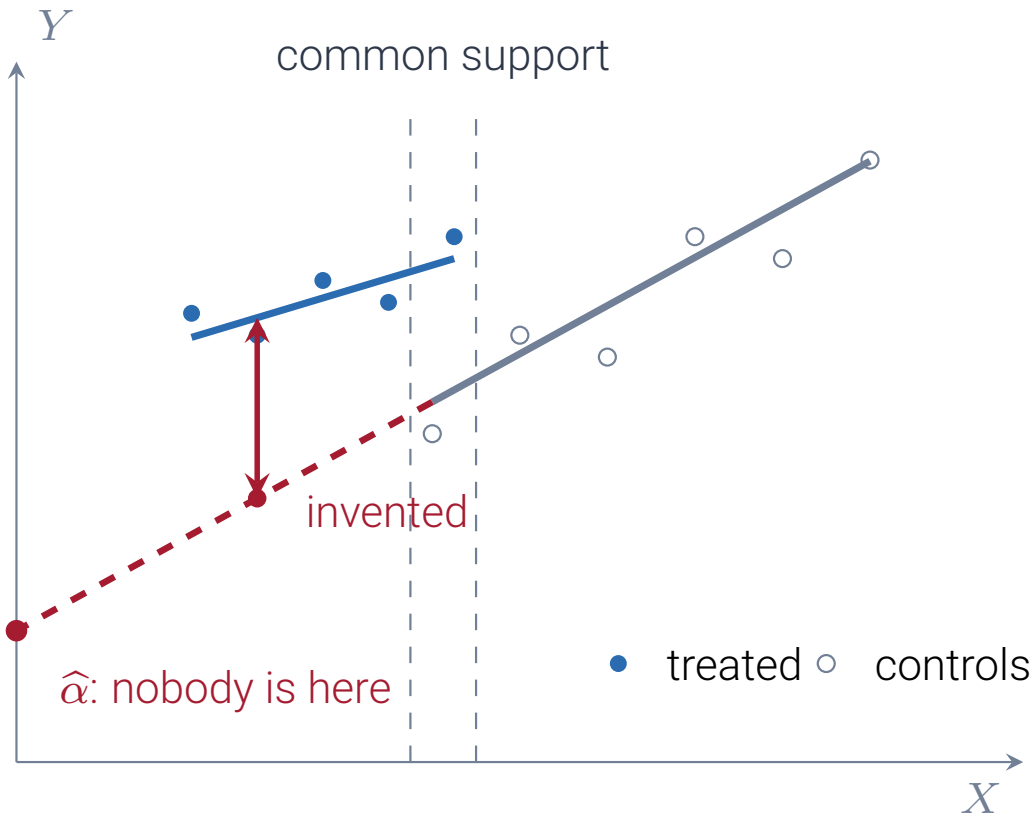
- `comparing.do` in `/Lab/Matching` runs all of it side by side
- Additive OLS, matching, bias-corrected matching, and regression adjustment
- Against a known $ATE = \$2{,}500$, $ATT = \$1{,}962$, $ATU = \$3{,}037$

But regression adjustment is not free either

- It still fits $\widehat{\mu}^0(X)$ where the treated live — often where no controls are
- That is extrapolation, and it comes from the functional form, not the data
- Matching refuses to do this. It just tells you it has no match
- Neither is safer. One hides the assumption, the other reports it

Losing common support is not a problem you escape by switching to a regression.

OLS doesn't stop at the edge of the data



- Solid where the group has data. Dashed is invented
- No control unit sits at $X = 2.2$ — that counterfactual is pure **slope**
- And $\widehat{\alpha}$ describes a person who isn't in the data

Why this happens: the curse of dimensionality

- One covariate and the gap is visible. You just saw it
- Add a second and you need overlap in the *rectangle*, not the margins
- k covariates, three bins each, and you have 3^k cells to fill

- Cells empty out long before you notice
- The propensity score *summarizes* those cells — it does not populate them
- So a balanced-looking weight can sit on top of cells with nobody in them

Two honest ways to lose

	Matching / weighting	**OLS / regression adjustment**
No match?	refuses, drops the unit	extrapolates from the slope
What you get	a smaller sample	a full-sample number
The assumption	visible	buried in functional form
Fails by	losing your estimand	quietly being wrong

- Both can be right. Both can be wrong
- **Neither one gives you back the data you don’t have**
- So go look at the cells before you pick

Concluding remarks

- Last years have shown a *particular* OLS specification yields biased estimates of all causal parameters when there exists heterogeneity in the treatment effects
  - → Goodman-Bacon (2021) for instance showed this with the common twoway fixed effects specification and dynamic treatment effects
- Turns out it was not a diff-in-diff problem only – it's an OLS problem more generally
  - → We saw it also in a paper by Goldsmith-Pinkham, Hull and Kolsar on contamination bias in linear models with multiple treatment effects and heterogeneity
- Here we see the perversity of the effects under unconfoundedness and heterogeneity which is the OLS coefficient counterinuititvely favors the smaller groups and therefore their causal parameters

Concluding remarks

- We mentioned work by Anna Aizer more likely is estimating coefficients that are more like the ATU than either the ATE or the ATT because of how large the treatment group is
- This has implications for analysis in which the proportion of groups differ in size due to population differences or over-representation of one group over another (i.e., Asians in criminal justice)
- The smaller the group, the larger their influence on the OLS coefficient
- Lesson here is that regression adjustment as well as nearest neighbor matching with bias adjustment may need to be your core models, not the additive in covariates OLS specification we are accustomed to

Roadmap

Introduction

Foundational ideas

Identification and Assumptions

Estimators

Advice

Design before analysis

- Almost all of the work happens *before* you touch the outcome
- Who chose the treatment? What did they know? Why did they do it?
- If you can’t answer that, you’re not ready to estimate anything
- The estimator is the last decision you make, not the first

Imbens and Xu (2024) put this first on their list too.

If you claim unconfoundedness, you owe us two pictures

- A **propensity score plot** by arm — so we can see the overlap
- A **balance table** with normalized differences — so we can see the imbalance

- Normalized differences, not t-stats
- Both are cheap. Both are observable. Their absence is a choice
- You must adjust for the confounders *insofar as they are imbalanced* — so show us which ones are

The real question: what causes Y^0?

- For the ATT, the thing you're missing is $E[Y^0|D=1]$
- So the covariates you need are the **causes of** Y^0
- Not "predictors of treatment." Not "whatever was in the extract"
- $X \rightarrow Y^0$ — that's the whole search

Heckman, Ichimura and Todd said the same thing in 1997: you want variables predictive of the untreated outcome.

How I actually find them

I use an AI agent to interview me. This prompt, more or less:

> "Interview me one question at a time using the idea that $X \rightarrow Y^0$ for $D = 0$ can be used to think about the covariates I need in my study. This is the common-sense set of things ordinarily responsible for the potential outcome Y^0. Ask me one question at a time, and at the end, suggest covariates."

- One question at a time is the important part
- It surfaces what I already know but hadn't written down

A second prompt I like even better

"Interview the world's leading expert on $X \rightarrow Y^0$. Then show me the table of contents of the book they're writing called $X \rightarrow Y^0$, where each chapter is a covariate."

- You get a structured list of the determinants of the untreated outcome
- Then ask which ones you have, and which ones you're missing
- **We know more than we think we know — or are willing to admit**

Levels today. Trends tomorrow

- **Unconfoundedness**: X has to explain the *level* of Y^0
- **Difference-in-differences**: X has to explain the *trend* in Y^0

- Different assumption, different covariates
- A variable that predicts where you start may say nothing about where you're going
- Worth knowing which one your design is actually asking for

Falsification: the pre-treatment period is free

- Same units, same confounders, same selection — but *no treatment*
- So any “effect” you estimate there is selection bias and nothing else
- Run your exact model where the answer must be zero

- Caveat: it measures the selection bias only *insofar as your model is right*
- A passing placebo doesn’t prove unconfoundedness
- A failing one does refute it — with the X you have

Imbens and Xu ran exactly this on LaLonde

- Placebo outcome: 1975 earnings, with 1975 dropped from the conditioning set
- Experimental benchmark ≈ 0, as it must be
- Every nonexperimental estimator returned a large negative placebo — diff-in-means on PSID was $-\$17{,}531$
- Still significant even after trimming

- Same test on the IRS lottery data (clear design, six lagged outcomes): **passes**
- That is the difference between a design you can defend and one you can't

Trimming is cheap — but it isn't free

- Imbens: trim for overlap. Crump et al. (2009) suggest $[0.1, 0.9]$
- It costs almost nothing in variance — Imbens and Xu dropped 99% of the CPS controls and the standard error rose about 30%

- But you are no longer estimating the ATT. You're estimating it *on the overlap region*
- With heterogeneous effects that is a different number, possibly very different
- Buyer beware — and say out loud what you dropped and who they were

Use more than one estimator

- Every review article of the last twenty years says this
- Matching, IPW, regression adjustment, doubly robust — they lean on slightly different assumptions
- So when they agree, that agreement is information

Imbens and Xu: once overlap is good, estimator choice stops mattering much. That is itself a diagnostic.

But agreement is not identification

- On trimmed LaLonde-CPS every estimator landed near the experimental ATT
- Encouraging. And still not enough

- The *same* data gave conditional effects ranging $-\$5{,}693$ to $\$7{,}364$
- Against an experimental range of $-\$329$ to $\$3{,}946$
- Getting the average right can hide getting everything else wrong
- Matching a benchmark can buy you unwarranted confidence

Imbens and Xu's checklist

Based on the lessons learned since LaLonde (1986) in nearly four decades, we offer the following recommendations to practitioners:

- Begin analyses of causal effects with an effort to understand the assignment mechanism. A clear grasp of the "design" is crucial for the credibility of the unconfoundedness assumption.
- Estimate the propensity score using a flexible method. Assess overlap by plotting the distributions of propensity scores for treated and control units. Trim the data based on the propensity score to make the groups more comparable.
- Apply modern methods, such as doubly-robust estimators, to estimate the average causal effects. Explore alternative estimands, such as the conditional average treatment effects and quantile treatment effects.
- Perform placebo tests, such as those using pretreatment outcomes, to validate unconfoundedness. Conduct sensitivity analyses to gauge the robustness of the findings.

We provide an online tutorial with `R` code to assist researchers in implementing these methods.

The curse of dimensionality bites before any estimator runs

- It happens at the top layer — in the raw cells of X, before you fit anything
- Propensity scores *summarize* the dimensions. They don't create matches
- If there's no one-to-one match within X, a balancing method can mask that rather than fix it

- When someone says they've handled it with a balancing weight, ask what the cells look like
- **Always go look at the dimensions**

Thank you

Northwestern, August 2026

- Slides, labs and data:
  github.com/Mixtape-Sessions/Causal-Inference-1
- Imbens and Xu (2024) tutorial and code:
  github.com/xuyiqing/lalonde

scunning@gmail.com